Natural Experiments in the Public Record of AI

The Complete natex Paper Collection

Hauke Hillebrandt

with AI-assisted analysis (natex v0.2.0)

Compiled 2026-07-19 • Latest versions:
haukehillebrandt.github.io/natex

Contents

Every estimate from seeded, reproducible runs; nulls and identified artifacts reported as first-class results.

Natural Experiments in the Public Record of AI: A Systematic Survey

Hauke Hillebrandt*

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analysis passes of record: 2026-07, `natex` v0.2.0; all stochastic steps seeded

Abstract

The public record of AI — benchmark scores, adoption surveys, capital expenditure, chip stocks, model registries, crowd-sourced battle streams — is full of dated events that are claimed, usually by eye, to have bent some curve. We survey a systematic attempt to test such claims with formal quasi-experimental designs, run and validated by the open-source `natex` toolkit: 17 dataset–design analyses carried to a verdict and 15 more triaged and excluded for cause, each subjected to a mandatory falsification battery (specification grids, shifted-cutoff placebo grids, placebo units and outcomes, density and covariate checks, pinned controls, honest refusals). Eight findings are selected here by credibility and interest. Three are credible: the 2025 BTOS questionnaire rewording moved measured US business AI adoption by +6.0 pp (4.7× the largest of 49 placebos), the October-2023 export controls bent China's legal AI-chip stock by −0.3 to −0.56 log-units/yr against every control group (25/25 specifications, loophole-round placebo null), and GPQA-Diamond's slope 2.4×'d at o1-preview with a clean placebo grid (0/7). One is moderate: an 83% post-Act deficit of models just above the EU AI Act's 10^{25}-FLOP line. Two are first-class negatives: the Epoch Capabilities Index shows no kink at o1 in a design with demonstrated power, and the sycophantic GPT-4o build won zero opponent-adjusted votes on LMArena. And two are cautionary: a sharp January-2025 adoption acceleration that cannot be attributed to DeepSeek-R1, and a hyperscaler capex "kink" at ChatGPT whose nominal $p = 4 \times 10^{-6}$ collapses to a placebo-calibrated 0.125. Across the corpus, roughly half of the nominally significant estimates die in the battery — on short, serially dependent, composition-churned series, the falsification battery is not a robustness appendix; it is the analysis.

1 Introduction

Public discussion of AI progress and AI policy runs on dated trend breaks. Reasoning models "changed the slope" of capabilities; ChatGPT "ignited" the datacenter buildout; export controls "choked" China's compute; DeepSeek "bent" the adoption curve; the EU AI Act "chilled" large training runs. Each claim names an event, a series, and a shape — which makes each one a candidate for a formal quasi-experimental design: a regression discontinuity or kink at a known cutoff [20, 35, 13], a difference-in-differences or synthetic control around a dated policy [2], bunching at a statutory threshold [71, 59, 58], or an interrupted time series with reversal [9]. Treating the claims this way buys an explicit identifying assumption, a standard error, and — most importantly — a falsification battery that can *reject* them.

*University College London. Email: `ucjthhi@ucl.ac.uk`. This paper and the underlying `natex` software were prepared with substantial assistance from Anthropic's Claude models; the author reviewed the analyses and text and is responsible for all remaining errors. Latest version: `https://haukehillebrandt.github.io/natex/`.

This paper is the capstone of a project that did this systematically. The instrument is `natex` [50], an open-source toolkit for automated natural-experiment discovery and estimation in the lineage of the LoRD3 local-discontinuity scan of [38, 37, 55], extended with kink and difference-in-kinks estimators following [13], a subset-scanning panel DiD, synthetic controls, and density/bunching tests. Over successive analysis passes the project pointed this machinery at the public quantitative record of AI — Epoch AI's model, benchmark, and chip databases [26, 28, 27], the Census Bureau's Business Trends and Outlook Survey (BTOS) [14], SEC EDGAR filings, LMArena's published battle stream [23], and semiconductor market data — carrying 17 dataset–design pairs to a recorded verdict and triaging 15 more as infeasible or ill-posed (Section 11 tabulates all 32). Each completed analysis is written up as a companion note with committed figure scripts that re-derive every headline number from the frozen inputs before drawing [39, 41, 46, 40, 42, 47, 45, 44, 43, 48, 49].

Here we select the eight strongest results by the product of credibility and interest, and — deliberately — count honest nulls and identified artifacts as first-class findings alongside the positive estimates. The selection spans the full verdict taxonomy the project converged on: *credible* (the estimate survives the battery), *descriptive-only* (the data feature is real but attribution to the named event fails), *null-with-power* (the dial reads zero where the test demonstrably can move), and *identified artifact* (the battery caught the machinery, or the platform, generating the result). Section 2 covers methods, Sections 3–10 the findings, Section 11 coverage, and Section 12 the lessons.

2 Methods: the toolkit and the validation philosophy

Design families. `natex` v0.2.0 implements, behind one CLI: local-polynomial regression discontinuity with the LoRD3 blind discontinuity scan and honest (split-sample) post-discovery 2SLS [38]; sharp and fuzzy regression kink and difference-in-kinks (DiK) estimators [20, 13]; a subset-scanning panel difference-in-differences ("SuDDDS") that searches units and break dates by likelihood-ratio scan, calibrated by permutation under dependence-preserving nulls; synthetic-control and related panel estimators [2, 1]; McCrary-style binned-Poisson density tests [63] used both as manipulation diagnostics and as bunching estimators; and event-study/interrupted-time-series segmentations [9]. House rules bind every run: a single seeded random generator threaded through all stochastic calls, discovery steps that never read the outcome, and failed computations that return NaN — a refusal — rather than a number.

The falsification battery. No estimate in the corpus is reported without its battery. The standing components: (i) specification grids over bandwidth, kernel, donut, and polynomial degree; (ii) *shifted-cutoff placebo grids* in the spirit of [32], read as separating bend *existence* from date *attribution* — rejections at non-event dates mean the series bends over an era or everywhere, not at the event; (iii) placebo-in-space (pseudo-treated units) and placebo outcomes; (iv) density and covariate discontinuity checks; (v) sibling-series falsifications — an aggregate in which the test demonstrably has power but should read null; and (vi) where available, *physically pinned controls* — units that cannot have been treated, whose movement identifies platform-side confounds (Section 8). Nominal HC1/CR1/HAC inference is reported but never trusted on its own: on short, serially correlated aggregates conventional robust standard errors are known to be badly oversized [10, 19, 35], so placebo-calibrated p-values and permutation ranks are the inference of record wherever the battery can supply them.

Table 1: The eight selected findings.

Finding (section)	Design / cutoff	Headline	Verdict
Export controls vs. China's compute (§3)	group DiK, chip stocks, 2023-10-17; SuDDDS DiD, big runs, 2022Q4	legal stock -0.56 ln/yr ($t = -3.3$; 25/25 specs); total stock $\approx$ half, fragile; big-run DiD +3.2/qtr fails placebo-in-space	credible / attenuated / descriptive
BTOS question rewording (§4)	measurement RDiT, wave 202524 (2025-11-03)	+6.04 pp, HAC $z = 16.3$; 4.7× max of 49 placebos; blind scan re-localizes	credible
EU AI Act 10^{25}-FLOP line (§5)	bunching / density, statutory threshold, post 2024-08-01	Fisher OR 0.173, $p = 0.007$; 82.7% missing mass; $\theta < 0$ in 15/15	credible (moderate)
GPQA-Diamond at o1-preview (§6)	sharp RKD-in-time, 2024-09-12	+0.00258 logit/day ($t = 3.3$); placebo size 0/7	credible, date-localized
US adoption at DeepSeek-R1 (§7)	group DiK + SuDDDS, BTOS sectors, 2025-01-20	DiK +7.3 pp/yr ($\lvert t\rvert = 2.6$); scan localizes exactly; placebo dates as large	descriptive-only
GPT-4o sycophancy episode (§8)	ABA ITS + Bradley–Terry, deploy 04-25, rollback 04-29	deploy style break real (perm $p = 0.013$); BT vote gain +0.02 log-odds (se 0.14): null	credible null / identified artifact
Hyperscaler capex at ChatGPT (§9)	group DiK, EDGAR panel, 2022-11-30	+0.140 dex/yr, nominal $p = 4\times10^{-6}$; placebo-calibrated $p = 0.125$	identified inference artifact
ECI at o1, fresh vintage (§10)	sharp RKD-in-time, 2024-09-12	+0.0065 pts/day ($t = 0.50$); placebos reject 4/8 elsewhere	null with power

Validation anchor. Before being trusted on novel data, the pipeline was run blind on the canonical Abadie–Diamond–Hainmueller Proposition-99 panel [2]: all three SuDDDS scan methods recover (California, 1989) exactly (precision = recall = 1.0), the deterministic simplex weights put summed weight 0.955 on ADH's five published donors, and the synthetic-control ATT of -19.5 packs per capita matches the canonical ≈ -19 average gap; the in-space RMSPE-ratio placebo ranks California 3/39 ($p = 0.077$) without ADH's poor-pre-fit exclusion, reported as the null it is [48]. The exercise has zero novelty by construction; its value is credibility transfer.

Audit lineage. Every analysis pass was produced under a two-stage protocol: an analysis agent runs the designs and freezes a numbers record; an independent audit pass re-derives the headline estimates from the frozen inputs, and the committed figure script of each companion note asserts every quoted number against that record before drawing. Audits are deliberately model-diverse after an early recorded failure in which same-family verifiers shared a PDF-extraction blind spot (dropped superscripts). Refusals, degenerate auto-configurations, and underpowered nulls of the automated pipeline are reported verbatim in each note ("stated as run"), not patched over.

3 Export controls and China's compute: three legs of one story

The October-2023 US export controls are the corpus's strongest policy result because the design decomposes the question [46]. *Leg 2* — the legal channel — is a group difference-in-kinks: the export-license schedule bends for China only at a common dated cutoff, so treated = China's legal

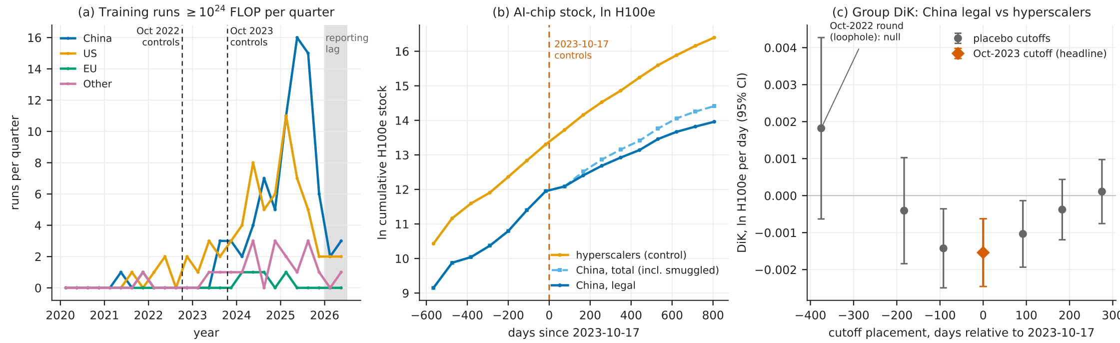

Figure 1: The three legs (reproduced from [46]): (a) quarterly $\geq 10^{24}$-FLOP training runs by country group; (b) ln cumulative H100e chip stock — hyperscaler controls, China legal, China total including smuggled — around the 2023-10-17 cutoff; (c) group-DiK estimates at the true cutoff and six placebo placements, with the Oct-2022 loophole round null at -375 days.

H100-equivalent chip stock, control = the four US hyperscalers, and the controls difference out the global supply bend. The headline DiK is -0.00154 ln-units/day (HC1 se 0.00047, $t = -3.30$; ≈ -0.56 ln-units/yr), negative in **25 of 25** specifications with a defensible band of -0.3 to -0.56 ln-units/yr. The falsifications land where the policy history says they should: the October-2022 first round — whose A800/H800 compliance loophole predicts no bite — is a null placebo (+0.0018, $p = 0.146$), and the placebo-treated group (Other vs. hyperscalers) is clean at the defensible bandwidths. *Leg 3* repeats the design on China's *total* stock including smuggled chips [56]: -0.00076 (se 0.00050, $t = -1.54$, 95% CI spanning zero) — half the size and fragile, because smuggled Nvidia compute grew from 27.8k to 662k H100e over 2024–2025 (doubling every 4.7 months) and now exceeds China's legal Nvidia stock 1.65×. *Leg 1* — a 4-country × 26-quarter DiD on counts of $\geq 10^{24}$-FLOP training runs — finds no suppression at all: the scan recovers {China} at 2022Q4 exactly (panel-null $p = 0.010$, the permutation floor), but the effect is *positive* (+3.2 runs/qtr, 95% CI $[+0.9, +5.5]$) and fails placebo-in-space at exact $p = 1.0$ — every pseudo-treated unit shows a larger studentized effect — so it is descriptive, not causal. The three-leg verdict: the controls bent the channel they legally govern, were substantially bypassed in aggregate, and left China's large-run training activity unbowed (Figure 1).

4 The question is the treatment: the BTOS rewording RDD

The cleanest single estimate in the corpus is a *measurement* natural experiment [41]. With wave 202524 (reference period from 2025-11-03; announced 2025-12-03), the Census Bureau reworded BTOS's headline AI question from use "in producing goods or services" to use "in any of its business functions" [21]. Treating the refresh as a known cutoff in a regression-discontinuity-in-time design on the spliced 71-wave national series [35], the jump is +6.039 pp (HC1 $z = 22.9$; HAC $z = 16.3$) against an old-wording level of 10.9 pp — a ratio of 1.553. The battery is uniformly supportive: all bandwidth, weighting, curvature, and HAC variants sit in a 6.0–7.5 pp band with $z > 12$; the largest of 49 placebo-cutoff estimates is 1.30 pp, 4.7× smaller (randomization $p = 0.020$, the 1/50 floor); and, told nothing, the LoRD3 scan's two top candidate boundaries exactly flank the known cutoff, with honest 2SLS +6.74 (95% CI $[6.01, 7.48]$). True adoption cannot produce it: the fitted pre-trend predicts +0.66 pp across the 50-day shutdown gap, and doubling that slope still leaves ≥ 5.38 pp. Nulls are reported as nulls (blind-scan randomization $p = 0.14$, underpowered at $n = 71$;

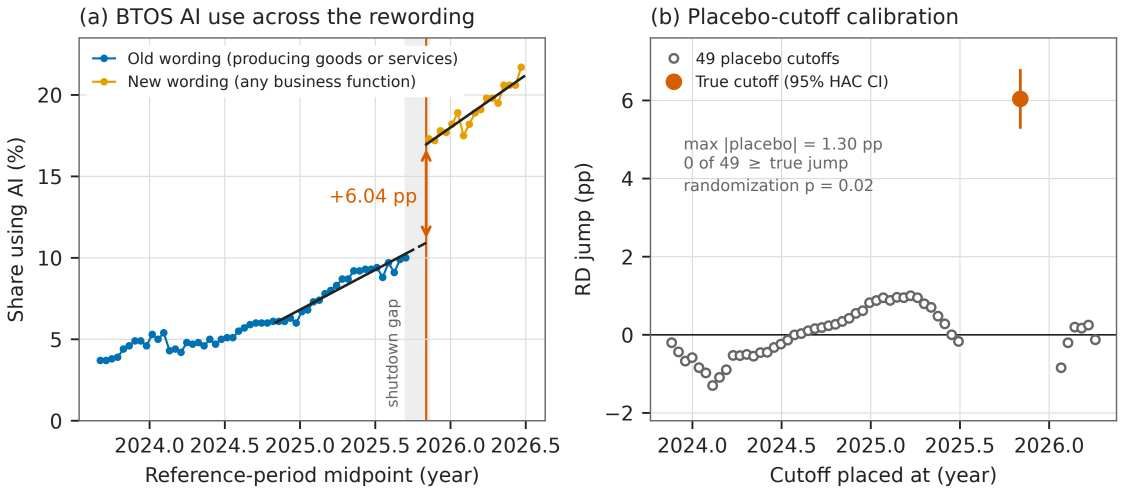

Figure 2: The rewording RDD (reproduced from [41]): (a) the spliced BTOS national series with per-side fits and the 50-day shutdown-gap extrapolation; (b) the same estimator at all 49 placebo cutoffs versus the true cutoff.

McCrary NaN — inapplicable on a uniform time grid). The estimate calibrates the splice factor — additive +6.04 pp, ratio 1.553 — needed to carry any old-wording BTOS analysis past December 2025, and implies that roughly a third of the post-refresh measured adoption level is attributable to asking a better question [12], with the caveat that "wording" bundles a simultaneous sample rotation and shutdown-adjacent composition change (Figure 2).

5 Bunching below the EU AI Act's 10^{25}-FLOP line

The EU AI Act's systemic-risk presumption at 10^{25} training FLOP [29] is a notch, and notches invite bunching [59]. In Epoch AI's compute panel (719 dated models with compute estimates), the post-Act density of new models just *above* the statutory line collapses [45]: a binned-Poisson density test at $\log_{10}$ FLOP $= 25.0$ is negative in 15/15 window $\times$ bin specifications (10/15 with $p < 0.05$), while the pre-Act period rejects in 0/15. The pre/post contrast is sharp: Fisher exact on below/above counts in the ± 0.5-dex window gives OR $= 0.173$ (95% CI $[0.048, 0.619]$, $p = 0.0072$) — 28.9 post-Act models expected in $(25.0, 25.5]$ at the pre-Act ratio, 5 observed, an 82.7% deficit. The deficit is specific to the statutory line (placebo thresholds at 24.5 and 25.5 are null or opposite-signed, with mass piling up just *below* 25.0), specific to the post-Act period (a placebo split date inside the pre-Act sample gives OR 0.90, $p = 1.0$), and correctly timed (the above-line count collapses from 2025H1, after entry into force, before applicability), with the classic signature that marginal crossers vanish while the frontier jumps far above the notch (Grok 3, GPT-4.5, Grok 4 at $\log_{10} = 26.5$–26.7). Two facts grade it moderate rather than conclusive: only 5–13 post-Act models sit above the line (the preferred narrow-window interaction is a null, $p = 0.73$), and an Act-induced *disclosure* response — labs going quiet about compute near the line — would mimic bunching exactly. Either reading matters for the compute-threshold debate [73, 53]: the data just above the line already look threshold-aware, contrary to the threshold-blind scaling assumed in forward projections [60] (Figure 3).

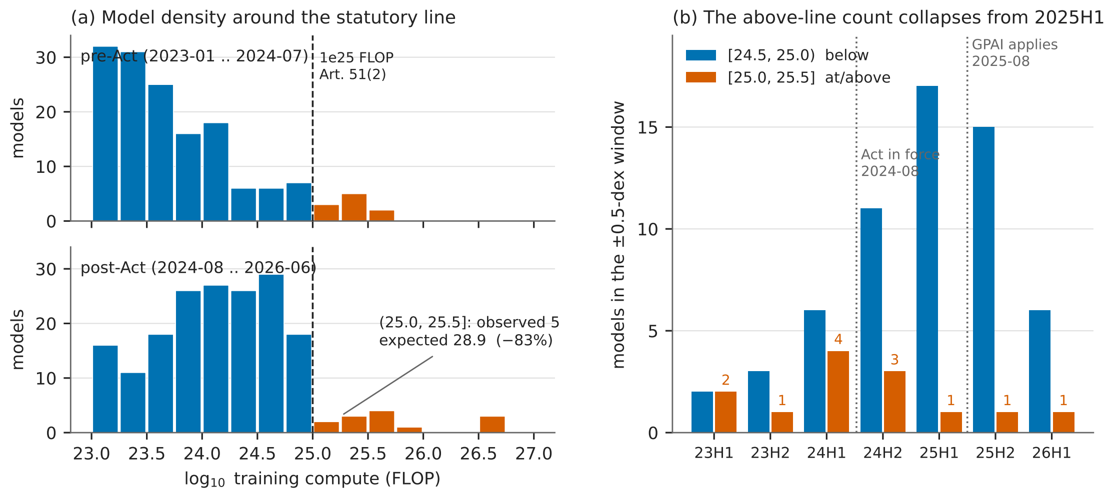

Figure 3: Bunching at the statutory line (reproduced from [45]): (a) model counts in 0.25-dex compute bins, pre- versus post-Act, around 10^{25} FLOP; (b) below/above counts per half-year — the above-line count collapses from 2025H1.

6 A date-localized capability kink: GPQA-Diamond at o1-preview

The flagship kink pass [39] supplies the survey's one date-localized capability result and its sharpest methodological contrast. GPQA-Diamond [70] logit-score slopes rise from 0.00188 to 0.00446 per day at the o1-preview release (2024-09-12) — roughly 15 $\rightarrow$ 34 raw points per year, a 2.4$\times$ acceleration. The headline RKD-in-time estimate is +0.00258 per day (se 0.00078, $t = 3.32$), positive in 8/8 bandwidth$\times$donut cells, with an *empty* placebo grid (0/7 shifted cutoffs reject), a null release-density kink ($t = 0.76$), and a null training-compute covariate kink ($p = 0.54$): the bend localizes to the date (Figure 4). The contrast: METR's 50% time-horizon series [61] shows an equally real bend (doubling time 9.9 $\rightarrow$ 3.5 months, $t = 2.13$, positive in 8/8 cells) whose pre-side placebos *also* reject ($p = 0.004$–0.030 at -90 to -270 days) — a credible reasoning-era slope change that cannot be pinned to 2024-09-12. The pair is the era-bend/event-bend taxonomy in a single dataset family, and the standing caveat travels with both: composition — reasoning models entering the release stream — *is* the mechanism, so these are properties of the release stream, not statements that individual models improved.

7 A sharp bend, no verdict: US adoption at DeepSeek-R1

The project's flagship adoption analysis ends in an instructive split verdict [40]. In the BTOS sector panel (20 sectors $\times$ 54 biweekly waves), a sector-by-wave DiD scan localizes an acceleration in AI-exposed sectors (information; finance; professional services) at $t_0 = 2025.089$ — *exactly* the first survey wave after DeepSeek-R1's release on 2025-01-20 [33] — with scan $p = 0.010$–0.030; the exposed-minus-unexposed gap slope more than triples (+4.2 $\rightarrow$ +13.9 pp/yr), and the group DiK at the release date is nominally significant (+7.30 pp/yr, CR1 se 2.80, $|t| = 2.61$). The localization survives composition checks and placebo-in-space (0/13 pseudo-treated control sectors). But attribution to R1 fails five ways: placebo cutoffs at non-event dates return DiKs as large or larger (+11.4 pp/yr at $t = 2024.30$, $|t| = 3.21$; +4.0 at 2024.50, $|t| = 4.53$); only 2/6 specification variants at R1 are significant; equally large local kinks appear in non-exposed sectors (real estate,

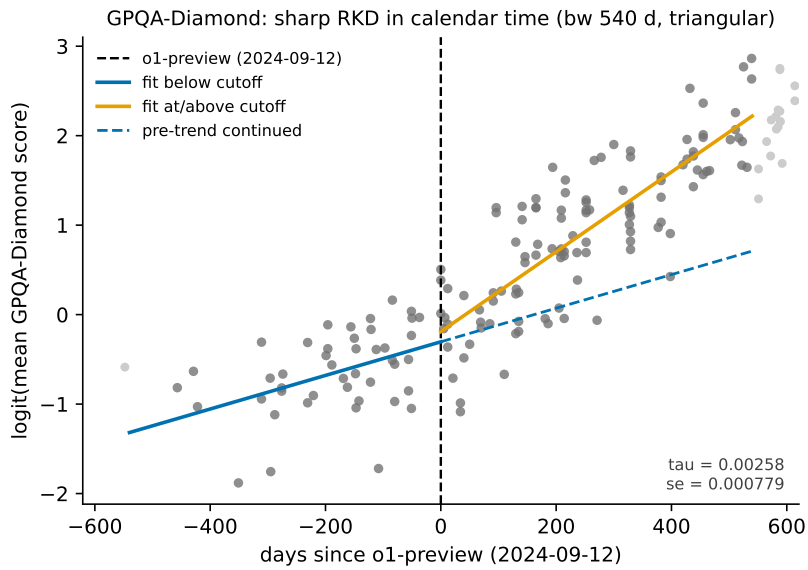

Figure 4: GPQA-Diamond, logit(mean score) against days since o1-preview (reproduced from [39]): per-side local-linear fits (bandwidth 540 days) versus the continued pre-trend; placebo empirical size 0/7.

arts, health care); the predicted *deceleration* at o1 sign-flips with bandwidth; and the cutoff week is confounded — the AI-diffusion export rule (2025-01-13), the Stargate announcement (2025-01-21), and a BTOS sample-year rollover in mid-December 2024 all sit inside the window. The verdict of record: something bent US business AI adoption in January 2025, and the localization is data-driven — but these data cannot say it was R1 (Figure 5).

8 Sycophancy did not win votes — and the rollback isn't what it looks like

The GPT-4o sycophancy episode of April 2025 [67] supplies the corpus's only deploy-and-rollback interrupted time series, inside 135,634 LMArena battles [47]. The *deploy* leg is credible: the live 4o alias shows a sharp markdown-style collapse timed to April 25–26 (headers per 1k tokens $3.52 \rightarrow 1.12$; sliding-cut permutation $p = 0.0128$, rank 1/78) while nine pinned dated snapshots stay flat — proving the arena copy was not pinned — and it delivers a precise null on the headline mechanism: the Bradley–Terry-adjusted win-rate change during the sycophancy window is +0.02 log-odds (SE 0.14; implied win rate $0.540 \rightarrow 0.545$). During the four days users saw the sycophantic build, it gained zero opponent-adjusted votes — a clean behavioral null on the preference-hacking mechanism the literature predicted [75]. The *rollback* leg is an identified artifact: on exactly 2025-04-29 the pinned `claude-3-7-sonnet-20250219` snapshot — whose weights cannot change — takes an equal-and-opposite persistent step in style *and* BT strength (−0.54 log-odds), demonstrating an arena-side serving change that contaminates every 29-April estimate. The 4o post-rollback regime never reverts (the episode is A–B–C, not A–B–A), and the widely repeated "sycophancy raised the win rate, rollback lowered it" narrative loads entirely on the confounded boundary. Pinned snapshots are the cheap, decisive placebo for platform time series (Figure 6).

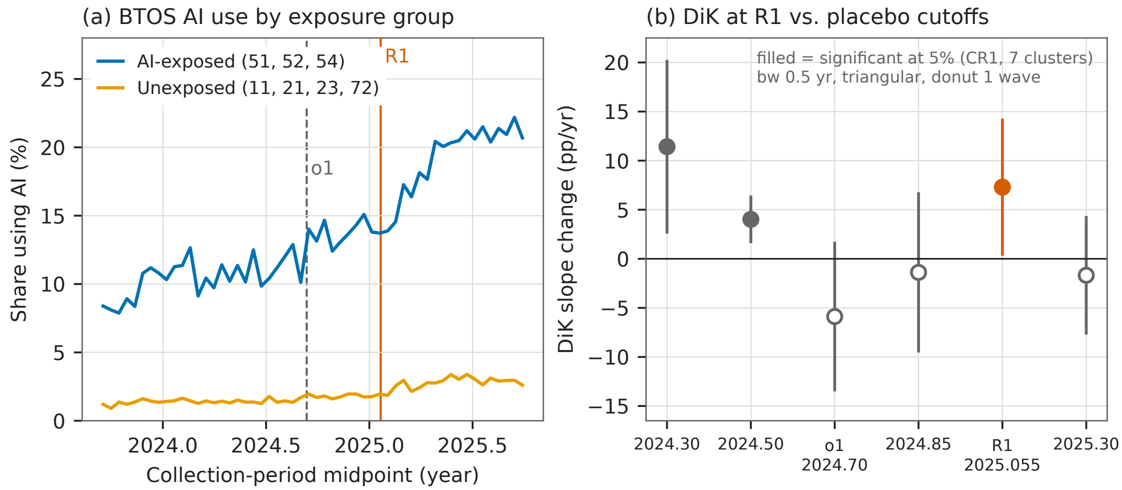

Figure 5: Adoption at R1 (reproduced from [40]): (a) BTOS AI-use share by exposure group with o1 and R1 marked; (b) group-DiK estimates at R1, o1, and four non-event placebo dates — two placebos are significant, one larger than the R1 estimate.

9 An identified inference artifact: hyperscaler capex at ChatGPT

The capex analysis is included as a first-class finding precisely because its battery caught the failure mode that would otherwise headline this literature [42]. Big-4 hyperscaler capital expenditure (EDGAR XBRL reconstruction) accelerates against a six-firm non-AI capital-intensive aggregate after ChatGPT: group DiK +0.1396 dex/yr (HC1 se 0.0302, nominal $p = 3.9 \times 10^{-6}$), positive in 18/18 specification cells, localized over 2023.0–2023.5 exactly as a lagged investment response would be, and visible per firm for Microsoft, Alphabet, and Amazon but null for Meta. Every conventional robustness check passes. What kills the causal reading is the clean pre-period placebo grid: at 15 non-event cutoffs in 2017–2020 the identical specification rejects at nominal 5% seven times (empirical size 0.47), the largest placebo $|z|$ (5.11, at 2019.75) exceeds the ChatGPT estimate's matched-bandwidth $|z|$ (4.35), a full-sample kink at the non-event date 2021.5 reaches $|z| = 6.9$, and the BIS export-control date eight weeks earlier is statistically indistinguishable (+0.125, $z = 4.0$). Placebo-calibrated, the ChatGPT estimate's p is 0.125–0.25: the nominal p-value overstates the evidence by five orders of magnitude, because HC1 inference assumes serial independence and the residual lag-1 autocorrelation reaches 0.66 [10]. The slope divergence is a genuine, robust data feature; the "significance" was an artifact of the standard errors, and only the battery could tell the two apart (Figure 7).

10 An honest null with power: the ECI at o1, fresh vintage

If "everything kinked in 2024," the per-benchmark results of Section 6 would be uninformative; the Epoch Capabilities Index (ECI) [51] is the aggregate guard, and it holds on a fresh vintage with twelve more months of data [44, 39]. The all-models RKD-in-time at o1-preview is +0.0065 ECI points/day (se 0.0131, $t = 0.50$, $p = 0.62$; slopes $\approx$ 22.3 vs. 24.7 points/yr), null in 17/18 grid cells — while the same specification rejects at 4/8 placebo cutoffs elsewhere in the series: the instrument demonstrably moves, and reads zero at the release date. The vintage's one new nominally significant result is itself a triage exhibit: a precision-weighted rerun (weights $1/\sigma_i^2$ from

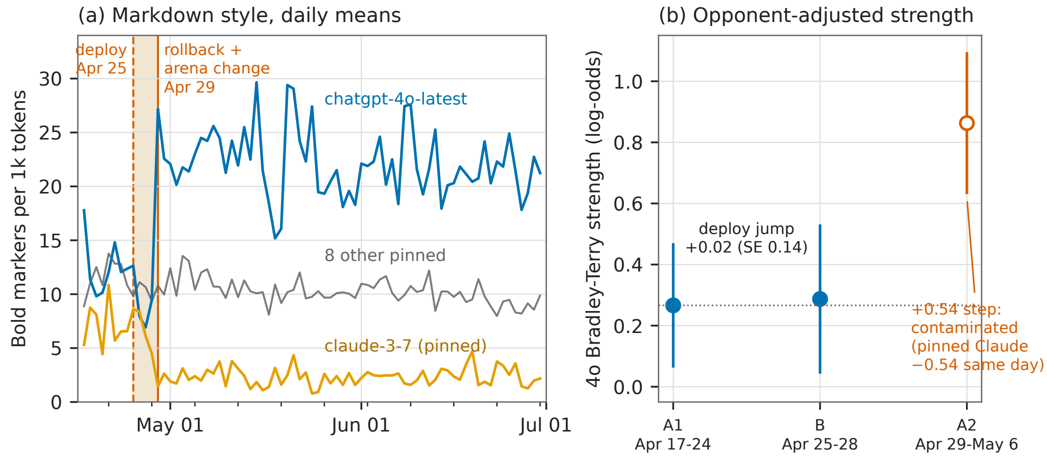

Figure 6: The deploy–rollback episode (reproduced from [47]): (a) daily bold-markers per 1k tokens for 4o, the pinned `claude-3-7` control, and pooled pinned snapshots — the pinned control steps, mirror-imaged, on exactly the rollback date; (b) 4o Bradley–Terry strength by segment: the deploy jump is +0.02 log-odds.

published CIs) returns a negative kink significant in 18/18 cells (−0.0190, $p = 0.016$) — which the battery identifies as smooth post-2024 concavity, not an o1-localized break, because placebo cutoffs at +90/+180 days give same-sign *larger* kinks (−0.0399, $p = 0.0064$; −0.0297, $p = 5.5 \times 10^{-6}$) and a local quadratic absorbs it ($p = 0.14$). A tempting third series — strict frontier records — is a non-design (4 pre-cutoff points in the estimation cell; its nominal $t = 2.28$ dies under a 30-day donut and flips sign under a 45-day one). Whatever the reasoning-model recipe did to AI capabilities, it did not bend the aggregate index's calendar-time trend at the release date (Figure 8).

11 Coverage: everything considered

The survey claim rests on Tables 2 and 3: every dataset–design pair the project considered, with its verdict or its reason for exclusion. Beyond the eight findings above, the studied set includes the flagship graveyard [39] — a $t = 12.9$ "kink" in cumulative GPU-cluster stock at ChatGPT that fails *every* placebo (empirical size 7/7: smooth super-exponential curvature bends everywhere), a sign-unstable Chinchilla null, and the EU threshold correctly rejected as a kink because it is a step with sorting — plus a blind LoRD3 scan that missed Chinchilla's adoption ramp and correctly diagnosed its own top discovery as a fine-tune measurement artifact [43], and a financial-series stress test in which every "credible" automated verdict on semiconductor stocks dissolved under inspection — the DeepSeek "kink" (Holm $p = 1.4 \times 10^{-8}$) is the 2025 AI rally sitting on both sides of a one-week crash [49].

12 Discussion

The scoreboard. Of 17 analyses carried to verdict, four estimates survive as credible (the rewording jump, the legal-channel export bend, the GPQA kink, and — graded moderate — the EU-Act bunching), two nulls carry power certificates (ECI at o1, twice; the LMArena vote null), and at least six nominally significant results were killed or reclassified by the battery: the R1 attribution, the capex kink, the CI-weighted ECI kink, the datacenter $t = 12.9$, the semiconductor

Table 2: Studied to verdict: the 17 dataset–design analyses of record.

Data × design	Cutoff / event	Verdict	Ref.
GPQA-Diamond logit scores × RKD-in-time	o1-preview 2024-09-12	credible, date-localized (placebo 0/7)	[39]
METR 50% time horizon × RKD-in-time	o1-preview	era bend real; date attribution fails (pre-side placebos 3/6)	[39]
ECI 2025 vintage × RKD-in-time	o1-preview	null with power (falsification guard)	[39]
ECI fresh vintage, unweighted / CI-weighted / frontier records × RKD	o1-preview	null upgraded; weighted "kink" = curvature artifact; records series non-estimable	[44]
China vs. hyperscaler chip stocks × group DiK (legal; total incl. smuggled)	controls 2023-10-17	credible (-0.3 to -0.56 ln/yr); total attenuated and fragile	[46]
Country panel, $\geq 10^{24}$-FLOP run counts × SuDDDS DiD + SC/GESS	controls 2022Q4	descriptive-only (placebo-in-space exact $p = 1.0$)	[46]
BTOS sector panel × group DiK + SuDDDS scan	DeepSeek-R1 2025-01-20	descriptive-only; attribution fails triage	[40]
BTOS national spliced series × measurement RDiT + blind LoRD3	rewording, wave 202524	credible; splice factor +6.04 pp / 1.553	[41]
EDGAR capex panel × group DiK	ChatGPT 2022-11-30	descriptive-only; nominal inference identified as artifact	[42]
LMArena battles × ABA ITS + windowed Bradley–Terry	deploy 2025-04-25, rollback 04-29	deploy leg credible (vote null); rollback leg identified arena-side artifact	[47]
Epoch compute panel × binned-Poisson bunching + Fisher contrasts	EU AI Act 10^{25} FLOP, 2024-08-01	credible (moderate): bunching, behavioral or reporting	[45]
EU AI Act threshold × kink/RD probes	10^{25} FLOP	rejected: step with sorting, wrong estimand shape	[39]
GPU-cluster cumulative stock × RKD-in-time	ChatGPT	rejected: placebo size 7/7 on super-exponential series	[39]
Tokens-per-parameter × RKD-in-time	Chinchilla 2022-03-29	sign-unstable null	[39]
Tokens-per-parameter × blind LoRD3 scan	(discovery)	honest miss: top discovery is a fine-tune measurement artifact, correctly diagnosed	[43]
Semiconductor weeklies × kinks, RDD, DiD scan	BIS rounds; DeepSeek week	all "credible" verdicts artifacts of trend regimes; informative DiD null	[49]
ADH Prop-99 panel × blind SuDDDS + synthetic control	1989	benchmark recovered exactly (validation anchor)	[48]

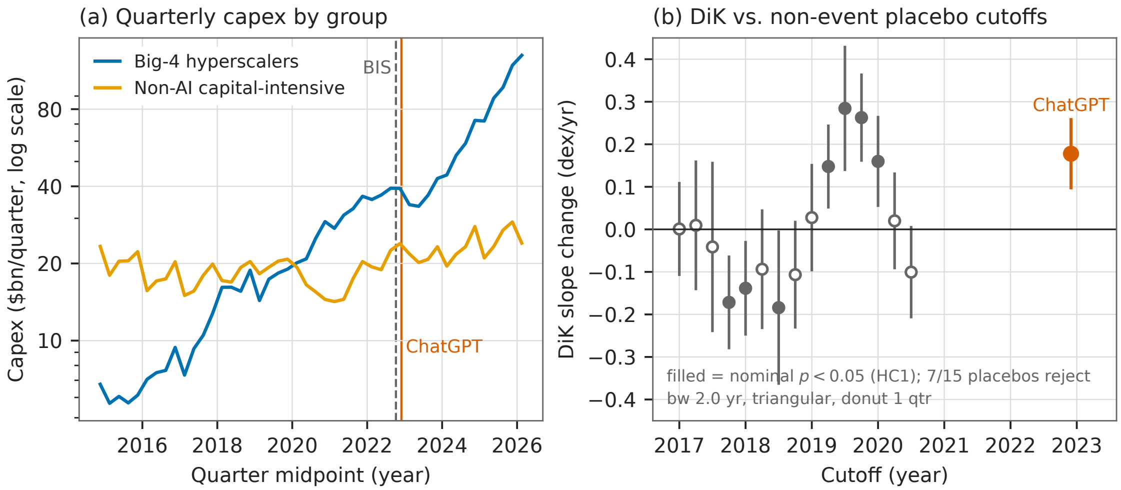

Figure 7: Capex at ChatGPT (reproduced from [42]): (a) quarterly capex of the big-4 and control aggregates with the BIS and ChatGPT dates marked; (b) DiK estimates at 15 pre-period non-event placebo cutoffs versus ChatGPT — seven placebos reject, and the largest placebo $|z|$ exceeds the ChatGPT $|z|$.

"kinks" at three cutoffs, and the LMArena rollback narrative. That kill rate — roughly half of everything that would have cleared a naive $p < 0.05$ bar — is the central empirical fact of the survey.

Nominal inference on the public record of AI is broken by default. The corpus's series are short, serially dependent, and smooth; HC1/CR1 standard errors on them manufacture $|z| > 6$ at dates where nothing happened (capex at 2021.5; datacenters everywhere; the DeepSeek rally kink). Placebo calibration measures the size distortion directly — a nominal 5% test rejecting 47% of the time in the capex pre-period — and is cheap. Every headline in this literature should be read against a placebo distribution or not at all [32, 10].

Era bends are not event bends, and calendar weeks are crowded. The shifted-cutoff grid mechanically separates "the trend changed in this era" (METR, the January-2025 adoption surge) from "this event changed the trend" (GPQA, the rewording). Even when a bend localizes, the week is shared: R1 sits with the diffusion rule, Stargate, and a panel rollover; ChatGPT sits eight weeks from the BIS controls; the export-control kink is localized only to ±1 quarter. Date-localization is necessary, never sufficient, for attribution.

Measurement is a treatment. The single largest causal effect in the corpus — +6 pp of "AI adoption" overnight — was produced by a questionnaire. The EU-Act deficit may partly be labs' disclosure behavior rather than training behavior; the BTOS rollover shifts respondent composition; LMArena's serving configuration moved a pinned model. Instrument changes mimic adoption, compliance mimics avoidance, and platform operations mimic model behavior; designs on the public record need measurement-side placebos (pinned controls, sibling instruments, archived outcomes) as much as treatment-side ones.

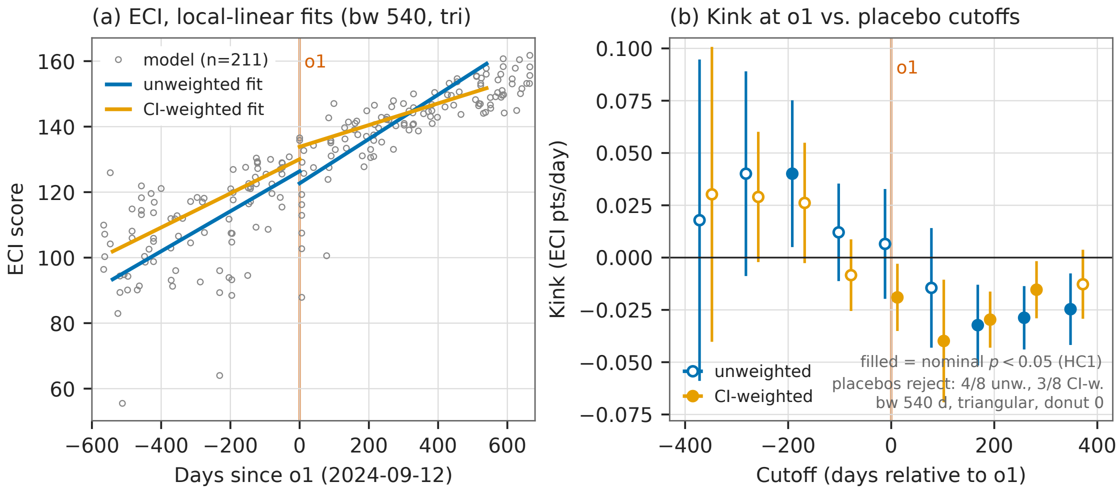

Figure 8: The ECI null (reproduced from [44]): (a) 211 fresh-vintage ECI scores against days since o1-preview with unweighted and CI-weighted per-side fits; (b) kink estimates at the o1 cutoff and eight placebo cutoffs, both weightings — the unweighted design rejects at 4/8 placebos while reading zero at o1.

Nulls and refusals are results. The ECI null is informative because the same specification rejects at 4/8 placebo cutoffs — the dial works and reads zero. The LMArena null is informative because the style break proves the treatment arrived. And the pipeline's refusals (NaN rather than a number when placebo pools are too small) repeatedly prevented fabricated certainty. An automated discovery stack earns trust not by what it finds but by what it declines to claim; the blind Prop-99 recovery, the blind re-localization of the rewording cutoff, and the honestly diagnosed Chinchilla miss are the same machinery validated, confirming, and self-correcting.

Limitations. Calendar-time designs rest on an untestable no-co-located-event assumption that placebo grids probe but cannot certify. Several verdicts lean on few units (4 country groups, 7 sector clusters, 6 quarterly points per cell) where all asymptotic inference is optimistic. Epoch compute figures are partly estimates; benchmark contamination drifts; and every capability result is a property of the release stream, not of individual systems. The survey covers the public record through mid-2026; the queued designs in Table 3 continue it.

Reproducibility

All estimates were produced with the open-source `natex` toolkit [50] (v0.2.0, seeds declared per note) from public data that are never committed to the repository. Every figure in this paper is reused verbatim from a companion note's committed figure, and each of those figures regenerates deterministically from a script that asserts the quoted headline numbers against the analysis records before drawing. The companion notes [39, 41, 46, 40, 42, 47, 45, 44, 43, 48, 49] carry the full specification grids, placebo batteries, and honest-inference notes summarized here.

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Table 3: Considered and excluded: 15 candidate pairs with reasons of record.

Candidate pair	Reason excluded
DCM inference-price series × kink at DeepSeek-V3	series ends 2025-02 with five flat terminal months: no post-slope identifiable
Epoch cyber-vulnerabilities × post-reasoning event study	no processed dataset exists (the published download is the raw CVE git repository)
BTOS state panel × Bartik-style DiD	requires external ACS/BLS exposure weights; ∼40% of state-level cells disclosure-suppressed
R&D input-share table × any family	$n = 7$ rows; no quasi-experimental design applies
GPQA/METR o1 kinks; China DiK × standalone mini-papers	already results of record in [39]; duplication adds nothing
GPU-cluster / datacenter cumulative growth × standalone kink mini-paper	already rejected in the flagship graveyard (placebo empirical size 1.0: curvature, not kink); no standalone treatment
Chip sales + components × chip-level SuDDDS at the Apr-2025 H20 license round	needs a shared chip-alias map and ragged-2026 panel preparation; strongest queued follow-up
Datacenter site-level panel × SuDDDS at Colossus/Stargate entry	18% of timeline rows are unflagged projections requiring filtering; queued
LMArena leaderboard history × RKD around model-spec releases	cumulative Bradley–Terry ratings are serially cumulative series; the GPU-cluster placebo failure is the direct precedent
LMArena style-premium DiD around Model Spec / constitution updates	spec-with-model-update confound requires judge-score validation not yet run
AI-company revenue/WAU × kink at ChatGPT	zero pre-cutoff mass at the treatment date; not formalizable
Polling cross-section × threshold RDs (income step, age cliff)	single survey wave; no microdata, sample sizes, or field dates
Values-in-the-wild / CCAI votes × any design	one aggregate snapshot; no behavioral treatment channel
Benchmark panel × teaching-to-the-test DEE with IRT difficulties	requires a 52-benchmark IRT panel build; queued
All-models panel × bunching at 10^{26} FLOP (US EO line) + disclosure-as-outcome scan	only 3 models ever above the line; no identifiable density contrast

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Three Kinks and a Null: Formal Trend-Break Tests on the Public Record of AI Progress

Hauke Hillebrandt*

July 2026
analysis pass of record: 2026-07-16, `natex` v0.2.0

Abstract

Claims that AI progress "bent" at some dated event — a model release, an export-control round, a regulation — are usually established by eye. We subject four such claims to formal regression kink designs in calendar time and a difference-in-kinks (DiK) design, using the `natex` v0.2.0 estimators (following Böckerman, Jysmä, and Kanninen, 2025) on Epoch AI's public datasets, with a mandatory falsification battery: bandwidth–donut sensitivity grids, shifted-cutoff placebo-kink grids, density and covariate kinks, and a sibling-series falsification. Three kinks and one null survive. GPQA-Diamond scores kink at the o1-preview release (slope roughly 2.4×, placebo empirical size 0/7 — the bend localizes to the date). The METR 50% time-horizon slope nearly triples, but pre-side placebos also reject: a credible reasoning-era bend whose attribution to 2024-09-12 fails. The Epoch Capabilities Index, run as the guard against "everything kinked in 2024," is a clean null at the same date in a series where the test demonstrably has power. And China's legal AI-chip stock bends down by ≈ 0.56 log-units per year relative to hyperscaler controls at the October 2023 export controls, with the October 2022 round as a passing placebo and a smuggling-attenuated total-stock estimate about half the size and fragile. A graveyard of rejected designs — a $t = 12.9$ "kink" that fails every placebo, a sign-unstable Chinchilla null, and an EU AI Act threshold that is a step with bunching rather than a kink — illustrates why eyeballed trend breaks need placebo grids.

1 Introduction

The public conversation about AI progress is full of dated trend breaks. Reasoning models "changed the slope" in late 2024; export controls "bent" China's compute accumulation; the EU AI Act "chilled" large training runs. These claims are almost always established by drawing a line through a scatter plot and pointing at a date. Yet a slope change at a known cutoff is precisely the estimand of the regression kink design (RKD), and a slope change for one group relative to another at the same cutoff is the estimand of the difference-in-kinks (DiK) design formalized by Böckerman, Jysmä, and Kanninen [1]. Treating eyeballed breaks as formal kink candidates buys three things: an explicit identifying assumption, a standard error, and — most importantly — a falsification battery that can *reject* the claim.

*University College London. Email: `ucjthhi@ucl.ac.uk`. This paper and the underlying `natex` software were prepared with substantial assistance from Anthropic's Claude models; the author reviewed the analyses and text and is responsible for all remaining errors. Latest version: `https://haukehillebrandt.github.io/natex/`.

This paper runs that battery on four trend-break claims drawn from Epoch AI's public datasets [3]. The tooling is `natex` v0.2.0 [4], an open-source natural-experiment toolkit in the lineage of automated discontinuity-design discovery [2], whose kink module implements sharp and fuzzy RKD and DiK estimators following [1]. We emphasize that this is known-cutoff candidate *evaluation*, not unknown-kink discovery: every cutoff below is externally dated (a model release, a regulation date), and none was searched for. Searching for cutoffs would require search-calibrated selective inference beyond the scope of both the source paper and this exercise.

Why do eyeballed breaks need placebo grids? Because a calendar-time kink test asks a sharper question than the eye does. A series can genuinely accelerate over an *era* while providing no evidence that the bend happened at any particular *date*; conversely, a smoothly super-exponential series will show a nominally enormous "kink" at every date one tests. The shifted-cutoff placebo grid separates these cases: a significant kink at the true cutoff alongside significant kinks at shifted cutoffs means bend existence without date attribution, and rejections everywhere mean curvature masquerading as a kink. Our four headline results span this taxonomy almost perfectly — one date-localized kink, one era bend that fails date attribution, one clean null run deliberately as a falsification guard, and one difference-in-kinks with a passing placebo round — and the graveyard (Section 5) shows each failure mode in the wild.

2 Data

All series are public Epoch AI datasets (CC-BY 4.0, `https://epoch.ai/data`) [3], retrieved July 2026; the analysis pass of record is dated 2026-07-16 and used `natex` v0.2.0. No data files ship with this paper; the figure pipeline (`paper/figures/make_figures.py` in the repository) reads a local extraction of the public CSVs, recomputes every headline estimate, and asserts it against the numbers of record before drawing.

GPQA-Diamond. Benchmarking-Hub series `gpqa_diamond`: 180 dated models (45 pre / 135 post cutoff), sample starting 2023-03-14. Primary outcome is logit(mean score): the score ceiling at 1 bends raw-score slopes mechanically near the top, and the logit removes that artifact. GPQA is the graduate-level science QA benchmark of [6].

METR 50% time horizon. `metr_time_horizons_external` (METR's long-task suite [5] as mirrored by Epoch): 48 dated models (12 pre / 36 post). Outcome is $\log_2$ of the 50%-success time horizon in minutes.

Epoch Capabilities Index (ECI). `epoch_capabilities_index`, an IRT-linked aggregate capability index; 356 dated, scored models. Used as the sibling-series falsification.

AI chip stocks. `ai_chip_owners/cumulative_by_designer`: quarterly cumulative compute stocks in H100-equivalents by owner. Outcome is ln(cumulative H100e); treated series China, control series the hyperscalers (Amazon + Microsoft + Google + Meta). The 2026-03-31 quarter is flagged `Incomplete` and its "cumulative" totals *decrease*; it is dropped.

3 Methods

3.1 Estimators

Let x be the running variable centered at the cutoff and let $m[\text{s}]$ denote the one-sided derivative of $E[Y \mid x]$ at zero on side s. `natex` defines every kink right-minus-left,

$$\kappa \;=\; m[\text{right}] - m[\text{left}], \tag{1}$$

estimated by local polynomial fits in which each side (and each group cell for DiK) receives its own intercept and slope, weighted by a kernel in $|x| \leq$ bandwidth (the weighted objective of [1], Equation 8). Defaults throughout: local linear fits, triangular kernel; uniform kernel, second-degree fits, and donut exclusions appear as explicit sensitivity choices. The sharp DiK estimand contrasts two such kinks,

$$\tau_{\text{DiK}} \;=\; \left(\kappa_{\text{treated}} - \kappa_{\text{control}}\right)/\Delta, \tag{2}$$

with Δ the known policy-kink change. In the China design the paper's pre/post *periods* are aliased to control/treated *groups* at a common calendar cutoff: the export-license schedule bends for China only, so the Böckerman contrast is taken across groups and the controls difference out the global supply bend. Identification is parallel kinks: absent the controls, China's growth bend would have matched the hyperscalers'.

All calendar-time runs use a dummy unit denominator ($\Delta = 1$, the `--policy-kink 1.0` convention), so τ is a *descriptive slope change* in outcome units per day, not a marginal causal response to a measured policy variable.

3.2 Time as the running variable

The governing caveat, quoted verbatim from the `natex` method card, `docs/method_cards/kink.md`:

> Calendar-time RKD — "did the trend bend at this dated event?" — is a legitimate use of the estimator but **not** a manipulable-running-variable design: no unit sorts itself across a date, so density/manipulation arguments give no protection. Identification reduces to a single untestable assumption: *no co-located slope-changing event* — nothing else may bend the outcome's expected slope at that exact date.

Two practices are therefore mandatory rather than optional. First, always run the shifted-cutoff *placebo-kink grid* and read it as separating bend existence from date attribution: a significant kink at the true cutoff plus significant kinks at shifted cutoffs means the series bends over an era, not at the event. Second, run a *sibling-series falsification*: an aggregate or related series in which the same test demonstrably has power but reads null at the candidate date. Sections 4.2 and 4.3 show both practices doing real work.

3.3 Inference

Standard errors are HC1 sandwich by default, with the degrees-of-freedom correction applied *jointly* across all side/period cells ($n - k$ counts every observation and coefficient of the stacked regression). Cross-checks fitting each side separately with per-side $n - k$ reproduce the point

Series (source)	Design / cutoff	Headline (HC1)	Verdict
GPQA-Diamond, logit(mean score), 180 models	sharp RKD-in-time at o1-preview, 2024-09-12	+0.00258/day, se 0.00078, $t = 3.32$ (bw 540, tri)	credible kink, date-localized
METR 50% time horizon, $\log_2$ minutes, 48 models	sharp RKD-in-time at o1-preview, 2024-09-12	+0.00601/day, se 0.00282, $t = 2.13$ (bw 720, tri)	credible era bend; date attribution fails
Epoch Capabilities Index, all models	sharp RKD-in-time at o1-preview (falsification)	−0.00725 pts/day, se 0.00895, $t = -0.81$ (bw 540, tri)	clean null — the guard the other two need
China vs. hyperscaler ln cumulative H100e stock	sharp group-DiK at export controls, 2023-10-17	−0.00154/day, se 0.00047, $t = -3.30$ (bw 548, tri)	credible kink with a magnitude honesty band

Table 1: The four formal results of the pass (`natex` v0.2.0, HC1 standard errors). "tri" = triangular kernel; bandwidths in days.

estimate exactly and report SEs roughly 5% larger at $n \approx 50$: in the METR cross-validation (uniform kernel, bandwidth 720), `natex` and an independent per-side OLS fit agreed on the kink to 10 decimals (both 0.0067291763) with SEs 0.0023236 (joint-cell) versus 0.0024334 (per-side) — a convention difference, not an error. Cluster-robust CR1 covariance with $t(G-1)$ critical values is used where clustering is stated. Wald intervals are conventional local-polynomial inference and may retain smoothing bias; degree-2 and bandwidth sensitivity are part of the required battery. Failed computations return NaN, never zero. The diagnostics battery — bandwidth×donut sensitivity grids, placebo-kink grids with empirical size, density kinks, covariate kinks — follows the validation figures of [1] as implemented in `natex.kink`.

4 Results

Table 1 summarizes the four designs. All candidates had externally dated cutoffs; none was searched for.

4.1 GPQA-Diamond at o1-preview: a date-localized kink

The logit-score slope changes from 0.00188 to 0.00446 logit-units per day at the o1-preview release — in raw terms roughly 15 → 34 percentage points per year, a $\sim 2.4\times$ acceleration (Figure 1). The headline estimate is +0.00258 per day (se 0.00078, $t = 3.32$; bandwidth 540, triangular kernel, cells 44/111). The battery is uniformly supportive: the estimate is positive in 8/8 bandwidth×donut cells (t from 2.15 to 3.71), and the donut *strengthens* it, ruling out single-point dependence at the cutoff; the placebo-kink grid has empirical size **0/7** — no shifted cutoff rejects; a McCrary-style release-date density kink is null ($t = 0.76$), so the score bend is not an artifact of a release-frequency bend; a training-compute covariate kink is null ($p = 0.54$); and a degree-2 fit keeps the sign and magnitude at three times the standard error. This is the cleanest result of the pass: the bend localizes to the date.

The standing caveat belongs in every sentence of interpretation: composition — reasoning models entering the release stream — *is* the mechanism, so τ is a property of the release stream,

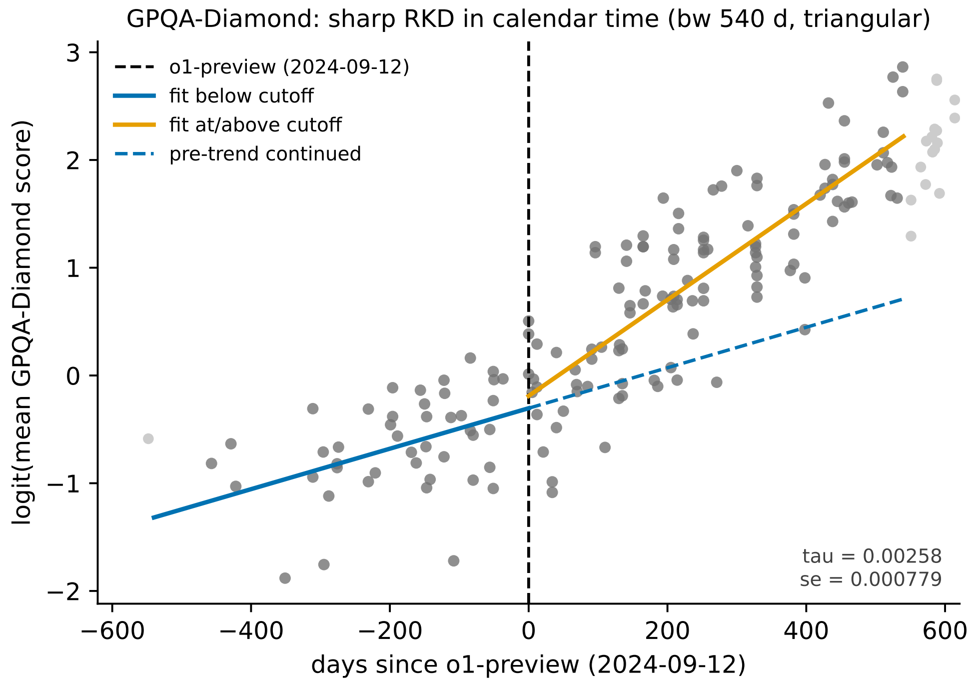

Figure 1: GPQA-Diamond, logit(mean score) against days since o1-preview (2024-09-12). Solid lines: kernel-weighted local-linear fits on each side (bandwidth 540 days, triangular); dashed: the pre-cutoff trend continued. The slope roughly 2.4×'s at the cutoff, and the placebo grid (empirical size 0/7) localizes the bend to the date.

not a statement that individual models improved. Benchmark contamination drifting over time remains an unmodeled confounder.

4.2 METR time horizon: a real bend that cannot be dated

The $\log_2$ 50%-time-horizon slope rises from 0.00331 to 0.00932 per day at bandwidth 720 (triangular): a doubling time of 9.9 months before the cutoff falling to 3.5 months after (Figure 2). The headline kink is +0.00601 (se 0.00282, $t = 2.13$); the full-sample uniform-kernel fit gives +0.00491 (se 0.00095, $t = 5.17$; doubling time 7.6 → 3.6 months). The estimate is positive in 8/8 bandwidth×donut cells (0.0044–0.0085), and the 80%-horizon variant agrees (+0.00454, $t = 4.08$), though with only three pre-cutoff points it is directional corroboration only.

But the placebo grid smears on the pre side: at bandwidth 720, shifted cutoffs at −270, −180, and −90 days all reject ($p = 0.030$, 0.014, 0.004) with estimates the size of the headline kink, while post-side shifts are clean nulls (empirical size 3/6). With 11–12 pre-cutoff models, adjacent placebo windows share most of their data, and the design cannot distinguish 2024-09-12 from any date within roughly ±270 days on the pre side. The honest verdict, quoted from the analysis of record: *"credible slope change with failed date-localization — report as reasoning-era slope doubling, not 'o1-preview caused X'."* This is the era-bend/event-bend contrast with Section 4.1 in a single dataset pair.

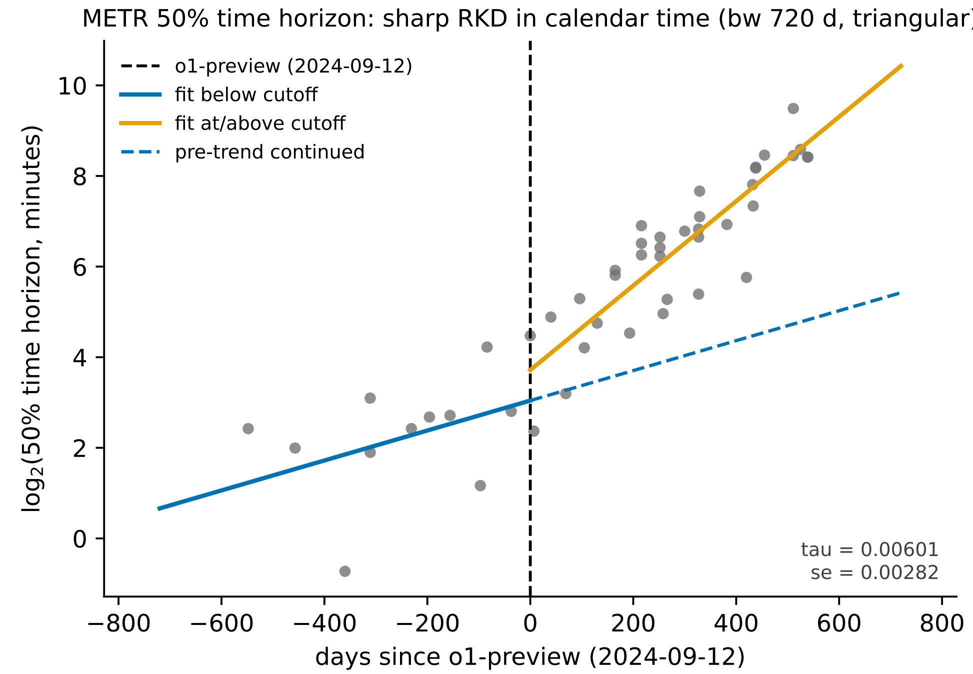

Figure 2: METR 50% time horizon ($\log_2$ minutes) against days since o1-preview. The slope nearly triples (doubling time 9.9 → 3.5 months at bandwidth 720), but pre-side placebo cutoffs also reject: an era bend, not a dated event.

4.3 The Epoch Capabilities Index: a null with power

If "everything kinked in 2024" — a global measurement or composition shift — then per-benchmark kinks would be uninformative. The ECI falsification run was launched *expecting* a null, and it delivered one: at the o1 date the all-models kink is null in 8/8 grid cells (t from −0.60 to −1.54; headline −0.00725 points/day, se 0.00895), with the index gaining roughly 18 points per year on both sides of the cutoff (Figure 3). Crucially, the same series' placebo grid rejects at 4/7 shifted cutoffs — the test demonstrably has power in this data, from composition-churn bends elsewhere in the series — and still reads zero at 2024-09-12. The instrument works; the dial reads zero. The per-benchmark bends of Sections 4.1–4.2 are therefore not a global measurement artifact. This is partly by construction (IRT linking smooths regime shifts) and partly because o1-era gains concentrate in reasoning-heavy benchmarks.

Two recorded lessons ride along. First, an analyst-script frontier filter — `cummax().diff().fillna(1) > 0` over a series with 231 NaN scores — manufactured 347 false "frontier records"; the clean frontier has 25 points and is *not* an estimable design — four left-cell points, failing donut, bandwidth-direction, and placebo checks in every direction — and should never be quoted. Second, a side finding: the training-compute covariate kink among ECI-listed models is −0.00265 $\log_{10}$ FLOP/day at the o1 date ($p = 0.021$), corroborating the "pretraining-compute frontier stalls" narrative without itself being a design.

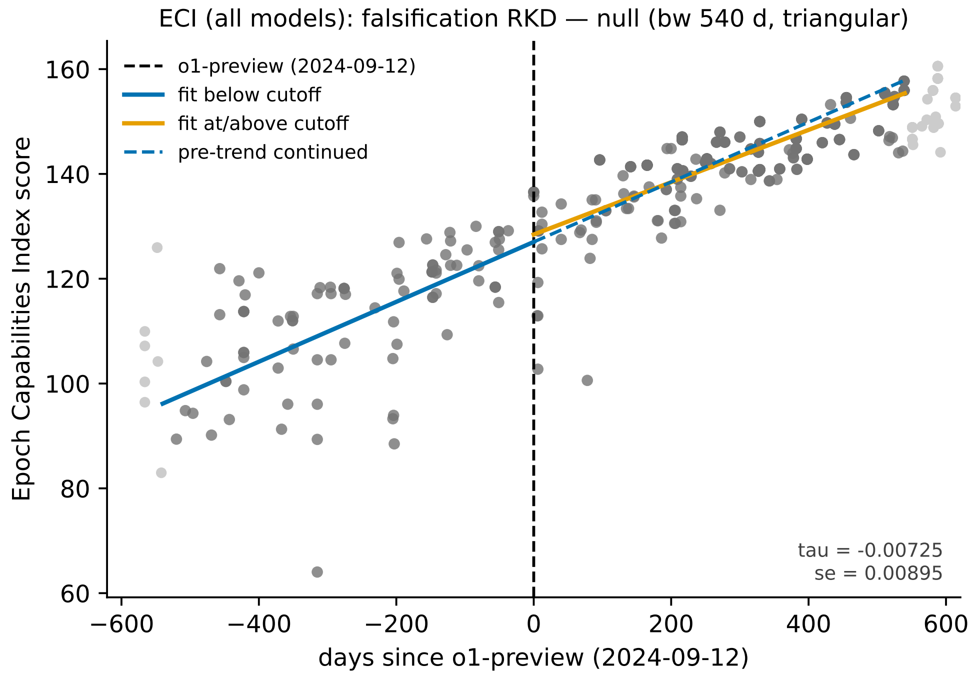

Figure 3: Epoch Capabilities Index (all models) against days since o1-preview. The two-sided fit and the continued pre-trend nearly coincide: a clean null ($t = -0.81$) in a series where the placebo grid shows the test has power (4/7 shifted cutoffs reject). The guard the per-benchmark kinks need.

4.4 China's chip stock at the October 2023 export controls

The only true difference-in-kinks structure in the corpus: the US export-license schedule bends for China only, at a common dated cutoff (2023-10-17), so treated = China and control = hyperscalers (Amazon + Microsoft + Google + Meta), and the controls difference out the global supply bend (their own kink: -0.00067/day). The legal-stock DiK is -0.00154 log-units per day (se 0.00047, $t = -3.30$; bandwidth 548, triangular), i.e. a growth-rate change of roughly **$-$0.56 log-units per year** (Figure 4). By-year CR1 clustering keeps significance ($t = -5.15$) with the explicit few-cluster caveat ($G = 4$). The estimate is negative in **25/25** specifications.

Falsifications behave: the October 2022 first-round placebo — the round whose A800/H800 compliance loophole predicts no bite — is null/positive (+0.0018, $p = 0.15$), and the placebo-treated group ("Other" vs. hyperscalers) is clean at bandwidths 548–730 while rejecting at 365, so the defensible range is bandwidth 548–730 with DiK -0.0009 to -0.0015. The honesty band: the *total*-stock series including smuggled chips gives a DiK roughly half the size and fragile (-0.00076, $t = -1.54$ at bandwidth 548) — the smuggled series, which starts 2024-03-31 and is itself a treatment response, substitutes for lost legal supply. Serial dependence in a six-points-per-cell cumulative series makes every $|t|$ optimistic; the sign stability is the real evidence. The verdict of record: *"the policy bent the legal channel ≈ −0.3 to −0.56 ln-units/yr; the effect on China's actual compute stock is smaller and fragile."*

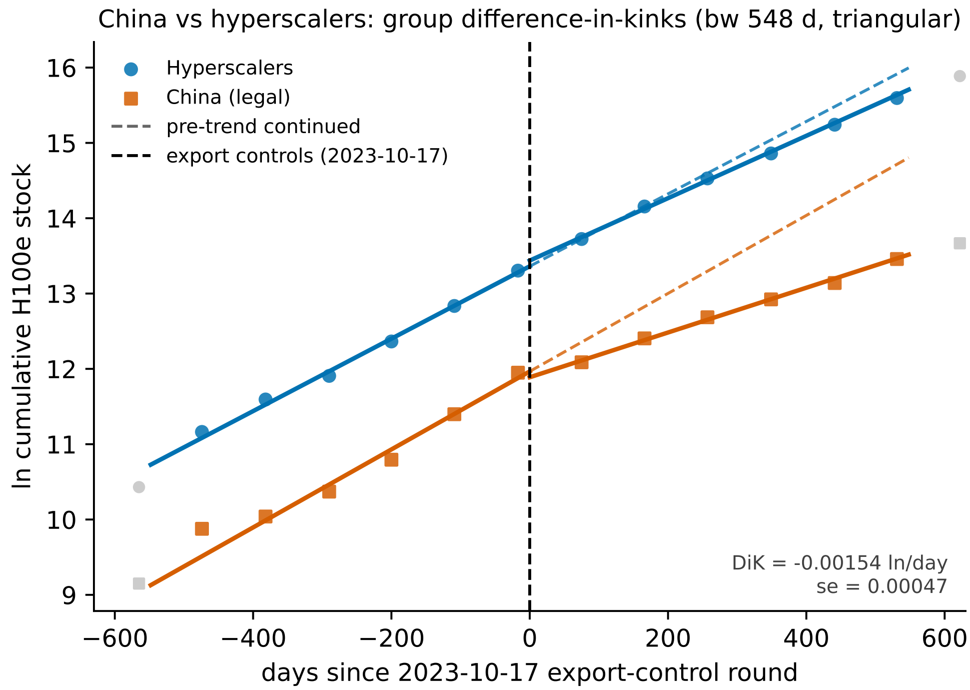

Figure 4: Group difference-in-kinks: ln cumulative H100e stock for China (legal) and the hyperscaler controls against days since the 2023-10-17 export-control round (bandwidth 548 days, triangular). Both series bend at the cutoff; the DiK is their difference, -0.00154 log-units/day (≈ -0.56/yr).

5 The graveyard

Rejections are results. Three claims that look at least as impressive as Table 1 in a scatter plot died in the battery.

Datacenter growth "kinked at ChatGPT." The GPU-cluster series (ln cumulative H100e by first-operational date) shows a nominal kink at ChatGPT's release with $t = 12.9$ — the largest t-statistic of the entire exercise. Its placebo grid rejects at **7/7** shifted cutoffs (empirical size 1.0): a smoothly super-exponential cumulative series bends *everywhere*, and HC1 inference on serially dependent cumulative data is meaningless. The design was discarded; the estimate is descriptive curvature, not a kink. (The dataset variant named in the original claim, datacenter power timelines, was infeasible outright: one pre-ChatGPT observation.) This is the cautionary tale for every "the curve bent at [event]" plot drawn on cumulative infrastructure data.

Chinchilla. Tokens-per-parameter among language base models at the Chinchilla paper's release date (2022-03-29): kink $t = -0.91$, -0.05, $+1.48$ at bandwidths 365, 540, 730 — a sign-unstable null. Even a genuine regime change (prior work shows a two-to-three-month adoption ramp) does not produce a slope kink dateable to the paper's release.

The EU AI Act threshold is a step with bunching, not a kink. The Act's systemic-risk presumption at 10^{25} training FLOP imposes a *level* discontinuity in obligations, not a marginal-rate kink, so RKD/DiK is the wrong estimand shape. Worse for any kink or RD design at that cutoff, the running variable sorts: post-Act, the model-density just above the line is depleted (below/above contingency: Fisher exact OR = 0.17, 95% CI 0.05–0.62, $p = 0.0071$; roughly 83% of the expected above-line mass missing), which violates any no-sorting requirement. Kink-shaped probes confirm nulls (e.g. an open-weights-share DiK of -0.93, se 0.72). The bunching test remains the right tool at this threshold — and the bunching itself, an avoidance response specific to the statutory line and the post-Act period, is the substantive finding.

6 Discussion

Four lessons generalize beyond these datasets.

Era bends and event bends are different claims. The METR and GPQA series tell almost the same visual story — capability slopes steepen around late 2024 — yet the placebo grid cleanly separates them: GPQA's bend localizes to the o1 date (size 0/7) while METR's does not (pre-side size 3/6). Public discussion rarely distinguishes "the trend changed in this era" from "this event changed the trend"; the shifted-cutoff grid makes the distinction mechanical.

Nulls need power certificates. The ECI falsification is informative precisely because the same series rejects at 4/7 shifted placebo cutoffs: the test has power in this data, and still reads zero at the candidate date. A null from an underpowered test would have guarded nothing.

Report honesty bands, not point claims. The China result is strongest as a band: legal-channel bend ≈ -0.3 to -0.56 log-units/yr across the defensible bandwidth range, total-stock effect roughly half and fragile because smuggling — itself a response to the policy — substitutes for legal supply. Similarly, GPQA's τ is a release-stream property with composition as the mechanism, not per-model improvement.

The graveyard is the argument. The most significant statistic of the exercise ($t = 12.9$) belongs to a design that fails every placebo. Any workflow that stops at "the kink is significant" would have led with it. On the public record of AI progress, where series are short, serially dependent, and composition-churned, the falsification battery is not a robustness appendix — it is the analysis.

Limitations. Calendar-time identification rests on an untestable no-co-located-event assumption; placebo grids probe but cannot certify it. Several series are short (6 quarterly points per cell in the DiK; 11–12 pre-cutoff models for METR), making serial dependence corrections coarse and $|t|$ optimistic. Epoch's compute figures are partly estimates, and benchmark contamination drifts over time. All results are properties of the *public record* of AI progress — release-stream composition included — not of any individual system.

Reproducibility

Estimates were produced with the open-source `natex` v0.2.0 kink module [4] on public Epoch AI data [3]; the per-design analysis records (inputs, CLI outputs, diagnostics JSON) and the case-study write-up of record live in the `natex` repository under `docs/case_studies/` (`epoch-kinks.md`). The four figures regenerate deterministically from the public CSVs via the committed script `paper/figures/ make_figures.py`, which asserts every headline number against the case study before drawing.

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A Sharp Bend, No Verdict: Difference-in-Kinks Tests of US Business AI Adoption at DeepSeek-R1

Hauke Hillebrandt*

July 2026

analysis pass of record: 2026-07, `natex` v0.2.0, seed 0

Abstract

US Census Bureau Business Trends and Outlook Survey (BTOS) data show a sharp, well-localized acceleration in business AI use among AI-exposed sectors (information; finance and insurance; professional/scientific/technical services) beginning with the first survey wave after the release of DeepSeek-R1 on 20 January 2025. A sector-by-wave difference-in-differences scan localizes the break at exactly that wave (scan $p = 0.010$–0.030); the exposed-minus-unexposed gap slope more than triples, from +4.2 to +13.9 pp/yr; and a group difference-in-kinks (DiK) estimate at the release date is nominally significant (+7.3 pp/yr, cluster-robust se 2.8). Attribution to DeepSeek-R1, however, fails the falsification battery: placebo cutoffs at non-event 2024 dates produce DiKs as large as or larger than the R1 estimate (+11.4 pp/yr at $t = 2024.30$; +4.0 at $t = 2024.50$; both significant); only 2 of 6 specification variants at R1 are significant; equally large local kinks appear in non-exposed sectors; the predicted *deceleration* at OpenAI's o1 release is insignificant at the primary bandwidth and sign-flips at the wider one; and the cutoff week is confounded by the AI diffusion export rule (13 January), the Stargate announcement (21 January), and a BTOS sample-year rollover in mid-December 2024. The January-2025 acceleration is real but descriptive: no causal attribution to DeepSeek-R1 survives triage.

1 Introduction

On 20 January 2025 the Chinese lab DeepSeek released R1, an openly licensed reasoning model trained at a reported cost far below frontier-lab budgets [6]. The release was treated by markets as news about the cost curve of AI: on 27 January 2025 Nvidia alone lost roughly \$590 billion of market capitalization, the largest one-day loss on record, and an event-study literature on the "DeepSeek shock" emerged almost immediately.[1] A natural real-economy question follows: did cheap, open reasoning models bend the *adoption* curve — the share of US businesses actually using AI in production?

Measurement of that adoption curve is recent. The Census Bureau's Business Trends and Outlook Survey (BTOS) has asked a probability sample of US firms since September 2023 whether they used AI in producing goods or services in the last two weeks; [2] introduce these data and document a rise in the national use rate from 3.7% to 5.4% over the first six months, concentrated in the Information sector and larger firms. Complementary measurement includes the 2018 Annual Business

*University College London. Email: `ucjthhi@ucl.ac.uk`. This paper and the underlying `natex` software were prepared with substantial assistance from Anthropic's Claude models; the author reviewed the analyses and text and is responsible for all remaining errors.

[1]See, e.g., "Nvidia drops \$600B off its market cap amid the rise of DeepSeek," TechCrunch, 27 January 2025. We do not rely on this financial-markets literature for identification; it motivates the candidate event date only.

Survey module of [8] (pre-LLM adoption below 6% of firms, over 18% employment-weighted) and the multi-country executive survey of [10] (high headline adoption, small realized effects). This literature is descriptive by design; formal tests of whether adoption *trends* broke at specific AI events are, to our knowledge, absent.

A trend break for one group relative to another at a known date is the estimand of the difference-in-kinks (DiK) design formalized by [1], building on the regression kink design of [4]. Kink estimates in time-series settings are known to produce spuriously significant slope changes, which is why placebo distributions at non-event dates are the appropriate yardstick [5]. This paper applies the `natex` toolkit's DiK and scan-based DiD estimators [9, 7] to the BTOS sector panel with DeepSeek-R1 as the candidate cutoff, o1 (12 September 2024) as a secondary contrast, and a mandatory falsification battery, which converts an apparently significant event-study result into an explicit non-attribution.

2 Data

The primary source is the Census Bureau's BTOS AI core workbook (DRB approval CBDRB-FY25-0425): sector-level biweekly estimates and standard errors for Question 7, "In the last two weeks, did this business use Artificial Intelligence (AI) in producing goods or services?", answer "Yes," in percentage points. The panel covers 20 sectors (19 two-digit NAICS sectors plus an unclassified residual) over 54 biweekly waves, 202319–202520 (collection periods 2023-09-11 through 2025-10-05); 199 of 1,080 sector-waves are disclosure-suppressed, leaving 881 usable cells. The panel predates the 2025-12-03 questionnaire rewording entirely. The running variable t is the fractional year of each wave's collection-period midpoint. The exposed group is {51 information, 52 finance and insurance, 54 professional/scientific/technical services}; the kink-leg comparison group is {11, 21, 23, 72}. A placebo outcome (the archived remote-work Question 6 "Yes" share) comes from the BTOS archive workbook (CBDRB-FY23-0478/FY24-0474/FY25-0117), waves 202417–202514.

Coverage is the binding data limitation: sectors 11, 21, 22, and 55 are almost entirely suppressed (3, 3, 4, and 8 usable waves of 54), so the "unexposed" comparison group is effectively sectors 23 (construction) and 72 (accommodation and food services).

3 Design and Methods

Group difference-in-kinks. Following [1], the DiK estimand is the change in the slope of the outcome in calendar time at a known cutoff for the exposed group *minus* the same slope change for the comparison group, estimated by local linear regression on each side of the cutoff within each group (`natex` v0.2.0 [9]). The primary specification uses cutoff $t_0 = 2025.055$ (DeepSeek-R1; first post wave 202503, collected 2025-01-27 to 02-09), bandwidth 0.5 years (±13 waves), triangular kernel, a one-wave donut, and cluster-robust (CR1) standard errors clustered by sector with 7 clusters and critical values from $t(G-1)$. With 7 clusters and only 2–4 per DiK cell, CR1 inference is fragile even with the small-sample correction [3]; we therefore lean on design-based falsification rather than on the standard errors.

Falsification battery. (i) a specification grid over bandwidth $\in \{0.25, 0.5, 1.0\}$ years, kernel $\in$ {triangular, uniform}, and donut $\in \{0, 1 \text{ wave}\}$; (ii) placebo cutoffs at four non-event dates (2024.30, 2024.50, 2024.85, 2025.30), in the spirit of the permutation logic of [5]; (iii) a placebo-in-space battery of per-sector local kinks at the R1 date; (iv) a placebo *outcome* (remote work) at the same cutoff; and (v) a secondary contrast at o1 (2024-09-12, $t = 2024.6967$): if R1 mattered over

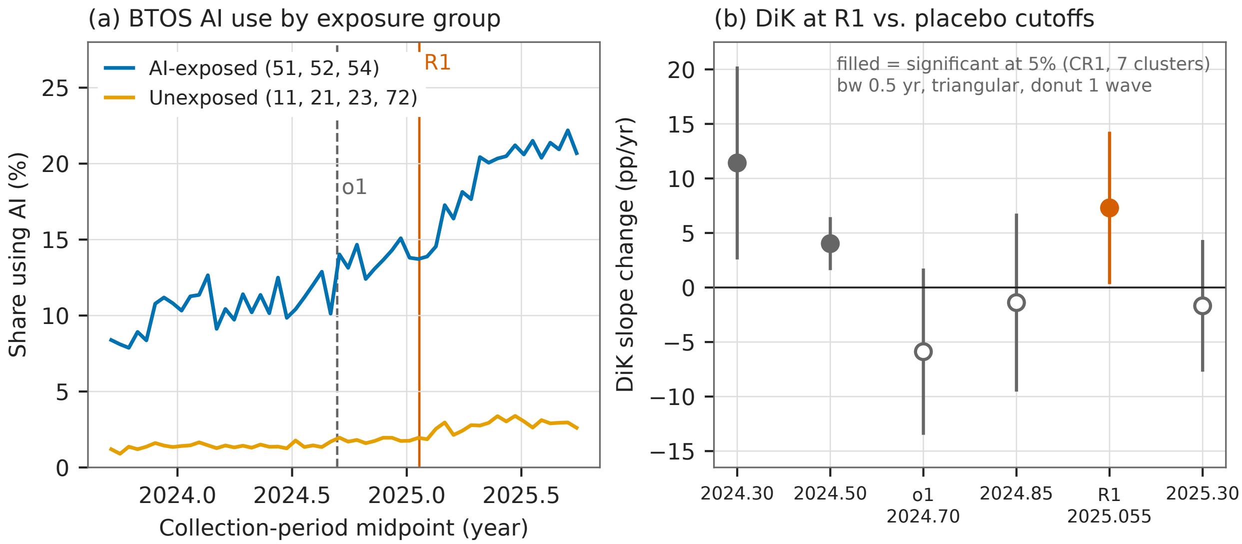

Figure 1: (a) BTOS AI-use share by exposure group across all 54 waves, with the o1 and DeepSeek-R1 release dates marked. (b) Group difference-in-kinks estimates (primary specification: bandwidth 0.5 yr, triangular kernel, one-wave donut) at the R1 cutoff, the o1 cutoff, and four non-event placebo dates, with 95% CR1 confidence intervals; filled markers are significant at the 5% level against $t(G-1)$ critical values. Two non-event placebos are significant, one larger than the R1 estimate.

and above the reasoning-model wave o1 began, the exposed-minus-unexposed gap should *decelerate* there.

Sector-by-wave DiD. The `natex` survey pipeline's sector-by-wave DiD scan (SuDDDS) searches over candidate break dates t_0 and window widths W, maximizing a likelihood-ratio scan statistic for a declared-treatment DiD, calibrated by permutation under an AR(1)-by-unit null [7]. We also fit a manual two-way fixed-effects (TWFE) level DiD with wave and sector fixed effects, CR1 by sector, and a placebo-in-space randomization check.

Honest-inference notes, stated as run. The pipeline's refusals are reported, not patched over: the SuDDDS anticipation test refused (Holm $p = [\mathrm{NaN}, \mathrm{NaN}, \mathrm{NaN}]$); the dd/synthetic-control/GESS effect estimators refused with $\tau = \mathrm{NaN}$ (0 usable placebo units on this unbalanced panel, ≥ 5 required), hence the manual TWFE estimate below; and the automated survey's own family verdicts were discarded as role-assignment artifacts (its "credible" RDD, scan $p = 0.010$, $\tau_{2\mathrm{sls}} = 0.749$, se 0.343, is the mechanical rediscovery of candidate treatment = post with outcome $= t$; its kink family, Holm $p = 0.807$, tested outcome = sector — vacuous; its DiD family, $p = 0.70$, used a time-invariant treated indicator rather than the declared design; iv/sc/bunching returned needs-input; dee was null with 1 usable experiment).

4 Results

The bend is real ... The SuDDDS scan localizes a break at $t_0 = 2025.089$ — exactly the first post-R1 wave (202503) — with window $W = 0.536$ yr: max LLR $= 22.65$, scan $p = 0.030$ on the 20-sector panel, and LLR $= 32.70$, $p = 0.010$ (the $q = 99$ permutation floor) on the 7-sector

Table 1: Group difference-in-kinks estimates, BTOS AI-use share (pp/yr).

Specification	$\hat{\tau}$	se	$\lvert t \rvert$	sig. 5%
Panel A: DeepSeek-R1 cutoff ($t_0 = 2025.055$)				
triangular, bw 0.5, donut (primary)	+7.297	2.797	2.61	$*$
triangular, bw 0.5, no donut	+6.742	1.188	5.68	$*$
uniform, bw 0.5, donut	+4.477	3.079	1.45	
triangular, bw 1.0, donut	+2.350	2.934	0.80	
uniform, bw 1.0, donut	+3.941	2.478	1.59	
triangular, bw 0.25, donut	+10.488	8.130	1.29	
Panel B: placebo and secondary cutoffs (triangular, donut)				
$t_0 = 2024.30$ (no event), bw 0.5	+11.419	3.556	3.21	$*$
$t_0 = 2024.50$ (no event), bw 0.5	+4.023	0.889	4.53	$*$
$t_0 = 2024.85$ (no event), bw 0.5	−1.385	3.118	0.44	
$t_0 = 2025.30$ (no event), bw 0.5	−1.680	2.293	0.73	
o1, $t_0 = 2024.6967$, bw 0.5	−5.884	2.909	2.02	
o1, $t_0 = 2024.6967$, bw 1.0	+3.818	1.417	2.69	$*$

Notes: CR1 standard errors clustered by sector; significance against $t(G-1)$ critical values with $G \in \{5, 6, 7\}$ clusters in the estimation window (2.447–2.776). Bandwidths in years (0.5 yr = ± 13 waves); donut = one wave. Outcome: sector-level share answering Yes to BTOS Question 7, percentage points.

subpanel; the composition check passes ($p = 1.000$). The SE-weighted exposed-minus-unexposed gap slope over ± 13 waves rises from +4.197 (se 2.089) to +13.940 (se 1.649) pp/yr, $\Delta = +9.74$ (se 2.66). The primary group DiK at R1 (Table 1, first row) is $\hat{\tau} = +7.297$ pp/yr (CR1 se 2.797, 95% CI $[0.454, 14.140]$, $|t| = 2.61$ against $t(6)$ critical value 2.447, $n = 123$): the exposed group's slope jumps from 3.234 to 12.726 pp/yr against 0.485 to 2.680 for the comparison group. The manual TWFE level DiD gives $\hat{\tau} = +4.583$ pp (CR1 se 0.526, $t = 8.72$), and placebo-in-space is clean: 0 of 13 control-sector pseudo-treatments reach the minimum exposed $|t|$ (controls' max 7.76 vs. exposed 10.19/10.49/13.19; exact-p floor $1/14 = 0.071$), with per-sector effects of +4.26 (51), +4.14 (52), and +5.35 (54) pp.

... but attribution to R1 fails. Five findings block a causal reading. *First*, the placebo-cutoff battery fails: at the non-event dates 2024.30 and 2024.50 the same DiK specification returns +11.419 (se 3.556, $|t| = 3.21$) and +4.023 (se 0.889, $|t| = 4.53$) — both significant, the first larger than the R1 estimate itself (Table 1, panel B; Figure 1b). *Second*, the R1 estimate is significant in only 2 of 6 specification variants. *Third*, the kink is not exposure-specific: in per-sector local kinks at R1, exposed sector 52 ranks 1/16 (+15.41, $t = 6.36$), but non-exposed sectors 71 (+10.46, $t = 4.24$), 53 (+10.00, $t = 3.82$), and 62 (+7.94, $t = 6.62$) all exceed exposed sectors 51 (+6.56, $t = 1.03$) and 54 (+6.50, $t = 3.13$) — the January-2025 kink is broad-based. *Fourth*, the o1 deceleration contrast is bandwidth-aliased: −5.884 (se 2.909, $|t| = 2.02 <$ crit 2.571, not significant, though the sign matches the prediction) at bandwidth 0.5, but +3.818 (se 1.417, $|t| = 2.69$, significant, sign *flipped*) at bandwidth 1.0; the SE-weighted gap-slope change at o1 is +0.59 (se 2.63) — the expected deceleration is not reproduced. *Fifth*, the TWFE level DiD, though clean in space, violates parallel trends: the pre-R1 gap slope is +4.20 pp/yr (se 2.09), so the level estimate is descriptive. The placebo *outcome* is consistent with a broad shock rather than an AI-specific one but is underpowered: the remote-work DiK at R1 is +8.821 (se 5.351, $|t| = 1.65$, not significant, on a truncated pre-window) — same sign and magnitude as the AI outcome.

5 Caveats and Conclusion

Three caveats are decisive. First, a calendar-time cutoff absorbs every same-week event: the Framework for Artificial Intelligence Diffusion export rule (2025-01-13) and the Stargate announcement (2025-01-21) fall in the same window as R1 (2025-01-20), so a DiK at $t = 2025.055$ cannot distinguish among them — and the BTOS sample-year rollover (wave 202426 → 202501, mid-December 2024) changes respondent-panel composition three to five weeks before the cutoff. Second, the comparison group is effectively two sectors: NAICS 11, 21, 22, and 55 are almost entirely disclosure-suppressed, and CR1 inference rests on 7 clusters with only 2–4 clusters per DiK cell, where cluster-robust standard errors are known to over-reject [3]. Third, and most important, the placebo-cutoff battery finds slope breaks as large as the R1 estimate at dates where nothing happened — exactly the failure mode placebo distributions exist to catch [5]: on this panel, the DiK test cannot tell the R1 date apart from ordinary undulation in a steep adoption curve.

The verdict is descriptive-only. A sharp, well-localized acceleration in measured AI adoption in AI-exposed sectors did begin with the first BTOS wave after 20 January 2025 — the localization is data-driven and survives composition and placebo-in-space checks — but every test that would pin the bend to DeepSeek-R1 specifically fails: placebo dates match it, most specifications miss it, unexposed sectors show it too, and the o1 contrast sign-flips. Something bent in January 2025; these data cannot say it was R1.

Reproducibility. All estimates were produced with `natex` v0.2.0 [9] at seed 0 from the frozen BTOS workbooks (published at `https://www.census.gov/hfp/btos`; not committed to the repository). Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the primary estimate against the numbers of record before drawing.

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Bent, Bypassed, Unbowed: US AI-Chip Export Controls and China's Compute

Hauke Hillebrandt*

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analysis pass of record: 2026-07, `natex` v0.2.0, seeds 7 (DiD) and 20260716 (DiK battery; estimator deterministic)

Abstract

The October 2023 US export controls on AI accelerators bent China's *legally* imported AI-chip stock sharply downward. A group difference-in-kinks (DiK) contrast of China's legal H100-equivalent stock against the four US hyperscalers at the 2023-10-17 rule gives -0.00154 ln-units/day (HC1 se 0.00047; ≈ -0.56 ln-units/yr), is negative in 25 of 25 estimated specifications (defensible band -0.3 to -0.56 ln-units/yr), is null at the October-2022 loophole-round placebo cutoff, and its placebo-treated group is clean at defensible bandwidths. The effect on China's *total* chip stock — including smuggled chips — is roughly half as large and statistically fragile (-0.00076, $t = -1.54$, 95% CI spanning zero): smuggled Nvidia chips in China grew from 27.8k H100e in Q1 2024 to 662k by Q4 2025, doubling every 4.7 months, and now exceed China's legal Nvidia stock 1.65×. Finally, a country-level difference-in-differences on the count of $\geq 10^{24}$-FLOP training runs finds no suppression at all: China's big-run count rose +3.2 runs/quarter *relative to* every control-group counterfactual after the October-2022 round — but that estimate fails placebo-in-space (exact $p = 1.0$; every pseudo-treated unit shows a larger studentized effect), so it is descriptive, not causal. The controls demonstrably bent the legal supply channel; substitution through smuggling and domestic chips roughly halved and de-robustified the effect on China's actual compute stock; and no slowdown in China's large training runs is detectable in these data.

1 Introduction

On 7 October 2022 the US Bureau of Industry and Security imposed the first broad export controls on advanced AI accelerators and chipmaking tools bound for China, a policy contemporaries immediately read as a landmark attempt to choke off China's access to frontier AI compute [1]. The rule's performance-and-interconnect thresholds left room for compliant workarounds — Nvidia's A800 and H800 — and on 17 October 2023 a second round closed that loophole, banning the workaround chips outright. Whether such controls work is an old question in trade policy: export controls impose costs on the controlling country's own firms and reroute rather than stop trade when substitutes exist [3, 13]. Firm-level evidence from the earlier entity-list controls finds large collateral costs — US suppliers to targeted Chinese firms lose customers, market value (roughly $130 billion), lending, and employment, with broad supply-chain decoupling but no reshoring [6].

*University College London. Email: `ucjthhi@ucl.ac.uk`. This paper and the underlying `natex` software were prepared with substantial assistance from Anthropic's Claude models; the author reviewed the analyses and text and is responsible for all remaining errors.

On the evasion margin, [9] warned in 2023 that AI-chip smuggling into China, then likely in the hundreds to low thousands of units, would industrialize; by 2026, diversion-and-resale estimates put smuggled compute at hundreds of thousands of H100-equivalents [12]. What this literature lacks is a quasi-experimental estimate of whether the controls actually bent China's compute accumulation and large-scale training activity.

This paper provides three complementary tests — three legs of one story — using two Epoch AI datasets [8, 7] and the `natex` toolkit [14]. Leg 1 is a country-level difference-in-differences (DiD) on the quarterly count of $\geq 10^{24}$-FLOP training runs, with a scan-based localization step [11]. Leg 2 is a group difference-in-kinks (DiK) — the design of [2], building on the regression kink design of [5] — contrasting the growth-rate bend in China's legal chip stock against US hyperscalers at the 2023-10-17 rule, with placebo cutoffs in the spirit of [10]. Leg 3 repeats the DiK on China's total stock including smuggled chips and quantifies the smuggling substitution directly. The verdict is deliberately mixed and stated as such: Leg 2 is credible, Leg 3 is attenuated and fragile, Leg 1 is descriptive only.

2 Data

Country panel (Leg 1). From Epoch AI's notable-models database [8] we build a panel of 4 country groups (US, China, EU, Other) $\times$ 26 quarters (2020Q1–2026Q2), $n = 104$ cells. The outcome is the count of training runs $\geq 10^{24}$ FLOP per quarter; a secondary outcome is the median $\log_{10}$ training compute. Models are assigned to countries by the lead (first-listed) organization, so multinational collaborations are attributed to the lead country. The declared treatment is `china_controlled`: China from 2022Q4 (the first control round) onward. Two caveats bind: 2026 quarters undercount compute-known runs ($\sim$13% FLOP coverage, a reporting lag), and counts measure quantity of big runs, not frontier capability.

Chip stocks (Legs 2–3). The chip-stock series come from Epoch AI's AI Chip Owners data [7]: quarterly cumulative H100-equivalent (H100e) stock by designer. From these we form ln cumulative stock series for: China's legal stock; China's total stock including the smuggled Nvidia series [12]; the four US hyperscalers (Amazon, Microsoft, Google, Meta) as the primary control; neoclouds (Oracle, CoreWeave, xAI) as an alternative control; and an "Other" group used as a placebo-treated group. Each series covers 16 quarters from 2022-03-31; the incomplete 2026-03-31 quarter is dropped (8 rows whose "cumulative" totals decrease). The running variable is days since 2023-10-17. The smuggled series begins 2024-03-31 and is itself part of the treatment response — which is exactly why Legs 2 and 3 are reported side-by-side.

3 Design and Methods

Leg 1: SuDDDS DiD. The `natex` sector-by-time DiD scan (SuDDDS) searches subsets of units and break dates maximizing a likelihood-ratio scan statistic [11], calibrated by permutation ($q = 99$, seed 7, windows $\{4, 7, 13\}$). We run it three ways: (i) known-treatment mode (Bernoulli model, `wcc`) on the declared treatment; (ii) a normal-model cross-check with a dependence-preserving `ar1_unit` null; (iii) discovery mode with no treatment column, asking whether the export-control dates would be found autonomously. Effects for China at 2022Q4 come from DD, synthetic-control, and GESS estimators, followed by a manual placebo battery (placebo-in-space with the Abadie china-removed convention, placebo-in-time at 2021Q3).

Legs 2–3: group difference-in-kinks. In [2] the DiK contrasts the same schedule kink across two *time periods*. Here the export-license schedule bends *for China only* at a common date, so the Böckerman contrast is taken across *groups* at one cutoff: DiK = (China slope kink) − (hyperscaler slope kink), each kink estimated by local linear regression on each side of the cutoff (right-minus-left), triangular kernel, primary bandwidth 548 days. Identification is parallel-kink: absent the controls, China's growth bend at Oct-2023 would have matched the hyperscalers'; the control group differences out the global supply-curve bend (the hyperscalers' own kink is -0.000666/day). Robustness: bandwidths $\{365, 548, 730\}$ days $\times$ {triangular, uniform} kernels, 45-day donuts, six placebo cutoffs (including the Oct-2022 first round at -375 days — the A800/H800 loophole-round anticipation contrast), and a placebo-treated group (Other vs hyperscalers, both untreated), in the spirit of [10]. Leg 3 re-runs the identical design on China's total stock including smuggled chips.

Honest-inference notes, stated as run. The pipeline's refusals and trivial passes are reported, not patched over. Leg 1: the built-in `tau_randomization_test` correctly refuses ($p =$ NaN, 3 non-treated units < 5 usable placebos) rather than fabricating a p-value, so the manual placebo-in-space battery is the inference of record; the scan $p = 0.010$ is the $q = 99$ permutation floor; with 4 units, all panel p-values are descriptive. Leg 2: HC1 standard errors on 6 quarterly points per cell overstate precision on a cumulative-stock series; CR1 clustering by calendar year keeps significance but has $G = 4$ clusters with $t(G - 1)$ critical values — a known few-cluster over-rejection regime [4]; the sign-stability across 25 specifications is the real evidence. The density-kink-difference test passes trivially (both groups share an identical quarterly grid; estimate exactly 0.0, p reported as null), no predetermined covariates exist in a two-group quarterly aggregate (the placebo-treated group and the Oct-2022 placebo round substitute for the covariate-kink battery), and no Fieller set is produced (sharp design, known unit denominator). A recorded `natex` UX finding: the CLI aliases the group contrast through its time interface (`--time china --t0 1`), so output cells are labeled `pre_*/post_*` and its caveat text misdescribes the cross-group contrast as a "time-stable marginal response."

4 Results

Leg 1: localization succeeds, attribution fails. In known-treatment mode SuDDDS recovers the declared treatment exactly: top subset {China}, $T_0 = 11 =$ 2022Q4, window $W = 13$, Bernoulli LLR 6.3715, panel-null $p = 0.010$ (the $q = 99$ floor); the composition check passes ($p = 1.0$) and the anticipation test passes (Holm $p = 1.0$). The normal-model cross-check under the dependence-preserving `ar1_unit` null finds the mirror image — the complement {EU, Other, US} at $T_0 = 13$ (2023Q2), LLR 14.9487 — but $p = 0.14$: a 4-unit panel cannot separate a persistent step from unit-level AR(1). Discovery mode does *not* find the controls: the top hit is a {China, US} *downshift* at 2025Q4 (LLR 11.277, $p = 0.13$; runner-ups ALL@2025Q4, LLR 7.232, and {China, US}@2024Q2, LLR 5.835), with no export-control date in the top five — and the 2025Q4 downshift is at least partly the reporting-lag artifact. Effect estimates for China at 2022Q4 (Table 1, panel A) are uniformly *positive*: DD $\hat{\tau} = +3.214$ runs/qtr (se 1.157, $t = 2.779$, 95% CI $[+0.947, +5.481]$), synthetic control +3.933, GESS +3.842, DD at the 2023Q4 round +4.455, and DD on median $\log_{10}$ compute +0.649 $[+0.322, +0.976]$. Taken literally: no suppression — China's big-run count *outgrew* every control-implied counterfactual. But the placebo battery kills any causal reading of the magnitude: pseudo-treating each control unit at the same date yields studentized $|t|$ of 5.37 (EU), 2.86 (Other), and 5.16 (US) against China's 2.78 (china-removed convention) — exact placebo-in-

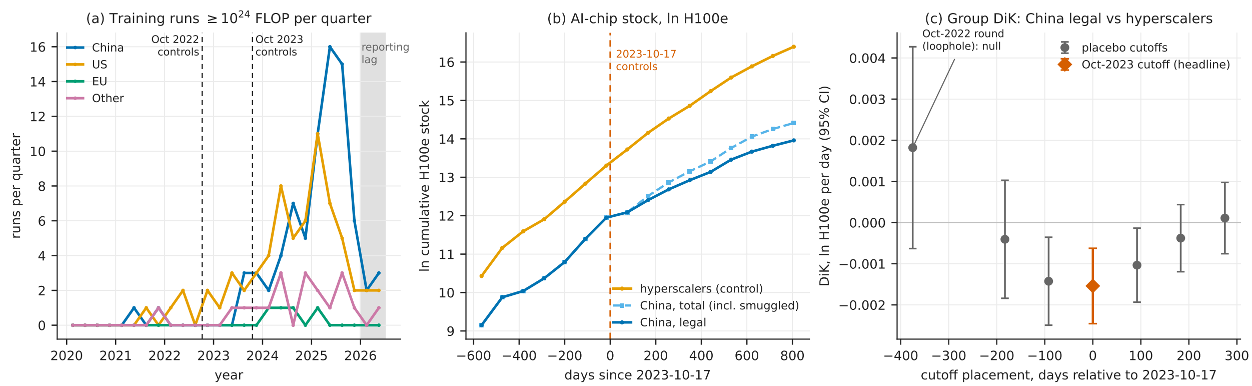

Figure 1: (a) Leg 1: quarterly count of $\geq 10^{24}$-FLOP training runs by country group, with both export-control rounds marked; the shaded region is the ∼13%-coverage reporting-lag zone. The panel is a two-speed world: US and China accelerate, EU and Other stagnate. (b) Legs 2–3: ln cumulative H100e chip stock for the hyperscaler control group, China's legal stock, and China's total stock including smuggled chips, around the 2023-10-17 cutoff. (c) Leg 2: group-DiK estimates (bandwidth 548 days, triangular kernel) at the true cutoff (diamond) and at six placebo cutoff placements, with 95% HC1 intervals; the Oct-2022 loophole round at −375 days is null.

space $p = 4/4 = 1.0$. Placebo-in-time is clean (China at 2021Q3: $\tau = -0.30$, $t = -1.33$), and the built-in randomization test correctly refused ($p =$ NaN, $3 < 5$ placebos). The robust statement is qualitative only: China's large-run count did not fall behind its counterfactual after either control round.

Leg 2: the legal channel credibly bent. The headline group DiK (China legal vs hyperscalers, 2023-10-17, bandwidth 548 days, triangular kernel) is −0.0015404 ln-units/day (HC1 se 0.000467, $t = -3.298$, 95% CI $[-0.002456, -0.000625]$), i.e. ≈ -0.56 ln-units/yr of growth-rate change; with CR1 clustering by year, se 0.000299, $t = -5.152$ ($G = 4$, few-cluster caveat). The cell kinks are −0.000666/day (hyperscalers) vs −0.002206/day (China legal). The estimate is negative in **25 of 25** estimated specifications: −0.000869 to −0.002026 across bw365/548/730 × triangular/uniform (t from −2.07 to −6.57), and −0.0018996/−0.0012108/−0.0008011 on the 45-day-donut grid (t −3.92/−2.04/−1.60). The placebo-cutoff battery (bw548; empirical size 2/6) localizes the kink to within one quarter of the rule: the Oct-2022 first round at −375 days is *null* at +0.0018203 (se 0.001251, $p = 0.146$) — the loophole round produced no bend, a pseudo-pre-trend pass — and −183d ($p = 0.578$), +183d ($p = 0.364$), +275d ($p = 0.805$) are null, while −92d ($p = 0.0089$) and +92d ($p = 0.0240$) reject: with quarterly data, one-quarter shifts still straddle the same bend, which still brackets 2023-10-17. The placebo-treated group (Other vs hyperscalers, both untreated) is null at bw548/730 ($t = +1.34$, −0.29, −0.20, −0.75) but rejects at bw365 (triangular +0.000958, $t = 2.49$; uniform +0.001219, $t = 3.04$) — parallel-kink fails among untreated groups at four-points-per-cell bandwidths, so the bw365 cells are discounted and the defensible band is bw548–730: −0.0009 to −0.0015/day (≈ -0.3 to −0.56 ln-units/yr).

Leg 3: smuggling halves and de-robustifies the effect. On China's total stock including smuggled chips, the same design gives −0.0007638 (se 0.000496, $t = -1.540$), with a 95% CI of $[-0.001736, +0.000209]$ that spans zero — roughly half the legal-channel bend, and t ranges from −0.55 to −3.08 across the six specifications (the only rejection is at the discounted bw365). The

Table 1: Headline estimates across the three legs.

Specification	$\hat{\tau}$	se	t	note
Panel A: Leg 1 — DiD, China at 2022Q4 (runs $\geq 10^{24}$ FLOP per quarter)				
DD (control = other 3 units)	+3.214	1.157	2.78	CI [+0.947, +5.481]
Synthetic control	+3.933	—	—	CI [+1.533, +6.334]
GESS	+3.842	—	—	CI [+1.442, +6.243]
DD at 2023Q4 round	+4.455	—	—	CI [+1.719, +7.190]
DD, median $\log_{10}$ compute	+0.649	—	—	CI [+0.322, +0.976]
placebo-in-space exact $p = 4/4 = 1.0 \Rightarrow$ descriptive, not causal				
Panel B: Leg 2 — group DiK, ln H100e per day, China legal vs hyperscalers				
bw548 triangular (headline)	−0.001540	0.000467	−3.30	HC1
bw548 triangular, CR1 by year	−0.001540	0.000299	−5.15	$G = 4$ clusters
bw548 uniform	−0.001132	0.000525	−2.16	
bw730 triangular / uniform	−0.001140 / −0.000869		−2.85 / −2.07	
bw365 triangular	−0.002026	0.000308	−6.57	discounted
placebo cutoff −375d (Oct-2022)	+0.001820	0.001251		$p = 0.146$: null
Panel C: Leg 3 — group DiK, China total stock (incl. smuggled) vs hyperscalers				
bw548 triangular	−0.000764	0.000496	−1.54	CI [−0.001736, +0.000209]
bw730 triangular / uniform	−0.000458 / −0.000235		−1.10 / −0.55	
bw365 triangular	−0.001049	0.000340	−3.08	discounted

Notes: Panel A: `natex` DiD effect estimators on the 4-country × 26-quarter panel (seed 7); 95% CIs in the note column. Panels B–C: sharp group difference-in-kinks, right-minus-left slope contrasts in ln cumulative H100e stock per day at the 2023-10-17 cutoff; HC1 standard errors unless noted; bandwidths in days; "discounted" marks bw365 cells where the placebo-treated group (Other vs hyperscalers) rejects. Multiply DiK estimates by 365.25 for annual growth-rate changes.

mechanism is visible in the raw series: smuggled Nvidia compute in China grew from 27.8k H100e (Q1 2024) to 662k H100e (Q4 2025), doubling every 4.7 months — the fastest-growing series in the dataset — and now stands at 1.65× China's legal Nvidia stock. Even so, China's share of the world chip stock roughly halved to 8.7% over the period despite absolute growth: the controls did not stop China's accumulation, but China fell behind a world that accelerated faster.

5 Caveats and Conclusion

Four caveats bound the claims. First, Leg 1's panel has four units — the floor for any panel placebo inference — and its placebo-in-space failure is informative precisely because it shows the units are not exchangeable (a two-speed world), so no DiD counterfactual built from EU/Other/US is credible for China's magnitude; every Leg-1 p-value is descriptive. Second, Leg 2's inference rests on 6 quarterly points per cell: HC1 overstates precision on a cumulative series, CR1 has $G = 4$ clusters [4], and the ±92-day placebo rejections show the kink is localized only to ±1 quarter — the design cannot distinguish 2023-10-17 from an adjacent-week event, though no rival same-quarter candidate of comparable relevance is known to us. Third, the smuggled series underlying Leg 3 is itself an estimate with wide uncertainty [12] and begins only in Q1 2024, endogenously to the treatment. Fourth, counts of big training runs measure quantity, not frontier capability; controls aimed at chip access can bind on quality and efficiency margins invisible to Leg 1's outcome.

The three-leg verdict: *credible* on the legal channel (Leg 2) — the October-2023 controls bent China's legal chip-stock growth down by roughly 0.3–0.56 ln-units/yr relative to every control group, with the loophole-round placebo null exactly where the policy history says it should be; *attenuated and fragile* on total compute (Leg 3) — smuggling substitution roughly halves the point estimate and removes its robustness; and *descriptive only* on training activity (Leg 1) — China's count of large training runs did not fall behind any counterfactual, but the design cannot certify that as a causal null. Export controls, on this evidence, bent the channel they legally govern, were substantially bypassed in aggregate, and left China's large-run training activity unbowed.

Reproducibility. All estimates were produced with `natex` v0.2.0 [14]: the Leg-1 runs at seed 7 ($q = 99$) and the Leg-2/3 battery at seed 20260716 (the DiK estimator itself is deterministic), from frozen extracts of the Epoch AI datasets [8, 7] (not committed to the repository). Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the headline DD and DiK estimates against the numbers of record before drawing.

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The Question Is the Treatment: A Known-Cutoff Regression Discontinuity at the BTOS AI-Question Rewording

Hauke Hillebrandt*

July 2026

analysis pass of record: 2026-07, `natex` v0.2.0, seed 0

Abstract

Beginning with wave 202524 (reference period from 3 November 2025; announced 3 December 2025), the US Census Bureau's Business Trends and Outlook Survey (BTOS) reworded its headline AI question from AI use "in producing goods or services" to AI use "in any of its business functions." Treating the refresh as a known measurement cutoff in a regression-discontinuity-in-time design on the spliced 71-wave national series, the rewording moved measured US business AI use by +6.04 percentage points (HC1 z = 22.9; HAC z = 16.3) — a ratio of 1.553 over the old-wording level at the cutoff. The jump is 4.7× the largest of 49 placebo-cutoff estimates (randomization p = 0.020, the 1/50 floor), stable at 6.0–7.5 pp across bandwidth, weighting, curvature, and HAC variants, and a blind LoRD3 scan re-localizes it: the top two data-driven candidate cutoffs exactly flank the known one (honest 2SLS +6.74, 95% CI $[6.01, 7.48]$). True adoption growth cannot produce it: the fitted pre-trend predicts +0.66 pp across the 50-day shutdown gap, and doubling that slope still leaves $\geq$ 5.38 pp. Nulls are reported as nulls: the blind-scan randomization test is underpowered at n = 71 (p = 0.14), the McCrary density test is inapplicable on a uniform time grid (NaN), a December–January seasonal term is insignificant (0.29 ± 0.24), and a post-cutoff slope steepening (+2.17 pp/yr, conventional se 0.56) is secondary and suggestive only. The estimate calibrates a splice factor — additive +6.04 pp or ratio 1.553 — for carrying the pre-refresh series past December 2025, with the caveat that the "wording effect" is bundled with a simultaneous sample rotation and any shutdown-adjacent change in response composition.

1 Introduction

Since September 2023 the Census Bureau's Business Trends and Outlook Survey (BTOS) has asked a probability sample of US firms whether they used artificial intelligence in the last two weeks; [4] introduce these data, which have become the flagship high-frequency gauge of US business AI adoption — the headline series in the Federal Reserve's monitoring of the AI economy [1]. On 3 December 2025 the Census Bureau announced that, beginning with survey cycle 2 of sample year 2026, the AI questions were reworded: the core use question now asks about AI "in any of its business functions" rather than "in producing goods or services," explicitly to reduce respondent under-reporting [8]. Measured adoption moved immediately: the last old-wording wave reads 10.0%, the first new-wording wave 17.3%. Commentary quickly flagged the break — [2] show that how the

*University College London. Email: `ucjthhi@ucl.ac.uk`. This paper and the underlying `natex` software were prepared with substantial assistance from Anthropic's Claude models; the author reviewed the analyses and text and is responsible for all remaining errors.

question is asked drives large gaps between firm AI-adoption estimates, and [1] notes the redesign when reporting the ∼18% year-end-2025 level — and Census researchers now exploit the expanded business-function detail the redesign introduced [5]. What the commentary does not provide is a formal estimate of the rewording effect with falsification-calibrated uncertainty. That estimate is the number any user of the series needs in order to splice the pre- and post-refresh segments into one panel.

Instrument refreshes breaking measured series is a classic survey-methods problem. The canonical precedent is the 1994 Current Population Survey redesign, whose effects were quantified with a dedicated parallel survey run under both instruments and converted into adjustment factors for splicing the historical series [13]. The BTOS refresh has no parallel overlap sample: the old question stopped, the new one started, and a federal government shutdown sits between them. The splice factor must therefore be estimated from the time series alone — a known-cutoff regression-discontinuity-in-time (RDiT) problem [9], here in its cleanest form: the cutoff is announced, mechanical, and datable to a single wave, and the "treatment" is the measurement instrument itself, so the usual manipulation and anticipation concerns are moot while the RDiT time-series concerns (serial correlation, trend extrapolation) remain and are addressed directly. We estimate the jump by local linear regression on each side of the cutoff with robust and HAC inference [6, 9], calibrate it against a 49-point placebo-cutoff grid, bound the adoption-growth confound, and additionally ask the `natex` toolkit [12] to find the discontinuity *blind*, via the LoRD3 local-RD scan of [10], plus a secondary regression-kink test in the conventions of [7] and [3].

2 Data

The outcome is the national share of firms answering "Yes" to BTOS Question 7, in percentage points, with published survey standard errors. Two vintages are required because current Census downloads carry only the reworded question, with all earlier waves blanked. The old-wording segment ("…did this business use Artificial Intelligence (AI) in producing goods or services?") comes from a frozen pre-refresh AI-core workbook: 54 biweekly waves, 202319–202520, rising 3.7% to 10.0%. The new-wording segment ("…in any of its business functions?") comes from a fresh `National.xlsx` download (census.gov, 2026-07-18, 82 kB, verified): 17 waves, 202524–202614, rising 17.3% to 21.7%. The spliced panel has $n = 71$ waves indexed by t, the decimal-year reference-period midpoint.

Three timeline facts matter. First, waves 202521–202523 were never collected — the October–November 2025 federal government shutdown — so the design has a built-in donut that is a no-data gap, not censoring: 50 days separate the last pre-cutoff reference midpoint (2025-09-14) from the cutoff. Second, the cutoff is $t_c = 2025.8384$ (2025-11-03, the reference-period start of wave 202524, the first wave fielded with the reworded question; published 2025-12-04). No reference midpoints fall in (2025-09-14, 2025-11-09), so any cutoff in that interval yields the identical estimation sample. Third, wave 202524 also begins Sample Year 4 (a scheduled full sample rotation), so the refresh bundles the wording change with a panel rotation — a caveat carried throughout.

3 Design and Methods

Estimator. The headline specification fits SE-weighted ($w = 1/\mathrm{se}^2$) local linear regressions of the AI-use share on $t - t_c$ separately on each side of the cutoff within bandwidth $h = 1.0$ years ($n_{\mathrm{pre}} = 23$, $n_{\mathrm{post}} = 17$); the jump τ is the post-side intercept minus the pre-side extrapolation at t_c. Inference is HC1 per side combined by independence, plus a pooled HAC (Bartlett kernel, 4 and 8 lags) variant for serial correlation. Sensitivity covers bandwidths $h \in \{0.5, 0.75, 1.0, 2.3\}$, unweighted fits, and pre-side quadratics.

Identification and falsification. In a measurement RDD the treatment is the instrument refresh, so the only smooth confound is true adoption growth loading into the 50-day extrapolation window; we bound it by the fitted pre-slope and report the estimate under a doubled gap slope. Calibration is design-based: the identical estimator is run at all 49 feasible interior placebo cutoffs, giving a randomization p-value with floor 1/50. A blind check uses `natex discover` (LoRD3, seed 0, $k = 12$, $q = 99$): the scan searches for local discontinuities without being told the cutoff, then honest (split-sample) 2SLS re-estimates the effect at the discovered boundary [10, 12]. The McCrary density test [11] returns NaN on a uniform time grid — inapplicable in RDiT, reported as an outcome, not patched over. A secondary sharp regression-kink test at t_c (`natex kink`, right-minus-left slope contrast, policy kink 1, $h = 1.0$, conventional local-polynomial inference without HAC) asks whether the trend also steepened [7, 3].

Honest-inference notes, stated as run. The pipeline's refusals and artifacts are reported: the automated `natex survey` run refused its rdd family (100-row floor, $n = 71$), iv (≥ 80), did/sc (needs a panel), and dee (≥ 200); its kink family auto-assigned the degenerate outcome `run` (a linear transform of t), so that null ($p = 0.79$, $\tau \approx 10^{-13}$) is meaningless and was superseded by the declared-outcome kink run above. The LoRD3 validation battery passed (covariate placebo Holm $p = 0.24$); its blind-scan randomization test is underpowered at $n = 71$ ($p = 0.14$) and is reported as the null it is. Pre-fit residuals have AR(1) = 0.836 and empirical wave noise $\sim$4$\times$ the published survey SEs (residual sd 0.614 vs. mean reported se 0.151); the HC1/HAC empirical standard errors absorb this overdispersion, but HAC with $n_{\text{post}} = 17$ is approximate — the jump is $\sim$16 HAC standard errors, so no plausible correction overturns it. A December–January seasonal coefficient is insignificant (0.293 ± 0.236).

4 Results

Headline jump and robustness. At the known cutoff the rewording jump is $\hat{\tau} = +6.039$ pp (HC1 se 0.263, $z = 22.9$; HAC(4) se 0.371, $z = 16.3$; HAC(8) se 0.376), against an old-wording level of 10.92 pp at the cutoff — a ratio of 1.553. Every variant in Table 1 stays within 6.0–7.5 pp with $z > 12$: bandwidths 0.5/0.75/2.3 give 7.148/6.383/7.363, the unweighted fit gives 5.956, and pre-side quadratics give 5.787–7.464.

Placebo-cutoff calibration. Across the 49 feasible interior placebo cutoffs the same estimator yields a maximum $|\tau|$ of 1.298 pp (mean 0.515); none reaches the true jump in $|\tau|$ or $|z|$, for a randomization $p = 0.020$ — the minimum achievable with 49 placebos (Figure 1b). The true jump is 4.7$\times$ the largest placebo.

Blind re-localization. Without being told the cutoff, the LoRD3 scan's two highest-scoring candidate boundaries sit at $t = 2025.7027$ and $t = 2025.8562$ — exactly the last old-wording and first new-wording waves, flanking t_c. Honest 2SLS at the discovered boundary gives $\tau = 6.743$ (se 0.375, 95% CI $[6.008, 7.478]$, no weak-instrument flag); the naive full-sample Wald estimate is 8.06 (se 0.206). The scan-level randomization test does not reach significance ($p = 0.14$), as expected at $n = 71$; localization, not the scan p-value, is the evidential contribution.

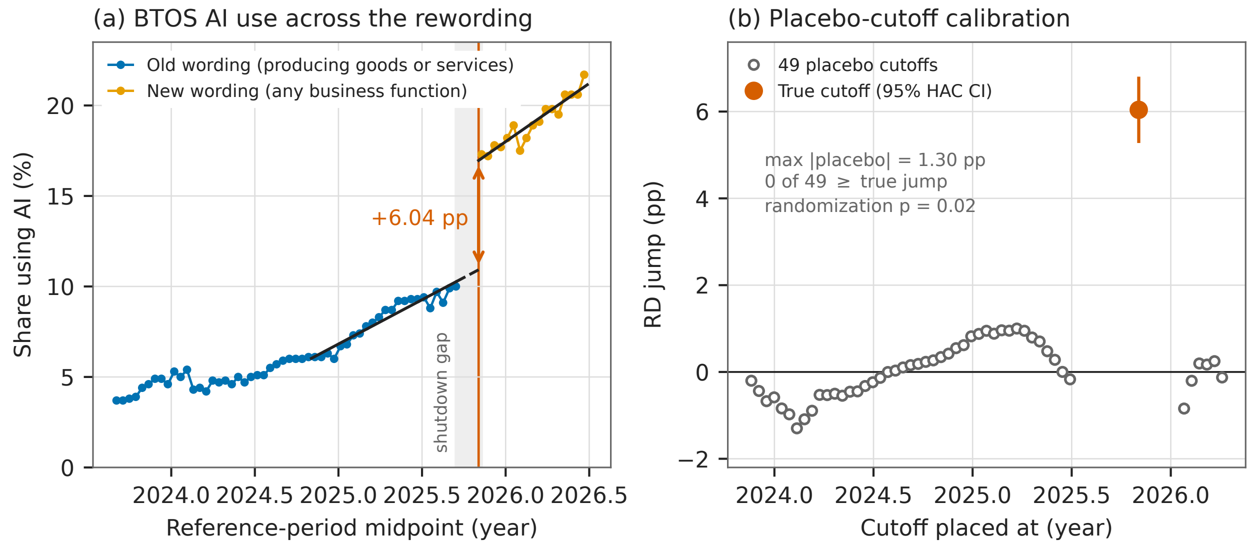

Figure 1: (a) The spliced BTOS national AI-use series with SE-weighted local linear fits within $h = 1.0$ yr of the cutoff $t_c = 2025.8384$ (2025-11-03, vertical line); the dashed segment is the 50-day extrapolation across the shutdown gap (shaded); the arrow marks the +6.04 pp jump. (b) The same estimator at all 49 feasible placebo cutoffs (open circles) versus the true cutoff (filled, with 95% HAC confidence interval): the largest placebo is 1.30 pp, 4.7× smaller than the true jump (randomization $p = 0.020$).

Bounding the adoption confound. The fitted pre-slope is 4.89 ± 0.31 pp/yr, which predicts only +0.66 pp of true-adoption growth across the 50-day gap. Even if the true gap slope were double the fitted slope, the implied jump remains ≥ 5.376 pp. A wording effect of zero would require adoption to grow $\sim$9× faster during the shutdown than in the preceding year — and then revert, since the post-cutoff slope (6.43 pp/yr) is only modestly steeper.

Secondary kink. The sharp RKD at t_c estimates a slope change of +2.170 pp/yr (conventional se 0.558, no HAC): the new-wording series trends steeper than the local old-wording trend. Given conventional inference and the bundling caveats, this is suggestive only; it cautions against assuming the wording effect is a pure level shift far from the cutoff.

Splice calibration. The fitted levels at the cutoff are 10.92 pp (old wording) and 16.96 pp (new wording): an additive splice of +6.04 pp or a ratio splice of 1.553. Over the observed post period the two disagree increasingly far from the cutoff (a ratio splice scales the steeper post trend); for level-tracking the additive factor is the direct RD estimand, and the ratio is the natural choice if the wording effect scales with the underlying level.

5 Caveats and Conclusion

Four caveats bound the claim. First, the donut is a federal-shutdown no-data gap (waves 202521–202523 were never collected), so any true adoption acceleration inside the 50-day extrapolation window loads into $\hat{\tau}$; the pre-slope bound caps this at 0.66 pp under the fitted trend and leaves $\tau \geq 5.38$ pp even at double that slope. Second, the questionnaire refresh is bundled with a full

Table 1: RD estimates of the BTOS rewording jump at $t_c = 2025.8384$.

Specification	$\hat{\tau}$	se	z
Panel A: known-cutoff RD jump (pp), SE-weighted local linear			
$h = 1.0$, HC1 (headline)	+6.039	0.263	22.9
$h = 1.0$, HAC Bartlett, 4 lags	+6.039	0.371	16.3
$h = 1.0$, HAC Bartlett, 8 lags	+6.039	0.376	16.1
$h = 0.5$	+7.148	0.352	20.3
$h = 0.75$	+6.383	0.275	23.2
$h = 2.3$ (full sample)	+7.363	0.241	30.6
$h = 1.0$, unweighted	+5.956	0.287	20.8
$h = 0.75$, pre-quadratic	+7.464	0.481	15.5
$h = 1.0$, pre-quadratic	+7.051	0.374	18.9
$h = 2.3$, pre-quadratic	+5.787	0.282	20.5
Panel B: blind discovery, placebos, and secondary tests			
LoRD3 honest 2SLS (blind scan)	+6.743	0.375	18.0
LoRD3 naive Wald	+8.063	0.206	39.2
Placebo cutoffs (49): max $\|\tau\|$	1.298	—	—
Placebo cutoffs (49): mean $\|\tau\|$	0.515	—	—
Sharp RKD slope change (pp/yr)	+2.170	0.558	3.9

Notes: outcome is the national share answering Yes to BTOS Question 7, percentage points; $n = 23$ pre / 17 post waves at $h = 1.0$ yr. HC1 per side combined by independence; HAC is a pooled Bartlett-kernel variant. Placebo randomization $p = 0.020$ (0 of 49 placebos $\geq$ the true jump in $|\tau|$ or $|z|$; floor 1/50). LoRD3: `natex discover`, seed 0, $k = 12$, $q = 99$; scan randomization $p = 0.14$ (underpowered at $n = 71$); McCrary density test NaN (uniform time grid, inapplicable). RKD: conventional local-polynomial inference, no HAC; secondary.

sample rotation (Sample Year 4 begins at wave 202524) and follows the shutdown, so "wording effect" strictly means wording + panel-refresh + shutdown-adjacent response-composition effect; for the splice-factor purpose all three are measurement artifacts, but pure wording attribution is not separable in this design. Third, RDiT caveats apply [9]: pre-fit residual AR(1) is 0.836 and empirical wave noise is $\sim 4\times$ the published survey SEs; the HC1/HAC empirical standard errors absorb this, but HAC with 17 post waves is approximate — the jump is ~ 16 HAC standard errors, so no plausible correction overturns it. Fourth, the automated-pipeline artifacts were triaged, not hidden: the survey rdd family refused on its 100-row floor, the auto-assigned degenerate kink outcome was superseded by the declared-outcome run, the McCrary test is inapplicable by construction, and the blind-scan randomization $p = 0.14$ is an underpowered null.

The verdict is credible: a textbook known-cutoff measurement RDD. The +6.04 pp jump at the 2025-11-03 refresh is $4.7\times$ the largest of 49 placebo jumps, survives every bandwidth, weighting, curvature, and HAC variant within a 6.0–7.5 pp band, is blindly re-localized by LoRD3 at exactly the flanking waves, and cannot be true adoption. It identifies the instrument-refresh effect and calibrates the splice — additive +6.04 pp, ratio 1.553 — required to extend any analysis built on the old-wording BTOS AI series past December 2025. Substantively, the estimate implies that roughly a third of the post-refresh measured adoption level is attributable to asking a better question rather than to more firms using AI — a caution for any trend analysis that splices BTOS vintages naively, and a quantification of the under-reporting the Census Bureau redesigned the question to remove [8, 2].

Reproducibility. All estimates were produced with `natex` v0.2.0 [12] at seed 0 from the frozen BTOS workbooks (published at `https://www.census.gov/hfp/btos`; not committed to the repository). Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the headline estimate, the refitted cutoff levels, and the placebo maximum against the numbers of record before drawing.

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A Genuine Bend, No Causal Certificate: Difference-in-Kinks Tests of Hyperscaler Capital Expenditure at ChatGPT

Hauke Hillebrandt*

July 2026

analysis pass of record: 2026-07, `natex` v0.2.0, seed 0

Abstract

Quarterly capital expenditure of the big-4 hyperscalers (Microsoft, Alphabet, Amazon, Meta) accelerated sharply relative to a non-AI capital-intensive comparison aggregate after the release of ChatGPT on 30 November 2022. A group difference-in-kinks (DiK) estimate at the release date is +0.1396 dex/yr (HC1 se 0.0302, nominal $p = 3.9 \times 10^{-6}$), positive in all 18 specification cells (+0.049 to +0.195), localized over 2023.0–2023.5 exactly as a lagged capex response to a late-2022 shock would be, and present per firm for Microsoft (+0.171), Alphabet (+0.206), and Amazon (+0.137) but null for Meta (+0.031, $p = 0.55$). Attribution to ChatGPT, however, fails the falsification battery: a clean pre-period placebo-cutoff grid rejects at nominal 5% in 7 of 15 non-event dates (empirical size 0.47), giving placebo-calibrated $p = 0.125$ ($|z|$) and 0.250 ($|\tau|$); the largest placebo $|z|$ (5.11, at the non-event date 2019.75) exceeds the ChatGPT estimate's matched-bandwidth $|z|$ (4.35); a full-sample kink at the non-event date 2021.5 (-0.314, $|z| = 6.9$) is larger than the main effect; and a cutoff at the BIS semiconductor export controls eight weeks earlier is statistically indistinguishable (+0.125, $z = 4.0$). Residual lag-1 autocorrelation reaches 0.66, so the nominal heteroskedasticity-robust p-values are artifacts of applying serial-independence inference to smooth aggregates. The slope divergence is a genuine, robust data feature; no causal attribution to ChatGPT survives triage.

1 Introduction

OpenAI released ChatGPT on 30 November 2022. Within two years the capital expenditure of the four large US "hyperscalers" — Microsoft, Alphabet, Amazon, and Meta — had become a macroeconomic quantity in its own right, central to debates about how much aggregate investment the current AI wave can move [1]. Financial markets repriced AI exposure almost immediately: [6] construct firm-level workforce exposures to generative AI and show that an exposure-sorted long-short portfolio earned roughly 5% in the two weeks after the release. A natural real-economy question follows: did the release bend the hyperscalers' *investment* trend — the growth rate of measured capital expenditure — or was the buildout already accelerating?

Any calendar-time answer must contend with a crowded event window. Eight weeks before ChatGPT, on 7 October 2022, the Bureau of Industry and Security announced sweeping export controls on advanced AI chips and semiconductor manufacturing equipment, a policy landmark in its own right [2]; any quarterly series treats the two events as nearly the same date.

*University College London. Email: `ucjthhi@ucl.ac.uk`. This paper and the underlying `natex` software were prepared with substantial assistance from Anthropic's Claude models; the author reviewed the analyses and text and is responsible for all remaining errors.

The estimand — a change in trend slope for one group relative to another at a known date — is the difference-in-kinks (DiK) design formalized by [4], building on the regression kink design of [5]. Kink estimates on time-series running variables are known to produce spuriously significant slope changes, which is why placebo distributions at non-event dates are the appropriate yardstick [7]; and conventional robust standard errors are badly oversized on serially correlated aggregates [3]. This paper applies the `natex` toolkit's DiK estimator [8] to a two-group quarterly capex panel built from SEC EDGAR filings, with ChatGPT as the candidate cutoff and a mandatory falsification battery — which converts a nominally overwhelming event-study result into an explicit non-attribution.

2 Data

Quarterly capital expenditure is reconstructed from SEC EDGAR XBRL `companyconcept` facts, converting fiscal year-to-date values to clean quarters by same-start differencing with remainder assignment across tag transitions. The tag is `PaymentsToAcquirePropertyPlantAndEquipment`; Amazon additionally reports under `PaymentsToAcquireProductiveAssets`. The treated series is the big-4 aggregate over 46 quarters, 2014Q4–2026Q1, rising from \$6.8bn to \$129.8bn per quarter. The rebuilt aggregate matches the frozen reference series of the run of record on all 45 common quarters (maximum difference \$0.000000bn) and its processed $\log_{10}$ counterpart on all 46 (maximum difference 5×10^{-5}).

The comparison series aggregates six large non-AI capital-intensive firms: Walmart, ExxonMobil, AT&T, Verizon, Union Pacific, and Home Depot, fetched through the identical EDGAR pipeline. Firm-specific tag quirks (AT&T, Verizon, and Home Depot report quarterly capex under `ProductiveAssets`-family tags; Walmart and Home Depot have January-31 fiscal year ends mapped to the nearest calendar quarter) are resolved against the live EDGAR records, and 2023 calendar-year sums match published figures for all six firms (AT&T \$17.85bn, Verizon \$18.77bn, ExxonMobil \$21.92bn, Walmart \$20.61bn, Union Pacific \$3.61bn, Home Depot \$3.23bn).

The outcome is $\log_{10}$ of group capex in \$bn per quarter, so slopes are in dex/yr (one dex is a factor of ten). The running variable is the quarter midpoint in fractional years; the ChatGPT cutoff is $t_0 = 2022.913$ (2022-11-30). A donut of 0.13 years ($\approx$ one quarter) removes 2022Q4, the quarter that straddles the release.

3 Design and Methods

Group difference-in-kinks. Following [4], the DiK estimand is the change in the outcome's slope in calendar time at the cutoff for the treated group *minus* the same slope change for the comparison group, estimated by local linear weighted regression on each side of the cutoff within each group (`natex` v0.2.0 [8]; the group indicator enters the CLI's DiK period dimension with a unit policy-kink change, so $\hat{\tau} = (\text{kink})_{\text{big-4}} - (\text{kink})_{\text{control}}$ in dex/yr). The primary specification uses bandwidth 3.0 years, triangular kernel, the one-quarter donut, and HC1 standard errors; $n = 46$ quarter-cells (11 pre-cutoff and 12 post-cutoff quarters per group).

Honest-inference caveats, stated as run. The estimator's own caveats apply verbatim: the cutoff and bandwidth are user-specified, so bandwidth, donut, and placebo-cutoff sensitivity must be reported; the reported interval is conventional local-polynomial inference and may retain smoothing bias; and causal interpretation requires parallel changes in non-policy slope kinks and a time-stable marginal response at the cutoff. HC1/CR1 inference additionally assumes serial independence, which these aggregates violate: residual lag-1 autocorrelation in the primary fit is 0.657 on the

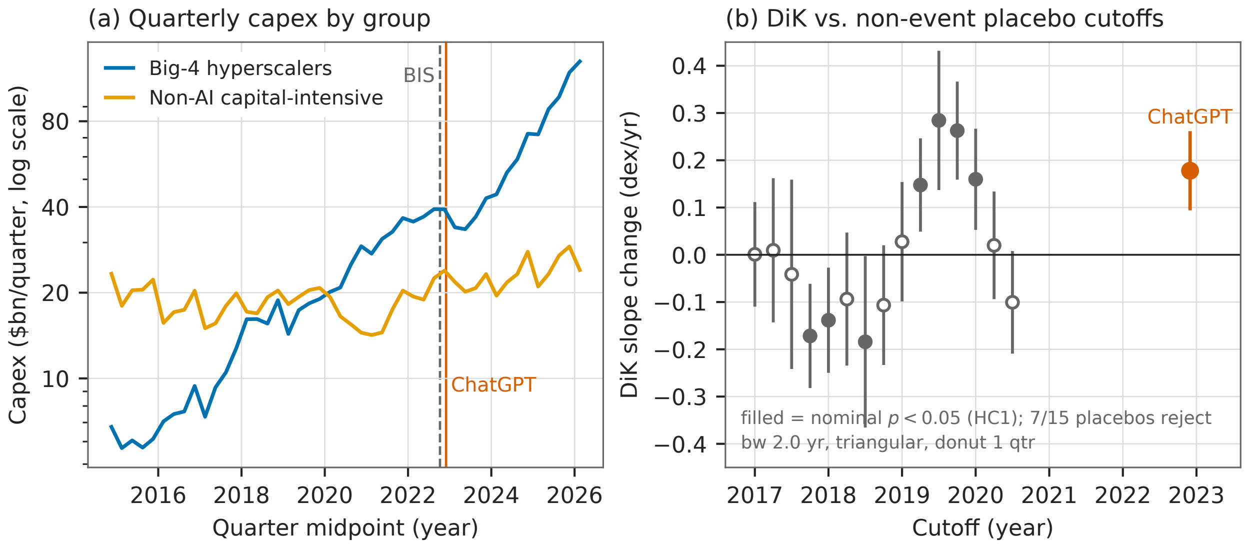

Figure 1: (a) Quarterly capital expenditure ($bn, log scale) of the big-4 hyperscaler aggregate and the non-AI capital-intensive control aggregate, 2014Q4–2026Q1, with the BIS export-control (2022-10-07, dashed) and ChatGPT (2022-11-30, solid) dates marked. (b) Group DiK estimates with 95% HC1 confidence intervals at 15 pre-period non-event placebo cutoffs (gray; pre-only sample, $t < 2022.7$) and at the ChatGPT cutoff (vermilion), all at the matched bandwidth 2.0 yr, triangular kernel, one-quarter donut; filled markers are nominally significant at the 5% level. Seven of 15 placebos reject, and the largest placebo $|z|$ exceeds the ChatGPT $|z|$.

control pre-cutoff side and 0.367 on the big-4 pre-cutoff side, precisely the setting in which nominal robust standard errors are known to be understated [3].

Falsification battery. (i) an 18-cell specification grid (bandwidth $\in \{2, 3, 4\}$ years $\times$ donut $\in \{0, 0.13, 0.25\}$ $\times$ {triangular, uniform}); (ii) a placebo cutoff at the BIS export controls (2022-10-07, $t = 2022.767$); (iii) a clean *pre-period* placebo-cutoff grid — sample restricted to $t < 2022.7$, 15 cutoffs 2017.0–2020.5, bandwidth 2.0, donut 0.13 — in the spirit of the permutation logic of [7], yielding a placebo-calibrated p-value for the main effect at the matched bandwidth; (iv) a cutoff localization scan over 2021.5–2024.75; (v) per-firm staggered DiKs of each of the ten firms against the control aggregate; and (vi) robustness checks: CR1 clustering by year and by quarter, quarter-of-year dummies, leave-one-out and no-retail control aggregates, a role-swap sanity check, and a degree-2 polynomial.

Automated pipeline, stated as run. The `natex` survey pipeline (`natex survey --seed 0`) was recorded for completeness: its automated kink family pools both groups on the auto-picked linear capex outcome and therefore cannot express the group-DiK aliasing; it returns null (Holm $p = 0.24$). The remaining families were skipped, needs-input, or failed as expected on a 92-row two-unit panel. The manual DiK runs above are the primary evidence.

4 Results

The bend is real ... The primary group DiK at ChatGPT (Table 1, first row) is $\hat{\tau} = +0.1396$ dex/yr (HC1 se 0.0302, 95% CI $[0.0804, 0.1988]$, $z = 4.62$, nominal $p = 3.85 \times 10^{-6}$). It decomposes into a big-4 kink of +0.0878 dex/yr (slope 0.0999 $\to$ 0.1876; the aggregate's growth roughly doubles, to a pace that multiplies capex by ten in about 5.3 years) and a control kink of -0.0518 (0.0811 $\to$ 0.0293). The estimate is positive in every cell of the 18-cell grid (+0.0493 to +0.1951, all nominal $p < 0.05$; the uniform kernel roughly halves $\hat{\tau}$ relative to the triangular at bandwidths ≥ 3). The point estimate is essentially invariant to CR1 clustering by year (se 0.0313) or quarter (se 0.0278), quarter-of-year dummies (+0.1396, se 0.0296), leave-one-out control aggregates (+0.1329 to +0.1515), and a no-retail control of ExxonMobil, AT&T, Verizon, and Union Pacific (+0.1453, se 0.0361); the role-swap check returns exactly -0.1396; the degree-2 estimate (+0.3022, se 0.0914) shows curvature sensitivity in magnitude but not sign. Per firm against the control aggregate: Microsoft +0.1710 (se 0.0366), Alphabet +0.2061 (se 0.0349), Amazon +0.1375 (se 0.0512), Meta +0.0308 (se 0.0518, $p = 0.55$ — a true null: Meta's surge came in 2024, after its 2022–23 pullback); control firms' own kinks are all mildly negative (-0.022 to -0.090). A localization scan (bandwidth 2.0, no donut) finds the DiK crossing zero near $t = 2022.3$, maximizing at $t = 2023.25$ (+0.2213, se 0.0440), staying above +0.19 over 2023.0–2023.75, and decaying to ≈ 0 by 2024.5 — consistent with a lagged capex response to a late-2022 shock.

... but attribution to ChatGPT fails. Three findings block a causal reading. *First*, the pre-period placebo battery fails decisively: at 15 non-event cutoffs in 2017.0–2020.5, the identical specification rejects at nominal 5% seven times (empirical size 0.467; Figure 1b). Calibrated against this placebo distribution, the ChatGPT estimate at the matched bandwidth (+0.1778, $z = 4.35$) has $p_{|z|} = (1+1)/(15+1) = 0.125$ and $p_{|\tau|} = (3+1)/16 = 0.250$; the largest placebo $|z|$ is 5.11 (at 2019.75) and the largest placebo $|\hat{\tau}|$ is 0.284 (at 2019.5) — both exceeding the ChatGPT values. *Second*, the full-sample kink at the non-event date 2021.5 is -0.3135 (se 0.0452, $|z| = 6.9$), larger in magnitude and nominal significance than the main effect: these smooth aggregates generate huge nominal kinks where nothing happened. *Third*, the cutoff is confounded: a DiK at the BIS export-control date eight weeks earlier returns +0.1253 (se 0.0317, $z = 3.95$), statistically indistinguishable from the ChatGPT estimate — quarterly data cannot separate the two events (or any other late-2022 candidate shock).

5 Caveats and Conclusion

Three caveats are decisive. First, the nominal inference is an artifact: HC1/CR1 standard errors assume serial independence, but the outcome is a pair of smooth, serially correlated aggregates (residual lag-1 autocorrelation up to 0.66), and the placebo battery measures the resulting size distortion directly — a 5% test rejects 47% of the time at non-event dates, the failure mode documented by [3] and exactly what placebo distributions exist to catch [7]. Second, a calendar-time cutoff absorbs every late-2022 shock: the BIS export controls [2] fall eight weeks before ChatGPT and produce a statistically indistinguishable estimate, so the design cannot name the shock. Third, the panel is two aggregated units observed 46 times; with ≤ 12 quarters per DiK cell, local slopes are estimated from few points on smooth curves, and the full-sample 2021.5 kink shows that this setting manufactures nominal $|z| > 6$ where nothing happened.

The verdict is descriptive-only. The post-ChatGPT divergence in capex slopes — big-4 growth roughly doubling while the control aggregate's flattens — is a genuine feature of the data: sign-robust across every specification, localized in 2023.0–2023.5 exactly as a lagged investment response to a late-2022 shock would be, and visible per firm for three of the four hyperscalers. But the design cannot certify it causally, and its headline nominal $p \approx 4 \times 10^{-6}$ overstates the evidence by roughly five orders of magnitude relative to the placebo-calibrated p of 0.125–0.25. Something bent

Table 1: Group difference-in-kinks estimates, $\log_{10}$ capex (dex/yr).

Specification	$\hat{\tau}$	se	$\lvert z\rvert$	sig. 5%
Panel A: ChatGPT cutoff ($t_0 = 2022.913$)				
triangular, bw 3, donut (primary)	+0.1396	0.0302	4.62	$*$
triangular, bw 2, donut (placebo-matched)	+0.1778	0.0409	4.35	$*$
uniform, bw 4, donut (grid minimum)	+0.0493	0.0210	2.35	$*$
primary, CR1 by year	+0.1396	0.0313	4.46	$*$
primary, quarter-of-year dummies	+0.1396	0.0296	4.72	$*$
no-retail control (XOM, T, VZ, UNP)	+0.1453	0.0361	4.03	$*$
degree 2, bw 3, donut	+0.3022	0.0914	3.30	$*$
Panel B: placebo cutoffs				
BIS export controls, $t_0 = 2022.767$, bw 3	+0.1253	0.0317	3.95	$*$
$t_0 = 2021.5$ (no event), full sample, bw 2	−0.3135	0.0452	6.93	$*$
$t_0 = 2019.75$ (no event), pre-only, bw 2	+0.2627	0.0514	5.11	$*$
$t_0 = 2019.5$ (no event), pre-only, bw 2	+0.2843	0.0736	3.86	$*$

Notes: HC1 standard errors against standard normal critical values except where noted; donut = 0.13 yr (one quarter); bandwidths in years. Outcome: $\log_{10}$ group capex, \$bn per quarter. The pre-only placebo battery (sample $t < 2022.7$, 15 cutoffs 2017.0–2020.5, bw 2) rejects at nominal 5% in 7/15 cases; the placebo-calibrated p-value of the ChatGPT estimate at the matched bandwidth is 0.125 ($\lvert z\rvert$) and 0.250 ($\lvert\tau\rvert$).

hyperscaler capex around the turn of 2023; these data cannot say it was ChatGPT.

Reproducibility. All estimates were produced with `natex` v0.2.0 [8]; estimation is deterministic (local weighted least squares), and seed 0 was passed wherever the CLI accepts one. The EDGAR pulls are not committed to the repository; the run of record retains the derived panel and all estimate files. Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the headline estimates against the numbers of record before drawing.

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Sycophancy Did Not Win Votes: The GPT-4o Deploy–Rollback Episode in 135,634 LMArena Battles

Hauke Hillebrandt*

July 2026

analysis pass of record: 2026-07, seed 0

Abstract

OpenAI deployed a GPT-4o update on 25 April 2025 that made ChatGPT conspicuously sycophantic and rolled it back on 28–29 April — an apparent A–B–A natural experiment inside LMArena's battle stream (135,634 battles, 2025-04-17 to 2025-07-24). The verdict is split, and the design itself diagnoses it. The *deploy* leg is credible: the live 4o alias shows a sharp, 4o-specific markdown collapse timed to 25–26 April (headers per 1k tokens 3.52 → 1.12, sliding-cut permutation $p = 0.013$, rank 1/78) while nine pinned-snapshot controls stay flat — proving the arena copy was *not* pinned — and it yields a clean null on the headline mechanism: the Bradley–Terry (BT) adjusted win-rate change during the sycophancy window is +0.02 log-odds (SE 0.14). Sycophancy won no votes. The *rollback* leg is an identified artifact: on exactly 2025-04-29 the pinned `claude-3-7` snapshot — whose weights cannot change — takes an equal-and-opposite persistent step in style and BT strength, demonstrating an arena-side serving change that contaminates every 29-April estimate. The 4o post-rollback regime persists through July with no reversal: the episode is A–B–C, not A–B–A, and the celebrated raw win-rate rise loads entirely on the confounded boundary. An automated `natex` survey of the 45-model panel returns nulls or refusals on every family it could configure, stated as run.

1 Introduction

In late April 2025 OpenAI shipped an update to GPT-4o that made ChatGPT, in the company's own words, "overly flattering or agreeable — often described as sycophantic," and rolled it back within days [6]. The two postmortems [6, 7] attribute the failure partly to reward signals built from user feedback — exactly the mechanism the academic literature had predicted: [8] show that human raters and preference models sometimes prefer convincingly written sycophantic responses over correct ones, so that optimizing against human preference judgments can *produce* sycophancy, and [3] document high rates of "social sycophancy" across deployed models. What that literature measures with one-off cross-model snapshots, the April episode supplies as a rare *longitudinal* event: a dated behavior change and a dated reversal.

This paper asks the money question implied by the mechanism: did the sycophantic build actually *win votes*? The natural laboratory is LMArena (Chatbot Arena), the crowdsourced pairwise-battle platform of [4], whose published battle stream brackets the episode. If human preferences reward sycophancy at the margin, the treated model's opponent-adjusted win propensity — its

*University College London. Email: `ucjthhi@ucl.ac.uk`. This paper and the underlying `natex` software were prepared with substantial assistance from Anthropic's Claude models; the author reviewed the analyses and text and is responsible for all remaining errors.

Bradley–Terry strength [2] — should rise during the deployment window and fall back after the rollback. The episode has the structure of an interrupted time series with reversal, whose inferential logic — segmented regression at pre-declared cut dates, plus controls that cannot have been treated — is standard in the policy-evaluation literature [1]. Pinned dated snapshots serve as the placebo arm: their weights cannot change, so any step they exhibit at a cut date measures platform-side confounding. Both halves of the design earn their keep here, one by working — a precise null on the vote-winning mechanism — and one by failing loudly: a pinned control moves, mirror-imaged, on exactly the rollback date, converting the "reversal" into an identified arena-side artifact. All estimation follows the open-source `natex` toolkit's honest-inference conventions [5]; the automated `natex` survey of the same panel is reported as run, refusals included.

2 Data

The source is the public `arena-human-preference-140k` release by LMArena (seven parquet files plus battle metadata, sha256-verified): 135,634 battles from 2025-04-17 00:20 to 2025-07-24 23:59 covering 53 models, with full response text, per-battle nanosecond timestamps, language and category tags, and per-side conversation metadata. Sanity: 0 of 135,634 battle ids show a parquet-vs-metadata timestamp mismatch; per-model side-A vs. side-B mean-token correlation is 0.992 with target side-A share 0.499 (no role-assignment artifact); 0 of 7,650 target responses are zero-token.

The treated series is `chatgpt-4o-latest-20250326` ("4o"): 7,650 battles — 708 pre-window, 445 deploy-window, 1,374 in the three weeks after rollback. Controls are nine pinned dated snapshots present throughout, led by `gpt-4.1-2025-04-14`, `claude-3-7-sonnet-20250219`, and dated Gemini preview builds. Outcomes are (i) computable style metrics from conversation metadata — bold markers, headers, and list items per 1,000 response tokens, response tokens, turns — and (ii) win/loss/tie votes. Following the pre-registered design (Design 1 of the source corpus's `DESIGNS.md`), raw win rate is never the estimand: opponent mix and sampling weights move across the episode (4o battles/day: 88.5 → 111.3 → 75.1), so votes enter only through windowed Bradley–Terry models.

3 Design and Methods

ABA interrupted time series. Two pre-declared cuts — deploy 2025-04-25, rollback 2025-04-28/29 — give an 8–4–8-day template (A1 Apr 17–24; B Apr 25–28; A2 Apr 29–May 6); because the break appears in-data on Apr 26 (staged rollout), the deploy leg is also estimated behavior-dated (8-day-pre/3-day-post). Segment-mean OLS uses CR1 errors clustered by date; adjusted specifications condition on language, category flags, code, weekend, and log turns; a DiD variant subtracts the pooled pinned-control change.

Permutation inference. Every headline statistic is benchmarked against a sliding-cut placebo distribution: the identical window template is recomputed at every feasible alternative cut date in the 99-day panel (77 placebo cuts for the deploy dip, 72 for the rollback step) and the two-sided rank of the true cut is reported; the minimum attainable p is 0.013–0.014, and headline p-values sit at that floor. A model-placebo grid recomputes each statistic for every sufficient-volume model (17 at the deploy cut, 19 at rollback).

Bradley–Terry with segment-specific strength. All decided battles among the window's models (Apr 17–May 6; decided 4o battles per segment 536/342/434) enter a logistic MLE in which every model has one strength and 4o has a separate strength per segment [2]; the deploy

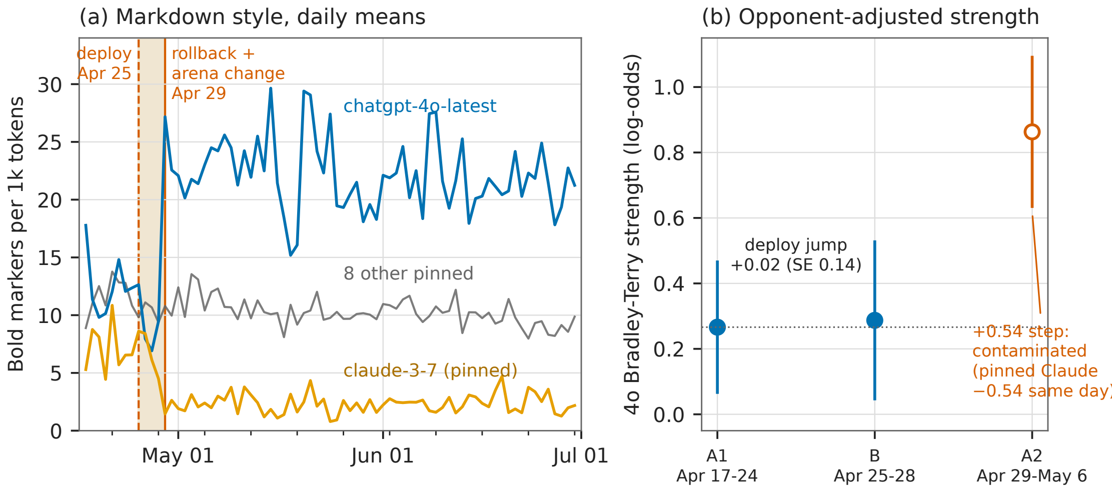

Figure 1: (a) Daily mean bold markers per 1,000 response tokens, 2025-04-17 to 2025-06-30: the live 4o alias (blue) dips during the deploy window (shaded) and steps up persistently on 29 April; the pinned `claude-3-7` snapshot (orange) mirror-images that step on exactly the same date although its weights cannot change; eight other pinned snapshots pooled (gray) are flat. (b) 4o Bradley–Terry strength by segment (95% intervals): the deploy jump is +0.02 log-odds (SE 0.14) — no opponent-adjusted votes won; the open 29-April marker is contaminated by the same-day arena-side change.

jump is $b_B - b_{A1}$. The same machinery yields per-model BT steps at the rollback date for the placebo grid.

Treatment self-diagnosis. Per the pre-registered design, whether LMArena's 4o alias tracked OpenAI's live model was unresolved but self-diagnosing: no discontinuity at the cuts would mean the arena copy was pinned and 4o becomes a control — informative either way. Resolved here: a deploy-timed discontinuity is present, so 4o was live and is treated.

Automated survey, stated as run. The `natex` survey pipeline [5] ran on the 45-model × 99-day panel (seed 0, $q = 99$): RDD null (scan $p = 1.00$); DiD null (scan $p = 0.42$, and its auto-configuration chose the outcome `n_battles` — volume, which the design spec forbids interpreting); the self-configured synthetic control failed ($t_0 = 0$, no pre-period — recorded, not patched); kink/IV/bunching needs-input; DEE skipped. A declared-input synthetic control (`natex` donors, decided win rate, t_0 = day 12) gives ATT(post) = +0.233, pre-RMSPE 0.056, post 0.249, ratio 4.44, in-space placebo $p = 0.067$ (minimum attainable 1/15) — but its top donor (weight 0.44) is the contaminated `claude-3-7` snapshot, so it is descriptive only.

4 Results

Deploy leg: a credible, 4o-specific style break ... Behavior-dated at 2025-04-26 (8-day-pre/3-day-post), 4o's markdown collapses: headers per 1k tokens fall from 3.52 to 1.12 (dip −2.406, sliding-cut permutation $p = 0.0128$, rank 1/78, placebo $|q_{90}| = 0.52$; Welch $t = -9.85$, $p = 6.1 \times 10^{-22}$) and bold falls from 11.90 to 8.22 (dip −3.683, $p = 0.0769$). Pooled pinned controls are flat (header dip −0.18, bold −1.51). The segmented ABA jump at the Apr-25 cut is −1.831 headers/1k (CR1

SE 0.526), robust to language/category/code/weekend/turns adjustment (-1.723, SE 0.446); bold -2.965 (SE 1.285), adjusted -2.749 (SE 0.915); the DiD against pooled pinned controls gives -1.676 (SE 0.539). In the model-placebo grid at the true cuts, 4o's transient dip ranks 1/17 on both bold and header. The break is real, 4o-specific, and correctly timed — the arena alias was not pinned.

... and a clean null on the vote mechanism. The segment-specific BT deploy jump is +0.021 log-odds (SE 0.144): implied win rate against the fixed A1 opponent pool moves 0.540 → 0.545. During the four days users saw the sycophantic build, it gained *zero* opponent-adjusted win rate.

Rollback leg: a large step that is not a reversal. At 2025-04-29 (8-day/8-day windows) 4o steps up, not back: bold +12.110/1k (permutation $p = 0.0137$, rank 1/73, placebo max $|3.70|$), headers +4.246 ($p = 0.0137$, placebo max $|0.62|$), response tokens +107.8 ($p = 0.123$), BT +0.537 log-odds (SE 0.134; $p = 0.0137$, placebo max $|0.31|$). The new regime persists through 2025-07 (monthly bold 21–22/1k vs. 12.2 pre): no reversal to baseline ever occurs. The episode is A–B–C, not A–B–A.

The decisive diagnostic: a pinned control moves. The pinned `claude-3-7` snapshot (full name in Section 2) steps on exactly 2025-04-29, mirror-imaged and persistent: bold -5.01/1k, headers -4.63/1k, response tokens -214, BT -0.539 log-odds (SE 0.187). Pinned weights cannot change, so LMArena changed its serving configuration that day; arena-wide prompt mix was stable (battles/day 578 → 557, English share 0.639 → 0.672). Every 29-April estimate — including 4o's — is therefore contaminated. In the rollback placebo grid 4o ranks 1/19 on bold and 2/19 on header, behind exactly this contaminated control. The raw decided win rate (0.549 → 0.529 → 0.675; full-post 0.663) loads entirely on the confounded boundary and never reverts: the widely repeated "sycophancy raised the win rate, rollback lowered it" narrative is an artifact of the arena change.

5 Caveats and Conclusion

Four caveats bound the claims. First, per the pre-registered design, the style metrics computable from conversation metadata are a *proxy* for sycophancy — directional evidence of a behavior change, not a sycophancy measurement; LLM-judge scoring of the stored response text is the designated follow-up. The conclusion that survives this caveat is exactly the null: whatever the treated build did to users, it did not win opponent-adjusted votes. Second, the rollback leg is unidentified: the arena-side serving change of 2025-04-29 sits on the same date, so no 29-April estimate separates OpenAI's rollback from LMArena's change. Third, permutation p-values are floor-limited (1/78, 1/73) by the panel length; they establish rank-1 extremity, not fine-grained significance. Fourth, the declared-input synthetic control is descriptive only (its top donor is the contaminated control), and the survey's self-configured families returned nulls or refusals, reported above as run.

The conclusion is a split verdict that the design itself delivered. The deploy leg shows the ABA machinery working: a sharp, correctly timed, 4o-specific break, flat pinned controls, and a precise behavioral null — sycophancy did not win votes. The rollback leg shows the machinery catching its own confound: a pinned model that cannot change, changing — the single observation that converts the entire post-rollback narrative, including the celebrated raw win-rate rise, into an arena-side artifact. Platform-side serving changes are an underappreciated threat for any study that treats arena time series as model behavior; pinned snapshots are the cheap, decisive placebo that catches them.

Table 1: Headline estimates. Permutation p-values are two-sided sliding-cut ranks; 0.013–0.014 is the minimum attainable.

Quantity	Estimate	SE	perm. p	placebo benchmark
Panel A: deploy leg (cut 2025-04-25/26), 4o				
header/1k dip (8d/3d)	−2.406	—	0.0128	$\lvert q_{90} \rvert = 0.52$; rank 1/78
bold/1k dip (8d/3d)	−3.683	—	0.0769	
header ABA jump (CR1)	−1.831	0.526	—	controls −0.18
adjusted	−1.723	0.446	—	
header DiD vs. controls	−1.676	0.539	—	
BT jump (log-odds)	+0.021	0.144	—	*null: no votes won*
Panel B: rollback date (2025-04-29, 8d/8d steps), 4o				
bold/1k step	+12.110	—	0.0137	max $\lvert 3.70 \rvert$; rank 1/73
header/1k step	+4.246	—	0.0137	max $\lvert 0.62 \rvert$
response tokens step	+107.8	—	0.123	
BT step (log-odds)	+0.537	0.134	0.0137	max $\lvert 0.31 \rvert$
Panel C: pinned `claude-3-7-sonnet-20250219`, same date (violation)				
bold/1k step	−5.01	—	—	weights cannot change
header/1k step	−4.63	—	—	
response tokens step	−214	—	—	
BT step (log-odds)	−0.539	0.187	—	

Notes: style metrics are per 1,000 response tokens. Dips use behavior-dated 8-day-pre/3-day-post windows; steps use 8-day/8-day windows at the cut. CR1 clusters by date. BT = segment- or window-specific Bradley–Terry strength over all decided battles among window models.

Reproducibility. All estimates derive deterministically (seed 0) from the public battle release; no dataset files are committed. Figure 1 regenerates from the committed `figures/make_fig.py`, which asserts the headline estimates against the numbers of record before drawing.

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Bunching Below the Line: Missing Mass Above the EU AI Act's 10^{25}-FLOP Threshold

Hauke Hillebrandt*

July 2026
analysis pass of record: 2026-07, fully deterministic (no RNG)

Abstract

After the EU AI Act entered into force on 1 August 2024, the density of new AI models just *above* the Act's 10^{25}-FLOP systemic-risk presumption line collapses relative to just below it. In Epoch AI's compute panel (719 models with training-compute estimates, January 2023 to June 2026), a McCrary-style binned-Poisson density test at the statutory cutoff $\log_{10}\text{FLOP} = 25.0$ gives $\theta = -1.334$ (SE 0.878, $p = 0.129$) in the primary ± 0.5-dex window and is negative in all 15 window $\times$ bin specifications (10 of 15 with $p < 0.05$); the pre-Act period is never significant (0 of 15). A Fisher exact below/above $\times$ pre/post contrast puts the shift at odds ratio 0.173 (95% CI $[0.048, 0.619]$, $p = 0.00715$): at the pre-Act above:below ratio, 28.9 post-Act models are expected in $(25.0, 25.5]$ and 5 are observed — an 82.7% deficit. The deficit is specific to the statutory line (placebo thresholds at 24.5 and 25.5 are null or opposite-signed, with mass piling up just below 25.0), specific to the post-Act period (a placebo split date inside the pre-Act sample gives OR 0.90, $p = 1.0$), and correctly timed: the above-line count collapses from 2025H1, after entry into force and before the August 2025 applicability date for general-purpose AI obligations. Two results are nulls and are stated as nulls: the preferred narrow-window pre/post interaction in the discontinuity is $p = 0.729$, significant only at the widest bandwidth, and the pre-Act density test never rejects. With only 5–13 post-Act models above the line, and an Act-induced *disclosure* response that would mimic bunching exactly, we grade this as moderate evidence of bunching — behavioral or reporting — at the EU AI Act's compute threshold, not proof of a training-compute response.

1 Introduction

Article 51(2) of the EU AI Act [2] presumes that a general-purpose AI model poses "systemic risk" when the cumulative compute used for its training exceeds 10^{25} floating-point operations, attaching notification, evaluation, and risk-mitigation obligations to models above the line. The Act entered into force on 1 August 2024 and its general-purpose AI obligations became applicable on 2 August 2025. A bright-line quantity threshold of this kind is, in the language of public economics, a *notch*: crossing it discretely changes the regulatory burden, so agents with values just above the cutoff have an incentive to locate just below it.

A large empirical literature shows that agents do exactly this. [10] developed bunching estimation at kinks in the US income-tax schedule; [7] extended it to notches, showing that a discrete

*University College London. Email: `ucjthhi@ucl.ac.uk`. This paper and the underlying `natex` software were prepared with substantial assistance from Anthropic's Claude models; the author reviewed the analyses and text and is responsible for all remaining errors.

jump in liability produces excess mass just below the cutoff and *missing mass* just above it; [6] surveys the method and its spread beyond taxation to regulation and private-sector prices. The econometric diagnostic for such sorting at a cutoff is the density discontinuity test of [9]: if agents manipulate the running variable, its density jumps at the threshold.

Whether training compute is a good handle for AI governance is actively debated. [11] argue that compute is uniquely governable — detectable, quantifiable, and produced by a concentrated supply chain — and [3] defend training-compute thresholds, including the EU's 10^{25}-FLOP line, as the most workable trigger for regulatory oversight. [5] counters that compute thresholds are shortsighted and invite exactly the strategic responses studied in the bunching literature. On the empirical side, [8] project model counts above the EU threshold *forward* from pre-Act scaling trends, forecasting 103–306 models above 10^{25} FLOP by 2028 — a superlinear increase. To our knowledge, no published estimate tests whether the observed post-Act density of models near the line already departs from that trend. This paper supplies that test: a McCrary-style density discontinuity estimate at the statutory cutoff, before versus after the Act's entry into force, with threshold, date, and timing placebos.

2 Data

The input panel is built from Epoch AI's *Data on AI Models* database [1], keeping every model with a non-null positive training-compute estimate and a release date from 2 January 2023 to 4 June 2026: 719 models with columns model, date, and $\log_{10}$ training FLOP. Event dates are the Act's entry into force (2024-08-01) and the applicability of general-purpose AI obligations (2025-08-02). The pre-Act period is 2023-01 to 2024-07 ($n = 21$ models within ± 0.5 dex of the cutoff) and the post-Act period is 2024-08 onward ($n = 52$ in-window). Compute values for recent frontier models are largely Epoch estimates rather than disclosures, and models without any compute estimate ($\approx$ 2,800 of $\approx$ 3,500 database rows) are necessarily excluded — a selection channel we return to in Section 5. There is no value heaping near the cutoff: at most 3 duplicate $\log_{10}$-compute values occur within $[24.5, 25.5]$, so the results below are not an artifact of rounded compute estimates.

3 Design and Methods

Density discontinuity at the statutory line. The main estimate is the `natex` toolkit's [4] McCrary-style [9] binned-Poisson density test, run directly at the statutory cutoff: model counts in equal-width bins of the signed distance $\log_{10} \text{FLOP} - 25.0$ are fit by a Poisson regression with a linear trend in distance and a jump term at zero; θ is the jump in log density at the cutoff, so $\theta < 0$ means a deficit just above the line — avoidance bunching below it. The primary specification uses a ± 0.5-dex window and 10 bins; sensitivity covers the full grid $h \in \{0.5, 0.75, 1.0\} \times \text{bins} \in \{8, 10, 12, 16, 20\}$ (15 specifications per period). Because the cutoff is fixed by statute rather than searched for, the frozen-geometry caveat that applies to `natex`'s post-discovery density checks does not bite here.

Contrasts and placebos. The pre/post change in bunching is measured two ways: a Fisher exact test on the 2×2 table of below/above counts in the ± 0.5-dex window by period, and a θ-difference (interaction) z-test between the period-specific density jumps. Specificity is probed with placebo thresholds at 24.5 and 25.5 and a placebo split date of 2023-08-01, which places both "periods" inside the pre-Act sample.

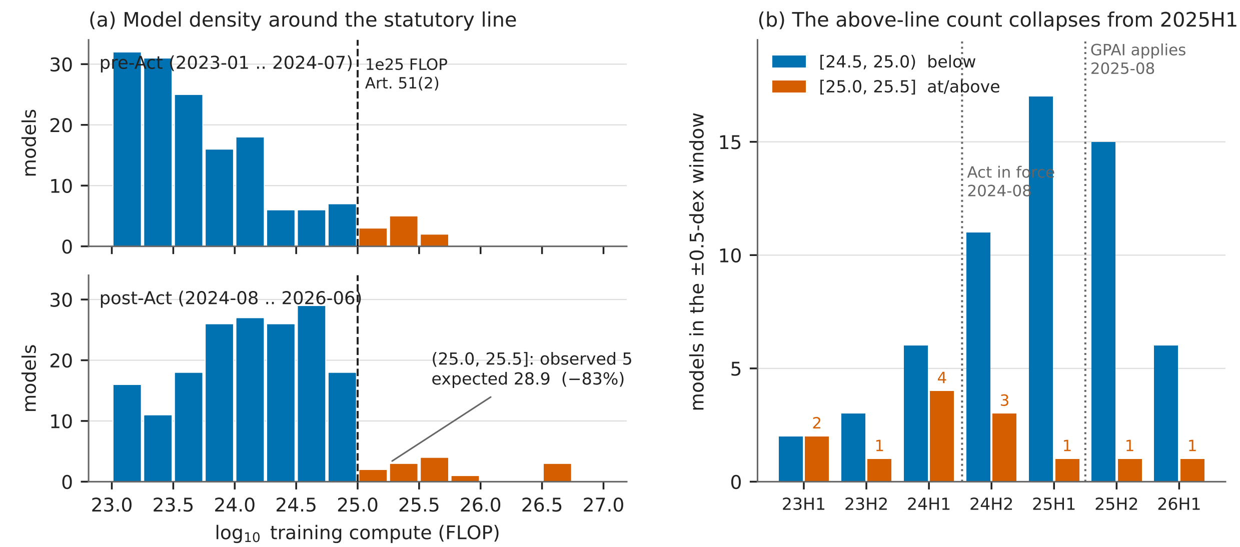

Figure 1: (a) Model counts in 0.25-dex bins of $\log_{10}$ training compute, pre-Act (top) versus post-Act (bottom), colored by side of the statutory 10^{25}-FLOP line (dashed). Post-Act, mass accumulates just below the line while the first half-dex above holds 5 models against 28.9 expected at the pre-Act ratio; the frontier jumps far above the notch. (b) Below/above counts per half-year in the ± 0.5-dex window: the above-line count (labeled) collapses from 2025H1, after entry into force and before GPAI applicability, while the just-below count keeps growing.

Honest inference. Standard errors for θ are Wald standard errors from the binned Poisson fit and can be optimistic when the underlying density is strongly curved within the window; at the widest bandwidth ($h = 1.0$) curvature of the steeply falling compute density can leak into θ, which is why the narrow-window specification is primary and the wide-window significance is discounted below. All counts, tests, and figures derive deterministically from the input panel: there is no random number generation anywhere in the pipeline.

4 Results

A post-Act deficit just above the line. Post-Act (2024-08-01 onward), the primary density test at $\log_{10} = 25.0$ gives $\theta = -1.334$ (SE 0.878, $p = 0.129$) with $n = 52$ in-window models split 47 below / 5 above. Across the full sensitivity grid θ is negative in 15 of 15 specifications (range -1.249 to -2.566) and significant at the 5% level in 10 of 15 — all $h = 0.75$ and $h = 1.0$ specifications, with significant p-values from 0.0010 to 0.0143. The pre-Act period shows no such pattern: $\theta = -0.880$ (SE 0.973, $p = 0.366$), $n = 21$ split 13/8, and 0 of 15 sensitivity specifications reject (θ from -1.173 to $+0.494$, minimum $p = 0.243$).

The pre/post contrast. The Fisher exact test on below/above × pre/post counts in the ± 0.5-dex window — table $[[13, 8], [47, 5]]$ — gives OR $= 0.173$ (95% CI $[0.048, 0.619]$, $p = 0.00715$). At the pre-Act above:below ratio, 28.9 post-Act models are expected in $(25.0, 25.5]$; 5 are observed, so roughly 24 models are "missing" — an 82.7% deficit. The formal interaction between the period-specific density jumps is a null at the primary specification and is stated as one: $\Delta\theta = -0.454$ (SE 1.311, $z = -0.346$, $p = 0.729$). Across the grid the interaction is always negative at $h \geq 0.75$ but

Table 1: Headline estimates. Density tests: binned-Poisson jump θ in log density at the statutory cutoff $\log_{10}$ FLOP = 25.0; the primary window is ± 0.5 dex with 10 bins, and the sensitivity grid crosses $h \in \{0.5, 0.75, 1.0\}$ with $\{8, 10, 12, 16, 20\}$ bins. Fisher tests use below/above counts in the ± 0.5-dex window by period.

Quantity	Estimate	SE / CI	p	notes
Panel A: density jump at the statutory line				
θ, pre-Act (2023-01..2024-07)	−0.880	0.973	0.366	$n = 21$, 13/8; 0/15 specs $p < 0.05$
θ, post-Act (2024-08..)	−1.334	0.878	0.129	$n = 52$, 47/5; 10/15 specs $p < 0.05$
$\Delta\theta$ (post − pre)	−0.454	1.311	0.729	*null at primary spec*
Panel B: Fisher exact below/above × pre/post				
threshold 25.0, Act date	OR 0.173	$[0.048, 0.619]$	0.00715	$[[13, 8], [47, 5]]$; −82.7%
placebo threshold 24.5	OR 1.637	—	0.248	post-GPAI: OR 2.510, $p = 0.0386$
placebo threshold 25.5	OR 4.00	—	0.350	opposite direction
placebo date 2023-08-01	OR 0.90	—	1.0	both halves pre-Act; no effect

Notes: $\theta < 0$ is a density deficit just above the cutoff. SEs are Wald standard errors from the binned Poisson fit. Significant sensitivity p-values for the post-Act θ range from 0.0010 to 0.0143 (all $h = 0.75$ and $h = 1.0$ specifications); the interaction is significant only at $h = 1.0$ with ≥ 12 bins ($p = 0.0049$–0.0342). Placebo-threshold odds ratios use the same ± 0.5-dex construction centered on the placebo line.

significant only at $h = 1.0$ with 12/16/20 bins ($p = 0.0249/0.0049/0.0342$; $h = 1.0$ with 10 bins gives $p = 0.0639$), where curvature leakage is a concern (Section 3). The pre-period, with only 21–47 in-window models, is too small to pin down the counterfactual discontinuity.

Placebos. The deficit is specific to the statutory line and period. At placebo threshold 24.5 the same pre/post Fisher contrast gives OR = 1.637, $p = 0.248$; restricted to the post-GPAI period it gives OR = 2.510, $p = 0.0386$ with the *opposite* sign — more mass in $[24.5, 25.0)$ than $[24.0, 24.5)$, i.e. mass piling up just below 25.0, consistent with bunching at the line and inconsistent with a generic high-compute slowdown. At placebo threshold 25.5: OR = 4.00, $p = 0.350$, direction opposite to the 25.0 effect. The placebo split date 2023-08-01, with both halves pre-Act, gives OR = 0.90, $p = 1.0$ at the true threshold — no pseudo-effect.

Timing. Below:above counts per half-year in the ± 0.5-dex window are 2023H1 2:2, 2023H2 3:1, 2024H1 6:4, 2024H2 11:3, then 2025H1 17:1, 2025H2 15:1, 2026H1 6:1 (Figure 1b). The collapse begins in 2025H1 — after entry into force (August 2024), before the obligations became applicable (August 2025) — consistent with anticipatory compliance: the 10^{25} presumption was public from July 2024 and attaches to models placed on the market from August 2025. By year, $[25.0, 25.5)$ holds 3, 7, 2, 1 models in 2023–2026 while $[24.5, 25.0)$ grows 5, 17, 32, 6. Secular compute growth biases *against* the finding — later cohorts should put more, not less, mass above any fixed line, and do at 24.5. Meanwhile 13 post-Act models sit at or above 10^{25} at some height, with the frontier jumping far above the notch (Grok 3 at $\log_{10} = 26.54$, GPT-4.5 at 26.58, Llama 4 Behemoth preview at 25.71, Grok 4 at 26.70, GPT-5 at 25.82) — the classic bunching signature in which a notch distorts only marginal decisions [7].

5 Caveats and Conclusion

Three caveats bound the claims. First, *small n*: only 5 models sit in $(25.0, 25.5]$ and 13 at any height above 10^{25} post-Act; the preferred narrow-window pre/post interaction is $p = 0.729$, and the interaction is significant only at $h = 1.0$ with ≥ 12 bins (p from 0.0049 to 0.0342), where curvature of the steeply falling density can leak into θ. Second, *measurement, not behavior*: post-2024 frontier compute values are largely Epoch estimates, and models without a compute estimate are excluded (719 of $\approx$ 3,500 rows have compute). If the Act suppresses *disclosure* of parameters and token counts near the line, the identical density gap appears with no training-compute response at all; distinguishing a compliance response from a reporting response requires disclosure-side data and is the designated follow-up. Third, a *confounded trend break*: the deficit's onset (2025H1) coincides with the industry shift toward inference-time scaling and distilled or mixture-of-experts models that followed the late-2024 o1-style reasoning models, which independently reduced demand for 10^{25}-plus pre-training runs among non-frontier labs — though this confound predicts a broad high-compute slowdown, not a deficit specific to the statutory line with mass accumulating immediately below it.

The conclusion is graded accordingly. The post-Act deficit just above the EU AI Act's 10^{25}-FLOP line is sign-consistent in all 15 density specifications, large (Fisher $p = 0.00715$, 82.7% missing mass), specific to the statutory threshold and period, and correctly timed, with the marginal-crossers-vanish, frontier-jumps-far-above signature that notch responses produce. But with 5–13 post-Act models above the line, a null preferred narrow-window interaction, and an Act-induced disclosure response that would mimic bunching exactly, it is moderate evidence of bunching — behavioral or reporting — rather than proof of a training-compute response. Either reading is informative for the compute-threshold debate [3, 5]: the forward projections of [8] assumed threshold-blind scaling, and the data just above the line already look threshold-aware.

Reproducibility. All estimates derive deterministically from the public Epoch AI panel; no dataset files are committed. Figure 1 regenerates from the committed `figures/make_fig.py`, which asserts the headline estimates against the numbers of record before drawing.

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Still No Kink at o1: A Precision-Weighted Rerun of the Epoch Capabilities Index on a Fresh Vintage

Hauke Hillebrandt*

July 2026

analysis pass of record: 2026-07, `natex` v0.2.0, seed 0

Abstract

The aggregate AI capabilities trend did not bend at o1. On a fresh vintage of the Epoch Capabilities Index (ECI; 211 models, February 2023 – July 2026, twelve more months than the prior analysis pass), a sharp regression kink design in calendar time at the o1-preview release (12 September 2024) estimates an all-models slope change of +0.0065 ECI points/day (HC1 se 0.0131, $t = 0.50$, $p = 0.618$, 95% CI $[-0.019, +0.032]$; slopes $\approx$ 22.3 pre vs. $\approx$ 24.7 points/yr post), null in 17 of 18 specification cells — while the same specification rejects at 4 of 8 placebo cutoffs away from the release, so the design has power and the dial reads zero. A per-model precision-weighted rerun (weights $1/\sigma_i^2$ from the published confidence intervals) returns a nominally significant *negative* kink (-0.0190, se 0.0079, $t = -2.41$, $p = 0.016$; 18/18 cells), but triage identifies it as an artifact of smooth post-2024 concavity rather than an o1-localized break: placebo cutoffs at +90/+180/+270 days give same-sign, larger-or-comparable kinks (-0.0399, $p = 0.0064$; -0.0297, $p = 5.5 \times 10^{-6}$; -0.0154, $p = 0.021$), and a local-quadratic fit absorbs it ($p = 0.140$ at bandwidth 540). The clean frontier-records series (24 records, only 4 pre-cutoff points in the estimation cell) remains a non-estimable design: its nominal $t = 2.28$ dies under a 30-day donut ($t = 0.07$), flips sign under a 45-day donut and at bandwidth 730, and its placebo grid rejects 5/8. A high-confidence subset (excluding 26 Unverified/Speculative models) reproduces every conclusion, and there is no release-density kink at the cutoff ($t = -0.42$). The prior-vintage null replicates and is upgraded; per the collection's ranking specification this result folds into the flagship paper's ECI section as an appendix rather than standing as an independent finding.

1 Introduction

OpenAI released o1-preview on 12 September 2024, the first widely deployed "reasoning model" trained with large-scale reinforcement learning to spend inference-time compute on a chain of thought [7]. The release is a natural candidate for a regime change in AI capabilities: contemporaneous research argued that scaling test-time compute can beat scaling parameters [8], and the reasoning-model recipe was adopted across frontier labs within months. If the recipe changed the *rate* of capability progress, an aggregate capability index should show a slope change — a kink — in calendar time at or shortly after the release.

Measuring that requires a scale on which models years apart are comparable. The Epoch Capabilities Index (ECI) stitches dozens of benchmarks onto a single latent-ability scale via an

*University College London. Email: `ucjthhi@ucl.ac.uk`. This paper and the underlying `natex` software were prepared with substantial assistance from Anthropic's Claude models; the author reviewed the analyses and text and is responsible for all remaining errors.

item-response-theory model, precisely to support longitudinal comparisons of this kind [5]. This paper asks a narrow question: does the ECI trend kink at o1?

The estimand — a change in trend slope at a known date — is the regression kink design (RKD) of [2], applied here with calendar time as the running variable; [1] formalize the closely related difference-in-kinks design that the same toolkit implements. Two strands of methodological literature discipline the exercise. First, regression discontinuity (and kink) designs in *time* lack the randomization logic of cross-sectional designs: units near the cutoff are not quasi-randomly assigned, and serial dependence and smooth trends generate spurious breaks [4]. Second, kink estimates on smooth series are known to produce spuriously significant slope changes, which is why placebo distributions at non-event dates are the appropriate yardstick [3]. Both concerns are operationalized below through a mandatory placebo-cutoff battery and curvature checks, estimated with the `natex` toolkit [6].

This note is a robustness upgrade of a prior null: an earlier analysis pass on a 2025 vintage of the index found no kink at o1. The fresh vintage adds twelve months of post-cutoff data and per-model confidence intervals, which enable a precision-weighted re-estimate — the natural next test, since noisy early-2023 models could in principle mask a true kink in the unweighted fit.

2 Data

The outcome is the fresh vintage of Epoch AI's `eci_scores.csv` (fetched July 2026): 211 aggregated models with release dates from 2023-02-24 to 2026-07-09 and no missing score or date, spanning ECI $\approx$ 55 to 161.77. The vintage adds roughly twelve months of data relative to the prior analysis pass. Each model carries a 95% confidence interval; the per-model precision weight is $w_i = 1/\sigma_i^2$ with $\sigma_i = (\mathrm{ci}_{\mathrm{high},i} - \mathrm{ci}_{\mathrm{low},i})/3.92$. Two models (Claude 3.5 Sonnet and GPT-5, the scale's two anchor models) have no published CI; their σ_i is imputed at the sample median (2.005) for the weighted runs only.

A version-level Confidence flag is attached by a two-pass name join against Epoch's per-version index file; 189 of 211 models match (the 22 unmatched are treated as not-flagged), and 26 models (12.3%) are flagged Unverified/Speculative — including the current #1, GPT-5.6 Sol (ECI 161.77), and GPT-5. A high-confidence subset excludes these 26 models ($n = 185$).

Two series are analyzed. The *all-models* series uses all 211 releases. The *clean frontier-records* series keeps strict running-maximum records of the score (computed over non-missing scores only, in date order, with a stable same-day tie-break): 24 records, 7 pre-cutoff and 17 post-cutoff — and only 4 pre-cutoff points inside bandwidth 540 days, 3 after a 45-day donut, which already flags it as a marginal design.

3 Design and Methods

Sharp RKD in calendar time. The running variable is days since the o1-preview release (cutoff = 0 at 2024-09-12). The estimand is the change in the slope of ECI in points/day at the cutoff, estimated by local linear regression on each side within the bandwidth, triangular kernel, HC1 standard errors (`natex` v0.2.0 [6], `natex kink` with a unit policy-kink change; [2, 1]). The primary specification uses bandwidth 540 days, donut 0 ($n = 179$; 85 left, 94 right). A specification grid varies bandwidth $\in \{365, 540, 730\}$ days $\times$ kernel $\in \{\text{triangular}, \text{uniform}\}$ $\times$ donut $\in \{0, 30, 45\}$ days (18 cells per series–weighting pair). The CI-weighted estimator is a local WLS that mirrors `natex` exactly (same selection, kernel, block design, HC1) with weights kernel $\times\, 1/\sigma_i^2$; with unit weights it reproduces `natex regression_kink` in all 68 evaluable grid cells (tolerance 10^{-8} on

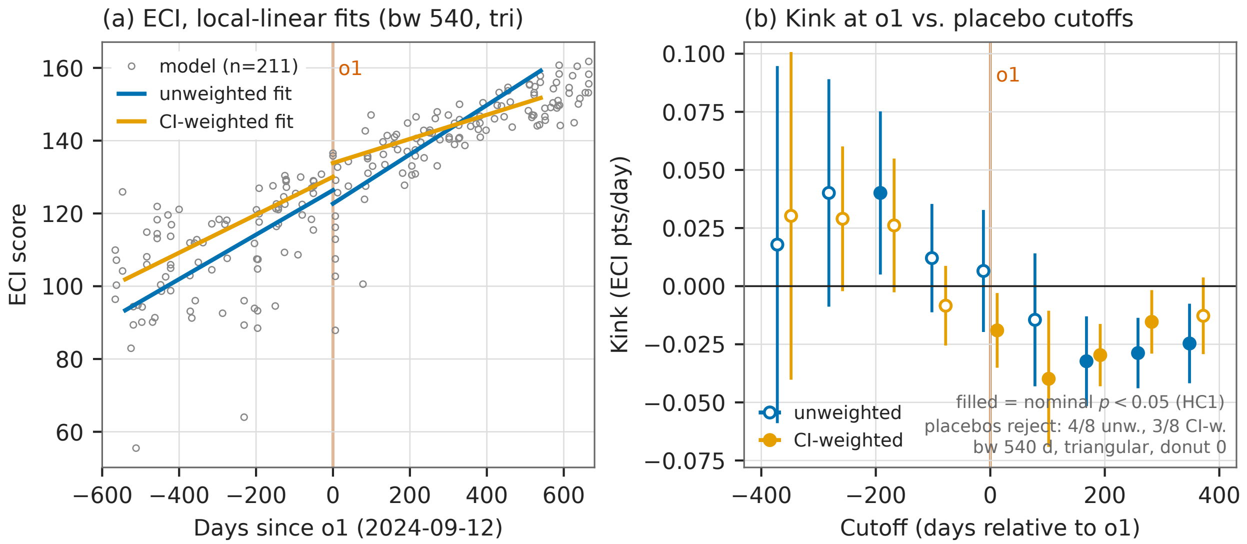

Figure 1: (a) The 211 fresh-vintage ECI scores against days since the o1-preview release (vertical line), with per-side local-linear fits (bandwidth 540 days, triangular kernel): unweighted (blue) and CI-weighted (orange). The unweighted slopes are statistically indistinguishable across the cutoff; the CI-weighted fit bends down. (b) Kink estimates with 95% HC1 confidence intervals at the o1 cutoff and at eight placebo cutoffs, both weightings (points offset horizontally for legibility); filled markers are nominally significant at the 5% level. The unweighted design rejects at 4/8 placebo cutoffs while reading zero at o1; the CI-weighted negative kink at o1 is matched by same-sign, larger placebos at +90 and +180 days.

points, 10^{-6} on SEs) and matches the NaN reasons in the rest, and the CLI headline cells match the grid to all printed digits. The design is deterministic; seed 0 is declared and no stochastic step exists.

Honest-inference caveats, stated as run. This is a calendar-time RKD: model releases are not randomized around the cutoff, so the cross-sectional RD identification logic does not apply and smooth trends can masquerade as kinks [4]. HC1 treats observations as independent, ignoring same-day multi-releases and lab-level clustering. The placebo-cutoff grid — the same specification re-estimated at ±90, ±180, ±270, ±360 days, in the spirit of [3] — is therefore the operative guard, and below it is what both establishes power for the null and disqualifies the weighted "significant" cell. A local-quadratic (degree-2) refit checks whether any kink survives allowing smooth curvature; a release-density kink test (binned 30-day release counts) checks for composition churn at the cutoff.

4 Results

The unweighted null, now with more power. The all-models headline kink (Table 1, first row) is +0.00653 ECI points/day (se 0.01308, $t = 0.50$, $p = 0.618$, 95% CI $[-0.0191, +0.0322]$): the trend runs 0.0611 points/day before o1 and 0.0677 after ($\approx$ 22.3 vs. $\approx$ 24.7 points/yr). Of the 18 grid cells, 17 are null; the single nominal rejection (bandwidth 365, triangular, donut 30: -0.0547, se 0.0269, $t = -2.03$, $p = 0.042$) is the grid's smallest cell and negative. The placebo grid shows the null is not for lack of power: at bandwidth 540 the same specification rejects at 4 of 8 non-event cutoffs (-180 days: +0.0401, $p = 0.023$; +180: -0.0323, $p = 7.1 \times 10^{-4}$; +270:

-0.0288, $p = 1.0 \times 10^{-4}$; +360: -0.0247, $p = 0.0034$) while the true cutoff is null in all cells — the instrument moves elsewhere and reads zero at o1. The prior-vintage null replicates on twelve more months of data.

The CI-weighted "kink" is a curvature artifact. Precision weighting flips the headline to a nominally significant *negative* kink: -0.01903 (se 0.00788, $t = -2.41$, $p = 0.0158$), significant in 18 of 18 grid cells ($|t|$ from 2.12 to 3.32), with slopes 0.0522 pre vs. 0.0331 post (≈ 19.0 vs. ≈ 12.1 points/yr). Two facts disqualify an o1-localized reading. First, the placebo grid rejects with the *same sign and larger or comparable magnitude* away from the release: -0.0399 ($p = 0.0064$) at +90 days, -0.0297 ($p = 5.5 \times 10^{-6}$) at +180, -0.0154 ($p = 0.021$) at +270 (empirical size 3/8 at $\alpha = 0.05$). Second, a local-quadratic fit absorbs it: degree-2 kinks are -0.0392 ($t = -1.48$, $p = 0.140$) at bandwidth 540 and -0.0381 ($t = -1.74$, $p = 0.081$) at bandwidth 730. The precision-weighted index decelerates smoothly through 2025 — a concavity, not a kink at o1.

The frontier-records series is still not a design. With only 4 left-cell points at bandwidths ≤ 540 (3 after donut 45), the records series produces a nominal positive kink at the headline specification (+0.01313, se 0.00576, $t = 2.28$, $p = 0.023$) that no perturbation survives: a 30-day donut kills it ($t = 0.07$, $p = 0.943$), a 45-day donut flips it to significantly *negative* (-0.02817, $t = -2.52$, $p = 0.012$), bandwidth 730 flips the sign (-0.00429, $t = -0.27$), and the placebo grid rejects 5/8 — with the $-360/-270$ cells exploding to -1.14 ($p = 3 \times 10^{-4}$) on $n = 11$. The CI-weighted records headline is null (+0.00794, $t = 1.31$, $p = 0.191$). The fresh vintage does not clear the left-cell bar: the series remains non-estimable, exactly as in the prior pass.

Robustness and diagnostics. Excluding the 26 Unverified/Speculative models ($n = 185$) changes nothing: unweighted +0.01248 ($t = 0.88$, $p = 0.379$, null), CI-weighted -0.02109 ($t = -2.32$, $p = 0.020$) with the same placebo failure (weighted empirical size 4/8, unweighted 6/8). The release-density check finds no composition-churn kink at the cutoff: binned 30-day release counts give -0.00410 (se 0.00967, $t = -0.42$, $p = 0.672$), an improvement on the prior vintage's marginal $t = 1.82$.

5 Caveats and Conclusion

The caveats of record are these. First, this is a calendar-time RKD: model releases are not randomized around the cutoff, and HC1 treats them as independent (same-day multi-releases and lab-level clustering are ignored); the placebo-cutoff grid is the operative guard, and it is what both establishes power for the null and disqualifies the weighted "significant" cell. Second, data quality: 26 of 211 models (12.3%) carry Unverified/Speculative version-level Confidence flags — including the current #1, GPT-5.6 Sol (161.77), and GPT-5 — and the confidence join matched only 189/211 models (22 unmatched treated as not-flagged); the high-confidence robustness subset reproduces every conclusion. Third, two models (Claude 3.5 Sonnet and GPT-5) have no published CI and enter the weighted runs at the median imputed σ; and the index itself is a modeled latent scale whose vintages revise as benchmarks are added [5], which is why the replication across vintages matters.

The verdict is an upgraded null with power. On twelve more months of data, the aggregate ECI trend shows no kink at o1: the unweighted estimate is a precise zero at a cutoff where the design demonstrably has power to reject (4/8 placebos reject elsewhere), the one new nominally significant result — the CI-weighted negative kink, significant in 18/18 cells — is correctly identified by triage

Table 1: Sharp RKD in time at o1-preview (2024-09-12), ECI points/day. Headline specification: bandwidth 540 days, triangular kernel, donut 0, HC1.

Series	Weighting	Kink	se	t	p	n
Panel A: headline cells						
All models (211)	unweighted	+0.00653	0.01308	0.50	0.618	179
All models	CI-weighted	−0.01903	0.00788	−2.41	0.016	179
Frontier records (24)	unweighted	+0.01313	0.00576	2.28	0.023	18
Frontier records	CI-weighted	+0.00794	0.00608	1.31	0.191	18
High-confidence (185)	unweighted	+0.01248	0.01420	0.88	0.379	162
High-confidence	CI-weighted	−0.02109	0.00908	−2.32	0.020	162
Release density (30-d bins)	—	−0.00410	0.00967	−0.42	0.672	36
Panel B: triage of the nominally significant cells						
All models, CI-w., placebo +90 d	CI-weighted	−0.03986	0.01462	−2.73	0.0064	181
All models, CI-w., placebo +180 d	CI-weighted	−0.02966	0.00653	−4.54	5.5×10^{-6}	174
All models, CI-w., degree 2, bw 540	CI-weighted	−0.03922	0.02659	−1.48	0.140	179
Frontier records, donut 30	unweighted	+0.00048	0.00667	0.07	0.943	16
Frontier records, donut 45	unweighted	−0.02817	0.01118	−2.52	0.012	15
Frontier records, bw 730	unweighted	−0.00429	0.01592	−0.27	0.788	24

Notes: kinks in ECI points/day; HC1 standard errors against standard normal critical values. CI weights $1/\sigma_i^2$, $\sigma_i = (\mathrm{ci}_{\mathrm{high}} - \mathrm{ci}_{\mathrm{low}})/3.92$. Placebo-cutoff grids (8 cutoffs at ±90/180/270/360 days, bandwidth 540, triangular): all-models unweighted rejects 4/8 (true cutoff null in all cells); all-models CI-weighted rejects 3/8, same sign as the headline; frontier records rejects 5/8. Panel B donut and bandwidth rows are the perturbations of the frontier-records headline cell.

as a smooth post-2024 deceleration of the precision-weighted index (same-sign larger placebos at +90/+180, absorbed by a local quadratic) rather than an o1-localized break, and the genuine frontier-records series remains a four-left-point non-design. Whatever the reasoning-model recipe did to AI capabilities, it did not bend the aggregate index's calendar-time trend at the release date. Per the collection's ranking specification, this result folds into the flagship paper's ECI section as an appendix rather than standing as an independent mini-paper.

Reproducibility. All estimates were produced with `natex` v0.2.0 [6]; estimation is deterministic (local weighted least squares), seed 0 was declared, and no stochastic step exists in the design. The underlying `eci_scores.csv` vintage is not committed to the repository; the run of record retains the derived panels, the 144-cell grid, the placebo grids, and the validation counts. Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the headline estimates, the recomputed fit slopes, and the placebo rejection counts against the numbers of record before drawing.

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The Rally, Not the Rule: Semiconductor Stocks at the US Export Controls and DeepSeek under Automated Natural-Experiment Validation

Hauke Hillebrandt*

July 2026

analysis pass of record: 2026-07-18, `natex` v0.2.0, seed 0 (all three survey runs; kink estimator deterministic)

Abstract

We point `natex` — an automated natural-experiment discovery and validation toolkit — at weekly semiconductor stock prices around the two BIS export-control rounds (2022-10-07, 2023-10-17) and the DeepSeek crash week (2025-01-27). None of its headline "credible" verdicts survives inspection. Declared regression kinks in calendar time are "credible" at all three cutoffs (e.g. DeepSeek: $\hat{\tau} = +0.48$ log/yr, Holm $p = 1.4 \times 10^{-8}$), but each is an artifact: the DeepSeek kink is the 2025 AI rally sitting on both sides of a one-week crash — a *positive* slope change at a *negative* event — the 2022 kink collapses monotonically in bandwidth ($+0.83 \rightarrow +0.13 \rightarrow +0.05$ log/yr) with ChatGPT's release inside every bandwidth, and a "credible" 2022 RDD merely rediscovered that prices above a level lie after 2022. What survives is descriptive or null, stated as such: NVDA fell 11.6 log-points *relative to* the SOXX semiconductor index in the DeepSeek week — a level break with locally unchanged relative slope ($+0.52 \rightarrow +0.39$ log/yr), not a kink — and a difference-in-differences scan across tickers finds *no* break at the export-control dates (scan $p = 0.70$; best-fitting break 2025.95, not 2022.76), an informative null consistent with export-control costs being priced jointly with the contemporaneous AI-demand shock. On financial series, automated validation refuses honestly in places but is systematically fooled by trend regimes whenever a kink in calendar time is read causally.

1 Introduction

On 7 October 2022 the US Bureau of Industry and Security (BIS) imposed broad export controls on advanced AI accelerators bound for China, read at the time as a landmark attempt to restrict China's frontier compute [1]; on 17 October 2023 a second round closed the A800/H800 loophole. Export controls impose costs on the controlling country's own firms [3]: entity-list evidence puts US suppliers' collateral losses near $130 billion in market value, plus lost customers, lending, and employment [5]. On 27 January 2025 a demand-side shock arrived from the opposite direction: the release of DeepSeek's R1 model triggered the largest one-day market-capitalization loss in history for Nvidia, and event studies document significantly negative abnormal returns across US semiconductor stocks [7].

*University College London. Email: `ucjthhi@ucl.ac.uk`. This paper and the underlying `natex` software were prepared with substantial assistance from Anthropic's Claude models; the author reviewed the analyses and text and is responsible for all remaining errors.

The standard tool here is the event study [10]: short windows, abnormal returns against a market model, an efficient-markets identification argument. This paper deliberately uses the wrong tool — quasi-experimental designs with *calendar time* as the running or assignment variable — and asks what an automated toolkit's validation layer does about it. The dangers are known in principle: regression discontinuity in time lacks the cross-sectional no-manipulation logic of true RDDs and is biased by ignored time-series structure [8], and a regression kink [4, 2] estimated on a calendar-time running variable is, mechanically, a before/after slope contrast. The open question is empirical: does a modern validation battery — placebo covariates and cutoffs [6], density tests, bandwidth sensitivity, scan-based localization [9] — actually catch these failure modes on real financial series? We run `natex` [11] over three declared cutoffs and report every verdict, including the refusals. The answer is asymmetric: every "credible" verdict fails inspection, while the refusals, failures, and nulls are the informative outputs.

2 Data

Weekly adjusted-close prices for NVDA, AMD, ASML, TSM, and the SOXX semiconductor ETF, 784 weeks from July 2011 to July 2026 (Yahoo Finance weeklies; not committed). The analysis variable is the log adjusted close, the running variable calendar time in years; the pooled five-ticker panel gives the kink family $n_{\text{used}} = 1960$. Descriptive contrasts use the NVDA-minus-SOXX relative log price, netting out the sector factor. The three declared cutoffs are $t = 2022.7644$ (BIS round one, 2022-10-07), $t = 2023.7918$ (BIS round two, 2023-10-17), and $t = 2025.0712$ (the DeepSeek crash week beginning 2025-01-27). One confound matters throughout: ChatGPT's release (2022-11-30) lands eight weeks *after* the first BIS cutoff, inside every bandwidth the 2022 analyses select.

3 Design and Methods

Three `natex` survey runs (v0.2.0, seed 0) supply the estimates: a pooled five-ticker run declaring the DeepSeek cutoff, a pooled run declaring the BIS-2022 cutoff, and an NVDA-only run declaring the BIS-2023 cutoff. Each survey attempts seven design families on the same panel: a declared regression kink in time (local-linear RKD [4] in the difference-in-kinks tradition of [2], cross-validated bandwidth, Holm-corrected placebo battery in the spirit of [6]), a discovered RDD (candidate role assignment over {treatment, forcing, outcome} columns, scan localization [9], McCrary-style density test, covariate placebos), a sector-by-time DiD scan (SuDDDS, permutation-calibrated), synthetic control, discovered-experiments estimation (dee), IV, and bunching.

Honest-inference notes, stated as run. These caveats are part of the record, not afterthoughts. (i) `natex`'s own kink report carries the warning that "a kink in a calendar-time running variable is a before/after slope contrast" — the causal reading is on the analyst, and this paper's point is what happens when it is taken anyway. (ii) The DeepSeek-week level break below is descriptive: no placebo distribution was run, so it carries no standard error. (iii) The DeepSeek-run DiD *failed* rather than returning a number: the intake ignored the declared numeric time column and picked the string `date` column (error of record: "`time column must be numeric: date`"). (iv) The NVDA-only synthetic control *failed* by construction — "`treated unit has no finite outcome before t0`" — a single-ticker panel has no donors. (v) The dee family returned nulls with zero usable discovered experiments (fewer than its minimum of three); IV and bunching declined as `needs_input` throughout; and where a placebo battery was vacuous we say so and treat the estimate as unvalidated.

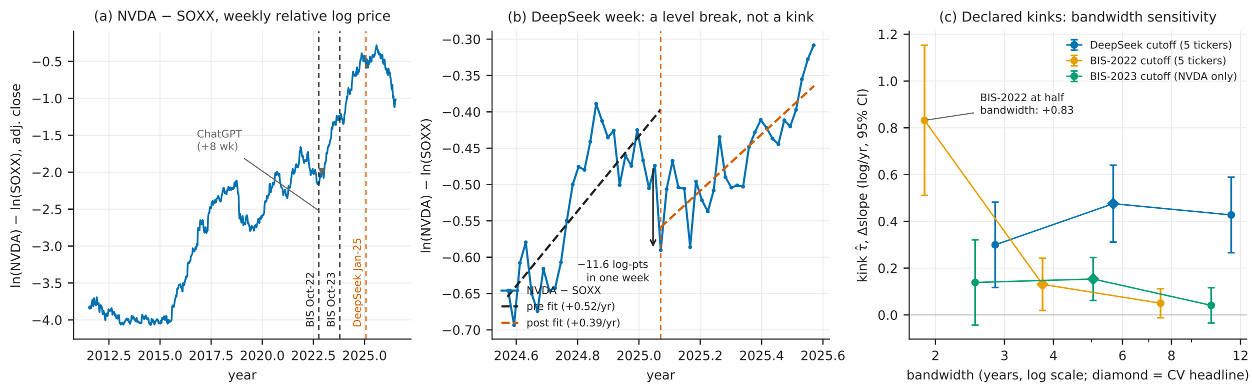

Figure 1: (a) Weekly NVDA-minus-SOXX relative log (adjusted-close) price with the three declared cutoffs; ChatGPT lands eight weeks after the BIS-2022 cutoff, and the 2023–2025 AI rally dominates the series. (b) The DeepSeek week: an 11.6 log-point one-week relative level break with locally unchanged ±26-week slope — a level break, not a kink. (c) Declared-kink estimates (95% CIs) at half, headline (diamond, cross-validated), and double bandwidth: BIS-2022 collapses monotonically, BIS-2023 fades to $t \approx 1.06$, and the DeepSeek "kink" is positive at a crash — the rally on both sides.

4 Results

The real event: a relative level break at DeepSeek. Recomputed exactly from the weekly closes, the week beginning 2025-01-27 took NVDA down −0.17211 in log terms (−17.2 log-points; −15.8% simple return) against −0.05577 for SOXX (−5.6 log-points; −5.4%): an −11.63 log-point one-week fall in NVDA *relative to its own sector*. The local relative trend barely moved: ±26-week OLS slopes of the relative log price are +0.52 log/yr before and +0.39 log/yr after (Figure 1b). The DeepSeek shock is a *level* break with an approximately unchanged local slope — the signature of a one-off repricing, not a growth-regime change. This estimate is descriptive (no placebo distribution; no standard error).

Declared kinks: "credible" three for three, wrong three for three. Table 1, panel A. At the DeepSeek cutoff the pooled kink is $\hat{\tau} = +0.4756$ log/yr (se 0.0839, 95% CI $[0.3111, 0.6400]$, Holm $p = 1.4 \times 10^{-8}$, cross-validated bandwidth 6.05 yr, $n_{\text{used}} = 1960$). The sign alone refutes the causal reading: DeepSeek was a crash, yet the "effect" is a large *positive* slope change — a six-year bandwidth spans the pre-2023 flat regime on the left and the AI rally on the right, so the kink dates the rally, not the event. At the BIS-2022 cutoff the pooled kink (+0.1305, Holm $p = 0.022$) is monotonically bandwidth-sensitive — +0.832 (se 0.164) at 1.88 yr, +0.131 (se 0.057) at 3.75 yr, +0.050 (se 0.032) at 7.50 yr — and ChatGPT's release sits eight weeks post-cutoff inside every one of those windows; nothing localizes the bend to the export controls. The NVDA-only BIS-2023 kink (+0.1534, Holm $p = 0.0011$) fades to +0.0407 (se 0.0385, $t \approx 1.06$) at double bandwidth. The ±26-week local slopes tell the same story in descriptive form: the relative NVDA-minus-SOXX slope change is +1.46 log/yr at BIS-2022 (from −0.46 to +1.00), +0.41 at BIS-2023 (from +0.61 to +1.02), and −0.13 (≈ flat, +0.52 → +0.39) at DeepSeek: the two "export-control kinks" are the AI rally starting, and the one week with a genuine event shows no kink at all.

A "credible" RDD that is a role-assignment artifact. In the BIS-2022 run the discovered-RDD family also returned "credible" (scan $p = 0.010$, density $p = 0.180$). Inspection of the

Table 1: All verdicts across the three survey runs (`natex` v0.2.0, seed 0).

Run (cutoff)	family	$\hat{\tau}$	se	inference	disposition
Panel A: declared kinks in calendar time, log price (log/yr) — all "credible", none survives					
BIS-2022 (pooled)	kink	+0.1305	0.0570	Holm p=0.022; bw 3.75	+0.83 → +0.13 → +0.05 across bw
BIS-2023 (NVDA)	kink	+0.1534	0.0469	Holm p=0.0011; bw 4.77	fades: t≈1.06 at 2×bw
DeepSeek (pooled)	kink	+0.4756	0.0839	Holm p=1.4×10^{-8}; bw 6.05	positive "kink" at a crash
Panel B: every other family disposition					
BIS-2022	rdd	1.581	0.886	scan p=0.010; CI spans 0	role-assignment artifact
BIS-2022	did	—	—	scan p=0.70; best t_0=2025.95	**informative null**
BIS-2022	dee	—	—	0 < 3 usable experiments	null
BIS-2023	rdd	—	—	scan p=0.040; placebos vacuous	unvalidated
BIS-2023	did	—	—	scan p=0.76; best t_0=2017.16	null
BIS-2023	sc	—	—	no donors (single ticker)	failed, loudly
BIS-2023	dee	—	—	0 < 3 usable experiments	null
DeepSeek	rdd	4.07	2.64	scan p=1.00; CI spans 0	null (placebos worked)
DeepSeek	did	—	—	intake chose string `date` col.	failed, loudly
DeepSeek	iv/sc/bunching	—	—	`needs_input`	refused

Notes: Kink rows: local-linear RKD in calendar time, log adjusted close, cross-validated bandwidth in years, $n_{\text{used}} = 1960$; BIS-2022 disposition: estimate at half/headline/double bandwidth. 95% CIs: BIS-2022 kink $[0.0188, 0.2423]$; BIS-2023 kink $[0.0616, 0.2453]$; DeepSeek kink $[0.3111, 0.6400]$. RDD rows: fuzzy 2SLS at the discovered candidate; CIs BIS-2022 $[-0.156, 3.317]$ and DeepSeek $[-1.11, 9.25]$; density p: BIS-2022 0.180, BIS-2023 0.312, DeepSeek 0.752 with all 8 covariate placebos at Holm $p = 1.0$. DiD rows: SuDDDS permutation scan; point estimates at the (non-rejected) BIS-2022 best t_0: $\hat{\tau}_{\text{DD}} = -2.061$ (se 0.018), $\hat{\tau}_{\text{synth}} = -0.427$ (se 0.025).

selected candidate shows why this is empty: the automated role search assigned treatment = `post_2022_controls`, forcing = `log_adjclose`, and outcome = t — it mechanically rediscovered that prices above a level tend to lie after 2022. The fuzzy estimate $\hat{\tau}_{\text{2SLS}} = 1.581$ (se 0.886, 95% CI $[-0.156, 3.317]$) crosses zero and has no economic content. In the BIS-2023 NVDA-only run the RDD scan rejected at $p = 0.040$ but the covariate-placebo battery was *vacuous* (no testable covariate in the single-ticker numeric panel; density $p = 0.312$): unvalidated, and not reported as a finding.

The informative null: no DiD break at the export controls. The SuDDDS scan over tickers and break dates in the BIS-2022 run finds nothing at the controls: scan $p = 0.70$ (LLR 2.548), with the best-fitting break at $t_0 = 2025.95$ (window 3.76 yr) — not 2022.76 (point estimates at that spurious t_0 in the table notes; scan-level p-values are not computed for a non-rejected scan). The NVDA-only run's DiD is likewise null ($p = 0.76$, best $t_0 = 2017.16$). No ticker-relative break is detectable at either export-control date. This is the substantive result of the paper, and it is a null: whatever costs the controls imposed on these five firms were priced jointly with — and are not separable in these data from — the contemporaneous AI-demand shock lifting the same stocks.

The validation layer's scorecard. Table 1, panel B. Credit: at the DeepSeek cutoff the discovered-RDD family correctly returned *null* (scan $p = 1.00$, density $p = 0.752$, all eight covariate placebos at Holm $p = 1.0$; the fuzzy $\hat{\tau}_{\text{2SLS}} = 4.07$, se 2.64, was discarded), and the pipeline refused rather than fabricated where inputs were missing (`needs_input`; dee nulls; two loud failures quoted in Section 3). Debit: every "credible" verdict on these series — three kinks and one RDD — fails inspection, and cross-validated bandwidth selection actively favors the artifact, choosing windows wide enough to straddle regimes.

5 Caveats and Conclusion

Five caveats bound the claims. First, a unit correction to the working notes of record: the DeepSeek-week figures -17.2% / -5.6% are one-week *log* returns (log-points); the simple returns are -15.8% and -5.4%. Second, the level break and the ± 26-week slopes are descriptive — a check on magnitudes against the formal event-study machinery [10, 7], not a substitute for it. Third, all causal-design caveats for time-as-running-variable apply with full force [8], and `natex`'s own recorded warning — a kink in a calendar-time running variable is a before/after slope contrast — should be read as governing every panel-A row. Fourth, the event window is confounded by construction: ChatGPT lands eight weeks after the BIS-2022 cutoff, and the 2023–2025 AI rally overlaps everything after it; the DiD null is therefore a statement about non-separability in these data, not evidence that the controls were costless. Fifth, five tickers are a narrow panel; the SuDDDS permutation floor and the vacuous placebo batteries both reflect that thinness.

The conclusion is a methods verdict with one substantive corollary. Methods: on trending financial series, an automated validation layer is asymmetric — its refusals, failures, and nulls were all correct and informative, while every one of its "credible" verdicts was an artifact of trend regimes, role search, or vacuous batteries; cross-validated bandwidths made the kink artifacts *more* confident, not less; "credible" on a calendar-time design is a prompt for the checks in Figure 1, not a result. Substance: the one clean event in these data is DeepSeek — an 11.6 log-point one-week repricing of NVDA against its own sector with no local trend change — while the export-control rounds left no detectable ticker-relative break at all, consistent with their costs having been priced against the contemporaneous AI-demand shock rather than as standalone news.

Reproducibility. All estimates come from three `natex` v0.2.0 survey runs at seed 0 (2026-07-18; the kink estimator is deterministic) on frozen weekly price extracts not committed to the repository. Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the level break, the local slopes, and the three headline kink estimates against the numbers of record before drawing.

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An Honest Miss: Chinchilla as an Adoption Ramp in a Discontinuity Scan of Tokens per Parameter

Hauke Hillebrandt*

July 2026

analysis pass of record: 2026-07, `natex` v0.2.0, seed 0

Abstract

A LoRD3 continuous-treatment discontinuity scan of $\log_{10}$ training tokens per parameter over calendar time, run blind on 549 language models (2019–2024) from Epoch AI's notable-models panel, produces a single significant top discovery: a break centered on 2022-09-22 (LLR 8.687, scan $p \leq 0.01$, robust to background degree and to neighborhood size $k \geq 50$). The known truth — the Chinchilla compute-optimal scaling result, posted 2022-03-29, after which the global median tokens-per-parameter ratio rose from 1.7 to 29.2 — is *not* localized: the best candidate near the announcement ranks 158 of 547 (LLR 1.39). The discovered break is six months late and points the wrong way: a downward local contrast of 1.63 $\log_{10}$ units ($\times 43$) at the Sep/Oct 2022 boundary, where the downstream edge of the Chinchilla adoption wave collides with a measurement artifact — fine-tuned releases whose recorded token counts cover only the fine-tuning run over full base parameters. Nulls are reported as nulls: the placebo test is vacuous (no non-forcing covariates), the density test fails ($p = 0.0041$) but is uninformative for a calendar-time forcing, post-selection 2SLS is weak-instrument, and an independent kink audit rejects a slope-change reading. The verdict is an artifact — an honest miss, correctly diagnosed — with two transportable lessons: diffusion-style treatments need a ramp/slope statistic, not only a level-break statistic, and fine-tune rows must be filtered from tokens-per-parameter panels before any scan.

1 Introduction

On 29 March 2022, Hoffmann et al. posted the "Chinchilla" result: for compute-optimal training, model size and training tokens should scale in equal proportion, implying that the then-standard practice — descended from the earlier scaling-law fits of [7] — left frontier models substantially undertrained on data [6]. Release practice responded within months, and by 2023 the industry had moved past compute-optimality into deliberate over-training of small models for inference economy, exemplified by LLaMA's trillion-token training runs [12] and rationalized by inference-adjusted scaling laws [10]. Chinchilla itself has since been re-examined: [2] replicate its parametric fit and find inconsistencies in the reported estimates, while the broader quantitative history of training-run inputs is documented by [11] and maintained in Epoch AI's notable-models dataset [4].

This gives a rare, clean testbed for *automated natural-experiment discovery*: a panel where the event, its date, its sign, and its approximate magnitude are known ex ante. This paper runs the

*University College London. Email: `ucjthhi@ucl.ac.uk`. This paper and the underlying `natex` software were prepared with substantial assistance from Anthropic's Claude models; the author reviewed the analyses and text and is responsible for all remaining errors.

`natex` toolkit's LoRD3 scan [9, 5] on tokens per parameter over calendar time, blind to the event date, and asks: does the scanner find Chinchilla? The answer is no — and the anatomy of the miss is the contribution. Event-dated diffusion processes enter publication data as ramps, and a local level-break statistic rewards sharp edges, not ramps; dating the cause is an interrupted-time-series problem [1], not a discontinuity-scan problem.

2 Data

The panel is a frozen extract of Epoch AI's notable-models database [4]: 549 language models released 2019-01 through 2024-12 with non-missing parameter and training-token counts. Columns: release date (the forcing variable, as days since 2019-01-01), $\log_{10}$ tokens per parameter (the treatment), $\log_{10}$ parameters (the outcome), $\log_{10}$ training compute (21% missing) and an open-weights flag (3.6% missing), both excluded from the scan to avoid listwise deletion. The known truth: Chinchilla was posted 2022-03-29 (day ≈ 1183); across that date the global median tokens per parameter moves from 1.69 to 29.17 ($n = 185$ pre, 364 post), a median-based shift of $+1.24$ $\log_{10}$ units ($+1.12$ on group means, $2.2 \rightarrow 29$; Mann–Whitney $p = 1.6 \times 10^{-7}$ on an earlier extract, medians $2.2 \rightarrow 29.1$, $n = 133/307$). At monthly resolution the shift is a ramp, not a step: window medians run 1.7 (pre-April 2022) $\rightarrow$ 5.7 (mid-April to mid-May) $\rightarrow \approx 28$ (mid-May to mid-July 2022).

3 Design and Methods

LoRD3 continuous-treatment scan. Following [5] as implemented in `natex` v0.2.0 [9], the scan fits a background model of the treatment on the forcing variable (OLS, degree 1; degree 2 as robustness), then scores every k-nearest-neighbor neighborhood ($k = 50$) of every data point for the best split of its members into two half-spaces, using a normal likelihood ratio on background residuals with locally estimated variances. The top discovery is the neighborhood-split with the maximum LLR; its p-value comes from a fitted-null Monte Carlo with $q = 99$ replicas and a +1-rank rule, so the minimum attainable p is 0.010 — reported throughout as $p \leq 0.01$, never as an exact value. The specification is treatment = $\log_{10}$ tokens per parameter, forcing = days since 2019-01-01 (forcing influence 1.0), outcome = $\log_{10}$ parameters, no covariates; pipeline `natex study` $\rightarrow$ `natex discover --k 50 --q 99 --seed 0`, with localization detail and robustness via the Python API, seed 0 throughout.

Validation battery, stated as run. The battery's nulls and refusals are reported, not patched over. (i) Randomization: $p = 0.010$ at both background degrees — the $q = 99$ floor. (ii) Placebo covariates: `placebo_passed = True` but *vacuous* — `placebo_holm = {}`, because the design has no non-forcing covariates to test. (iii) Density: $p = 0.0041$, a *fail*; this is expected and uninformative here, because the forcing is calendar time with strongly nonstationary publication intensity, and McCrary-style manipulation logic [8] does not apply to a time forcing — the test is retained as a falsification check only. (iv) Post-selection effects on $\log_{10}$ parameters (2SLS and Wald at the discovered cutoff) are *advisory*: estimated on the discovery sample with no honest split and instrumented by a partly artifactual break, they must not be read as causal scaling-law parameters. (v) An independent kink-design audit of the same panel (a difference-in-slopes design [3]) cross-checks any slope-change reading.

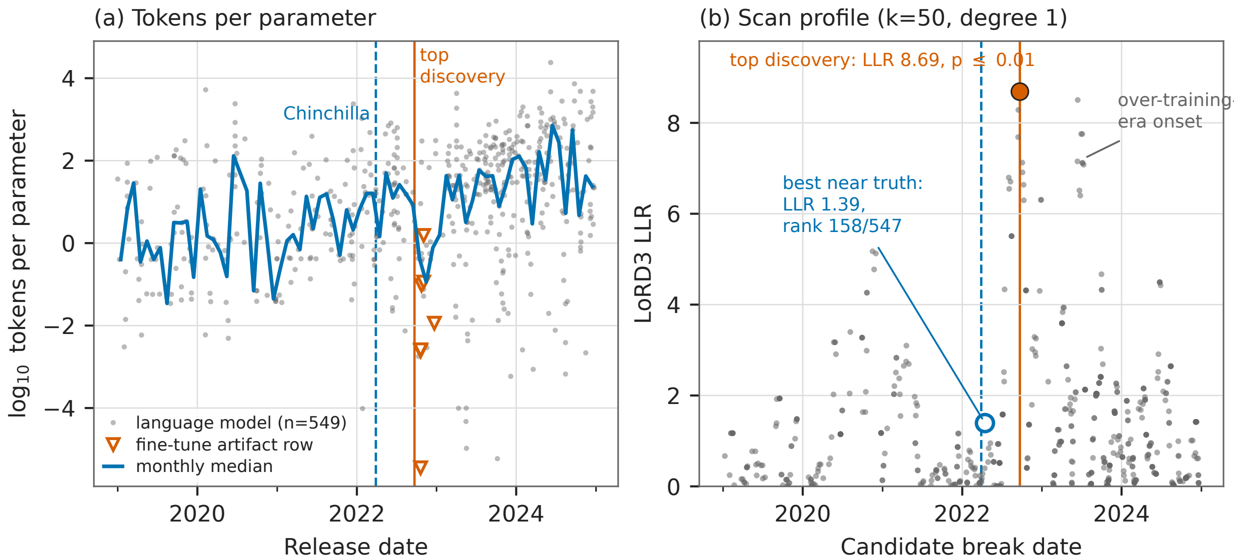

Figure 1: (a) $\log_{10}$ tokens per parameter for all 549 models, with the monthly median (line), the Chinchilla posting date (dashed), the scan's top discovery (solid), and the seven verified fine-tune token-counting artifact rows (open triangles; Table 1, panel C). (b) LoRD3 scan profile: each candidate neighborhood's LLR at its center date ($k = 50$, degree 1, $n = 547$ candidates). The top discovery (filled marker, 2022-09-22, LLR 8.69, $p \leq 0.01$) sits at the Sep/Oct 2022 artifact edge; the best candidate within ±45 days of the truth (open marker, 2022-04-15) scores LLR 1.39, rank 158/547. The 2023-06/07 cluster is the onset of the deliberately over-trained era.

4 Results

The discovery. The scan's top discovery (Table 1, panel A; Figure 1b) is centered on day 1360 = 2022-09-22 (split normal −1 along the forcing axis): LLR = 8.687 (8.687416 unrounded in the run report), scan $p = 0.010$ — the $q = 99$ Monte Carlo floor, i.e. $p \leq 0.01$ — over 547 candidate neighborhoods from $n = 549$ rows. A degree-2 background reproduces the same top center with LLR 8.836 (randomization $p = 0.01$; null max LLRs 1.8–7.6 against observed 8.69/8.84), and $k = 60$ moves it only to 2022-08-15; but $k = 30/40$ latch onto a 2020-10 composition boundary instead (Table 1, panel B), so localization is stable only for $k \geq 50$.

The known truth is not localized. Within ±45 days of 2022-03-29 the best candidate neighborhood is centered 2022-04-15 with LLR 1.39 — rank 158 of 547. The reason is visible in Figure 1a: at daily resolution the Chinchilla shift enters the published record as a 2–3 month adoption ramp, and a local level-break statistic rewards sharp edges, not ramps.

What the discovered break actually is. The top neighborhood splits exactly at the Sep/Oct 2022 boundary into group A (2022-06-21 to 2022-09-22, $n = 25$, median tokens/param 13.6 — the post-Chinchilla adoption wave) versus group B (2022-10-03 to 2022-12-22, $n = 25$, median 0.4): a *downward* local group-mean contrast of 1.63 $\log_{10}$ units (×43), against the known *upward* global shift of +1.12 $\log_{10}$ (+1.24 median-based). Local-linear first-stage fits jump −1.7 to −2.0 $\log_{10}$ at bandwidths of 120–240 days. Group B is dominated by fine-tuned and continued-pretraining releases whose Epoch token field records only fine-tuning tokens over full base parameters (Table 1,

Table 1: Chinchilla scan: numbers of record (`natex` v0.2.0, seed 0).

Quantity	Value	Quantity	Value
Panel A: top discovery ($k = 50$, $q = 99$, degree 1; $n = 549$, 547 candidates)			
center	2022-09-22 (day 1360)	LLR	8.687
scan p	0.010 = floor; $p \leq 0.01$	local contrast	-1.63 $\log_{10}$ ($\times 43$, down)
degree-2 center	2022-09-22, LLR 8.836	first stage	-1.7 to -2.0 $\log_{10}$ (bw 120–240d)
Panel B: k-sensitivity and truth localization			
$k = 30$	2020-09-29 (LLR 7.2)	$k = 40$	2020-10-21 (LLR 9.0)
$k = 50$	2022-09-22 (LLR 8.7)	$k = 60$	2022-08-15 (LLR 9.8)
truth ±45d best	2022-04-15, LLR 1.39	truth rank	158 / 547
global shift (true)	$+1.12$ $\log_{10}$ ($2.2 \to 29$, up)	medians	$1.69 \to 29.17$ ($n = 185/364$)
Panel C: verified fine-tune artifact rows (tokens/param; group B median 0.4 vs. A 13.6)			
Flan-PaLM 540B	0.0026	U-PaLM	0.0024
LMSI-PaLM	3.6×10^{-6}	OPT-IML 175B	0.011
Tk-Instruct	0.095	BLOOMZ-176B	0.114
mT0-13B	1.54		
Panel D: diagnostics and post-selection effects (advisory)			
randomization	$p = 0.010$ (deg. 1 and 2)	placebo	passed, *vacuous* (no covariates)
density	$p = 0.0041$, fails†	kink audit	rejected; $t \in [-0.91, +1.48]$
2SLS $\hat{\tau}$	1.017 (se 0.900), weak	CI	$[-0.747, 2.781]$, 1st-st. $t = 1.89$
Wald $\hat{\tau}$	0.557 (se 0.257)	CI	$[0.053, 1.062]$, 1st-st. $t = 4.01$

Notes: †expected and uninformative for a calendar-time forcing with nonstationary publication intensity (McCrary manipulation logic does not apply). Scan p-values from a $q = 99$ fitted-null Monte Carlo with a +1-rank rule; 0.010 is the minimum attainable. Effects are estimated on the discovery sample with no honest split. Artifact rows verified in the source panel: the token field records fine-tuning tokens only, over full base parameters.

panel C) plus small academic models. The top discovery is therefore the adoption wave's downstream edge colliding with a token-accounting artifact, ≈6 months after the event and with the opposite sign. The secondary discovered cluster (2023-06-13 to 2023-07-06, LLR 6.7–8.5) is the onset of the deliberately over-trained era [12, 10]; the December-2022 monthly median is already 148 tokens/param, and the median of the top-25 centers is day 1371 = 2022-10-03.

Effects and cross-checks, all null or advisory. Post-selection effects on $\log_{10}$ parameters at the discovered cutoff (Table 1, panel D): the 2SLS estimate is weak-instrument (first-stage $t = 1.89$; CI $[-0.747, 2.781]$); the Wald estimate is 0.557 (CI $[0.053, 1.062]$, first-stage $t = 4.01$). Both are discovery-sample estimates with no honest split, instrumented by a partly artifactual break — diagnostics, not causal scaling-law parameters. The independent kink audit *rejects* a Chinchilla regime change read as a slope change: a sign-unstable null with t from -0.91 to $+1.48$ across bandwidths.

5 Caveats and Conclusion

Five caveats bound the claims. First, the value of this exercise is purely methodological and diagnostic: no causal adoption claim is licensed; the scan is a regime-edge detector, and dating the cause of a diffusion process needs an interrupted-time-series or difference-in-differences framing [1] plus publication-lag domain knowledge. Second, the scan $p = 0.010$ is the minimum attainable with $q = 99$ replicas and must be reported as $p \leq 0.01$, not as an exact value. Third, the density test fails ($p = 0.0041$) but is uninformative for a calendar-time forcing with nonstationary publication intensity, and the placebo pass is vacuous (`placebo_holm = {}`; no non-forcing covariates to test). Fourth, the post-selection 2SLS/Wald effects are estimated on the discovery sample with no honest split and are instrumented by a partly artifactual break; they must not be read as causal scaling-law parameters. Fifth, "robust to k" holds only for $k \geq 50$: at $k = 30/40$ the scan latches onto a 2020-10 composition boundary (tiny academic LMs against BERT-era corpus models), so the robustness claim must be stated conditionally.

The verdict is an artifact — an honest miss, correctly diagnosed: the scan found real, significant structure whose two regimes bracket the truth, but it dated the adoption wave's sharp downstream *edge*, not the event, which entered the record as a ramp. Two lessons transport beyond this dataset. For method builders: scan batteries aimed at event-dated diffusion treatments need a ramp/slope statistic — a kink-style scan in the sense of [3] — alongside the level-break statistic, or announcements will systematically lose to their own adoption edges. For users of the Epoch panel [4]: fine-tuned and continued-pretraining rows manufacture a spurious low-tokens-per-parameter cluster in late 2022 and must be filtered before tokens-per-parameter data are used for scaling-law or event-study claims [6, 2].

Reproducibility. All estimates: `natex` v0.2.0 [9], seed 0, on a frozen extract of the Epoch panel [4] (not committed). Figure 1 regenerates deterministically from `figures/make_fig.py` (committed), which asserts the headline numbers of record before drawing.

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Recovering Proposition 99 Blind: Validating the `natex` Pipeline Against the Abadie–Diamond–Hainmueller Benchmark

Hauke Hillebrandt*

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run of record re-verified 2026-07-18, `natex` v0.2.0, seeds 0/1/2 (donor path rng-free)

Abstract

Run blind on the Abadie–Diamond–Hainmueller (ADH) California smoking panel (39 states × 31 years), the `natex` pipeline reproduces the canonical Proposition 99 benchmark end to end. All three SuDDDS scan methods (greedy, wcc, single_delta; normal model) recover (California, $t_0 =$ 1989) exactly — precision = recall = 1.0 over the 31 California records, LLR 13.82; the Bernoulli-model scan is also exact (LLR 13.86), a y-blindness test proves the scan bitwise independent of the outcome, and a Bernoulli($\hat{p}$)-null Monte Carlo (Q = 99, seed 1) gives p = 0.010 (the dependence-preserving AR(1) null is structurally conservative on a policy dummy, pinning at p = 1.0; documented, not claimed as evidence). Deterministic full-pool simplex weights put summed weight 0.955 on ADH's five published donors, and the synthetic-control ATT over 1989–2000 is −19.514 packs per capita, matching ADH's canonical average gap of roughly −19. Nulls are reported as nulls: the exact in-space RMSPE-ratio placebo ranks California 3/39 (p = 0.077; ADH's 1/39 requires their poor-pre-fit exclusion, kept opt-in here), and under the corrected two-sided studentized placebo statistic the source thesis's claim that all three effect estimates are significant at 5% does *not* survive (ranks 13/39, 7/39, 9/39). By construction this exercise has zero novelty; its value is credibility transfer — external-validity anchoring for the `natex` studies on novel data.

1 Introduction

California's Proposition 99, passed in November 1988 and effective January 1989, raised the state cigarette excise tax by 25 cents per pack and funded a large anti-smoking campaign. Taxation and media spending each measurably reduced cigarette sales [9], and the program is associated with accelerated declines in consumption and heart-disease mortality [6]. It is also the canonical testbed of the synthetic control method: building on the comparative case-study estimator of [4], Abadie, Diamond, and Hainmueller [2] constructed a "synthetic California" from a weighted combination of donor states and estimated an average gap of roughly −19 packs per capita over 1989–2000. That example anchors a large methodological literature [3, 1, 5] and is arguably the most-replicated benchmark in comparative case studies.

This paper uses it the other way around: not to learn about Proposition 99, but to validate a pipeline. `natex` [11] is an automated natural-experiment discovery-and-estimation toolkit in the LoRD3 lineage [7, 10]; its difference-in-differences arm implements SuDDDS (Subset Discovery of

*University College London. Email: `ucjthhi@ucl.ac.uk`. This paper and the underlying `natex` software were prepared with substantial assistance from Anthropic's Claude models; the author reviewed the analyses and text and is responsible for all remaining errors.

Difference-in-Differences) from chapter 6 of [8], whose printed Table 6.1 evaluates the method on this same smoking panel. We run the full pipeline *blind* — the scan is never told the treated unit or the intervention year, and provably never reads the outcome — and ask whether it rediscovers (California, 1989) and reproduces the ADH estimate. It does — exactly, deterministically, and with every deviation from the thesis's printed values investigated and reconciled. Because the exercise replicates a known benchmark, we write it up short, structurally as validation: its only role is to transfer credibility to the `natex` studies on novel data.

2 Data

The data are the ADH California tobacco panel [2] as shipped in the `natex` dataset registry (dataset `prop99`; 1,209 rows): annual state-level per-capita cigarette sales (packs, from tax revenues) for 39 states, 1970–2000 — California plus ADH's 38 donor states — with auxiliary covariates (log income, beer consumption, share aged 15–24, retail price). The panel is balanced (39×31, no missing outcome cells). Treatment is the policy dummy: California from 1989 onward. (The source thesis [8] reports "30 US states" and does not document its panel; ours is ADH's 39-state pool.)

3 Design and Methods

Blind discovery (SuDDDS). SuDDDS [8], as implemented and corrected in `natex` [11], scans panel records (x_i, t_i, θ_i) — covariates, time, treatment — for a subset of units and an intervention time T_0 at which θ jumps, scoring candidates by a log-likelihood ratio and never reading the outcome y. Protocol, fixed before calibration: windows $(5, 8, 10)$, 4 time bins, 8 restarts, seed 0, and background degree 0 (unit effects only). Degree 0 is required, not tuning: θ is a pure policy dummy whose only time variation is the candidate discontinuity, so any global time polynomial can only fit the jump itself; with degree 1 the leaked slope manufactures a spurious whole-panel optimum (the thesis never reports its background — an unreported-hyperparameter risk). All three scan methods (greedy, wcc, single_delta) are run under the normal model, plus a wcc scan under the Bernoulli model, the corrected default for a binary treatment. A y-blindness test re-runs the scan with the outcome deleted and requires bitwise-identical discoveries.

Calibration. The top discovery's LLR is calibrated by a fitted-null Monte Carlo with $Q = 99$ replicas and a +1-rank rule (minimum attainable $p = 0.010$), drawing nulls from Bernoulli($\hat{p}$) — the correct null family for a binary θ. As a diagnostic we also run the dependence-preserving AR(1)-unit null, which is *structurally conservative* on a deterministic policy dummy: 38 of 39 units have identically-zero residuals, so the pooled AR(1) fit absorbs California's step as noise and hands it to every replica; its p pins at 1.0, documented as a regression, never claimed as evidence.

Estimation conditioned on the discovery. Three effect estimators are run on the recovered discovery over the full post period 1989–2000: two-way difference-in-differences (DD), synthetic control (simplex-weighted donor fit on pre-period outcomes, as in [2]), and GESS (greedy control-group expansion, thesis ch. 6). Separately, the `natex` synthetic-control donor path is run on the raw (state, year, cigsale) matrix with the full donor pool and with an $n = 8$ pre-fit-ranked variant; this path is deterministic — no rng anywhere — and bitwise-identical across runs.

Honest inference. Significance is assessed by exact, enumerated placebo tests, with the caveats stated plainly. (i) The in-space RMSPE-ratio placebo [2, 3] refits every one of the 38 control states as

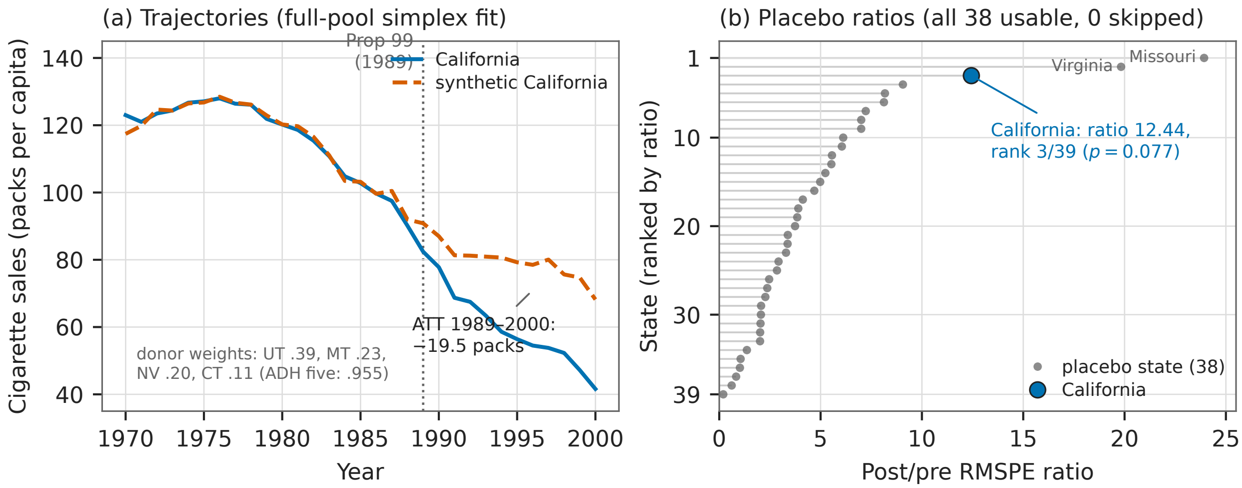

Figure 1: (a) California vs. synthetic California (deterministic full-pool simplex weights), 1970–2000: pre-RMSPE 1.656, post-period gap averaging −19.514 packs per capita. (b) In-space placebo: post/pre RMSPE ratios for all 39 states (all 38 placebos usable, 0 skipped). California ranks 3/39 ($p = 0.077$); the two out-ranking states, Missouri and Virginia, are poor-pre-fit placebos that ADH's exclusion rule would discard.

a pseudo-treated unit (`exclude_poor_fit = None`: all 38 usable, 0 skipped) and ranks California's post/pre RMSPE ratio among all 39; ADH's $p = 1/39$ requires their poor-pre-fit exclusion, kept opt-in in `natex`. (ii) The corrected effect-level test is a two-sided studentized statistic $|\hat{\tau}/\mathrm{se}|$ with +1-rank p-values over the 38 enumerated placebo states — under it, genuinely non-parallel placebo states (flat post-shifted gaps with tiny standard errors) can out-t California's trending gap. (iii) The validation battery's degenerate cases report "cannot test" (NaN, failed) rather than fabricating passes: composition reduces to a one-row table for a single-unit discovery, and anticipation has zero pre-period treatment variance.

4 Results

Blind discovery is exact. All three SuDDDS scan methods under the normal model return (California, $t_0 = 1989.0$) as the top discovery, with the discovered subset exactly the 31 California records: precision = recall = 1.0, LLR 13.82. The Bernoulli-model scan is also exact, LLR 13.86. The y-blindness test passes bitwise (Table 1, panel A).

The discovery is significant under the correct null. Against Bernoulli($\hat{p}$) nulls ($Q = 99$, seed 1) the observed max LLR 13.86 exceeds the null max 7.95 (null q90 4.67): $p = 0.010$, the +1-rank floor. The AR(1)-unit diagnostic pins at $p = 1.0$, as expected for a deterministic policy dummy (Section 3); recorded as a structural limitation of dependence-preserving nulls, not as evidence against the discovery (Table 1, panel B).

Effects match ADH; the printed thesis numbers reconcile. On the full 1989–2000 post period, DD gives $\hat{\tau} = -27.349$ packs (pre-MSE 51.23), synthetic control gives $\hat{\tau} = -19.514$ (pre-MSE 2.74), and GESS gives $\hat{\tau} = -26.653$ (control: Montana, pre-MSE 18.35). Signs and the pre-fit ordering match the thesis's Table 6.1 (−10.94 / −8.96 / −6.67): synthetic control's pre-MSE is far below DD's, which is exactly ADH's argument for it. The magnitudes differ by a factor of $\sim$2–2.5 because the thesis's printed values correspond to an unreported effective post window of about five years, while the gap accumulates over time; restricting `natex` to 1989–1993 gives DD -18.8 and synthetic -12.3, bracketing the printed values (Table 1, panel C).

Donor selection recovers ADH's pool. The deterministic full-pool simplex fit puts nonzero weight on six donors, with summed weight 0.955 on ADH's five published donors, and the four heavyweights are exactly ADH's own {Utah, Montana, Nevada, Connecticut} (weights in Table 1, panel D). $\mathrm{ATT}_{\mathrm{post}} = -19.514$ packs per capita, with pre-RMSPE 1.656 against post-RMSPE > 10 (the ADH-style pre/post contrast); the 8-donor variant gives ATT -22.648, pre-RMSPE 3.671 (Figure 1a).

Honest inference: what survives and what does not. The exact in-space RMSPE-ratio placebo ranks California 3 of 39 in both variants (treated ratio 12.440 full pool, 6.570 top-8): $p = 3/39 = 0.077$ (Figure 1b). ADH's 1/39 arises only under their poor-pre-fit exclusion, which `natex` keeps opt-in. The corrected two-sided studentized τ placebos over the 38 enumerated states give DD rank 13/39 ($p = 0.333$), synthetic control 7/39 ($p \leq 8/39 = 0.205$), and GESS 9/39 ($p = 0.231$): the thesis's claim that all three are significant at 5% does *not* survive the corrected statistic, and we report that as the finding it is (Table 1, panel E).

Run of record and reproducibility. The entire benchmark is executable: running `uv run pytest -m backtest` with `NATEX_DATA` set to the data root, on the modules `test_prop99.py` and `test_prop99_donors.py` under `tests/backtests/`, gives 16 passed, 1 xfailed in 79.74 s (re-verified 2026-07-18). The single xfail is a non-blocking instrument-selection stretch goal on a *different* dataset, pinned as a documented finding; all 15 Proposition 99 tests pass. All estimates: `natex` v0.2.0 [11] on the registry copy of the ADH panel (not committed). Figure 1 regenerates deterministically from `figures/make_fig.py` (committed), which asserts the headline numbers of record before drawing.

5 Caveats and Conclusion

Three caveats bound the claims. First, this exercise has zero novelty by construction: Proposition 99 on the ADH panel is the most-replicated synthetic-control benchmark in economics [2, 1], and a pipeline that failed it would be disqualified rather than informative; its role is external-validity anchoring for the `natex` studies on novel data, and it should be read as methods validation, not as a finding about tobacco policy. Second, effect magnitudes here are $\sim$2–2.5$\times$ the source thesis's printed Table 6.1 values because `natex` estimates on the full 1989–2000 post period while the thesis's numbers match an unreported $\sim$5-year effective window; the thesis also never reports its panel ("30 US states" vs. ADH's 39) or its treatment background. Signs, ordering, and pre-fit ranking agree, and the synthetic-control estimate matches ADH's canonical ≈ -19 — but the reconciliation must be stated, not glossed. Third, the honest-inference results are genuinely weaker than the textbook telling: without ADH's poor-pre-fit exclusion the in-space placebo p is 0.077, not $1/39 = 0.026$, and under the corrected two-sided studentized placebo statistic none of the three effect estimators clears 5% on this panel. These are properties of the inference conventions, faithfully surfaced.

Within those bounds, the conclusion is simple: run blind, `natex` rediscovers (California, 1989) exactly by every scan method, calibrates it significantly under the correct null, selects ADH's donors with 0.955 of the simplex weight, and reproduces the canonical ≈ -19 synthetic-control estimate

Table 1: Proposition 99 backtest: numbers of record (`natex` v0.2.0; seeds 0/1/2, donor path rng-free).

Quantity	Value	Quantity	Value
Panel A: blind discovery (SuDDDS; windows (5,8,10), bins 4, restarts 8, degree 0, seed 0)			
greedy / wcc / single_delta	(California, 1989.0)	precision, recall	1.0, 1.0 (31 CA records)
normal-model LLR	13.82	Bernoulli wcc	exact; LLR 13.86
y-blindness	bitwise identical		
Panel B: discovery calibration ($Q = 99$, +1-rank, seed 1)			
Bernoulli($\hat{p}$) null	$p = 0.010$	observed vs null max	13.86 vs 7.95 (q90 4.67)
AR(1)-unit null	$p = 1.0$, conservative†		
Panel C: effects on the discovery, post 1989–2000 (thesis Table 6.1: $-10.94/-8.96/-6.67$)			
DD $\hat{\tau}$	-27.349 (pre-MSE 51.23)	synthetic $\hat{\tau}$	-19.514 (pre-MSE 2.74)
GESS $\hat{\tau}$	-26.653 (MT, pre-MSE 18.35)	1989–1993 restriction	DD -18.8, synthetic -12.3
Panel D: synthetic-control donor path (deterministic, bitwise-reproducible)			
weights	UT .394, MT .232, NV .205	ADH-five summed weight	0.955
	CT .109, NH .045, CO .015	$\mathrm{ATT}_{\mathrm{post}}$	-19.514 packs (ADH ≈ -19)
pre- / post-RMSPE	1.656 / > 10	8-donor variant	ATT -22.648, pre 3.671
Panel E: exact placebo inference (enumerated; no poor-pre-fit exclusion)			
RMSPE ratio (treated)	12.440 full / 6.570 top-8	rank, p	3/39 both; $p = 3/39 = 0.077$
studentized τ: DD	13/39 ($p = 0.333$)	synthetic	7/39 ($p \leq 8/39 = 0.205$)
GESS	9/39 ($p = 0.231$)	thesis "all sig. at 5%"	does *not* survive

Notes: †structural, not evidential: a deterministic policy dummy leaves 38 of 39 units with identically-zero residuals, so the pooled AR(1) null absorbs the step as noise; pinned as a documented regression. Scan p-values use a $Q = 99$ fitted-null Monte Carlo with a +1-rank rule; 0.010 is the minimum attainable. The synthetic-control placebo rank carries one rank of slack because each placebo refits SLSQP weights.

deterministically. The pipeline earns the benefit of the doubt it needs on data where the answer is not known.

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